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Minimal Hypersurfaces in Closed Riemannian Manifolds via the Allen-Cahn Energy

Semester paper

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Abstract

The existence of minimal hypersurfaces in arbitrary closed Riemannian manifolds is a classical result originally established by the Almgren–Pitts min-max theory. Due to the technical complexity of geometric sweepouts in the space of varifolds, an alternative PDE-based framework was developed using the Allen–Cahn equation. By replacing the singular area functional with the analytically tractable Allen–Cahn energy, min-max methods can be applied to obtain unstable critical points whose transition layers converge to minimal hypersurfaces as the phase parameter approaches zero. This written project provides a comprehensive exposition of this phase-transition approach. We detail the analytical and geometric tools developed by Padilla, Hutchinson, Tonegawa, and Wickramasekera, culminating in Guaraco's 2015 proof of the existence theorem.

Notation

Set and topology

$\mathrm{int}(A)$	interior of the set A
$\overline{A}$	closure of the set A
$V \Subset U$	$\overline{V}$ is compact and $\overline{V} \subset U$
χ_A	indicator function of the set A
$d(\cdot,\cdot)$	distance between points and/or sets
$B_r(x)$	open ball with center x and radius r

Euclidean space and differentiation

$\boldsymbol{B}, \boldsymbol{v}$	bold font indicates vector or tensor quantities
ω_n	volume of the unit ball $B_1(0)$
$\langle x, y \rangle$	Euclidean inner product of x and y
$\lvert x \rvert$	Euclidean norm of x
∂_i	partial derivative with respect to the i-th coordinate
∇f	gradient of f
$D^2 f$	Hessian matrix of f
Δf	Laplacian of f
$\mathrm{div}\, X$	divergence of a vector field X

Measures and varifolds

$\mathcal{L}^n$	n-dimensional Lebesgue measure
$\mathcal{H}^s$	s-dimensional Hausdorff measure
$\mathrm{spt}\, \mu$	support of the measure μ
$\llcorner$	restriction of a measure
$\Theta^{*,k}(\mu, x)$	upper k-dimensional density of the measure μ at x
$\Theta^k_*(\mu, x)$	lower k-dimensional density of the measure μ at x
$\Theta^k(\mu, x)$	k-dimensional density of the measure μ at x
$\eta_{x,r}$	dilation map $y \mapsto \frac{y-x}{r}$
$f_\# \mu$	pushforward of the measure μ by the map f
$G(k, n)$	Grassmannian of unoriented k-dimensional subspaces in $\mathbb{R}^n$
$G_n(\Omega)$	Grassmannian bundle over Ω
∂^*_{ij}	derivative with respect to the (i, j) coordinates of a projection matrix in the Grassmanian
$\boldsymbol{V}_k(\Omega)$	space of k-varifolds in Ω
$\boldsymbol{RV}_k(\Omega)$	space of rectifiable k-varifolds in Ω
$\boldsymbol{IV}_k(\Omega)$	space of integral k-varifolds in Ω
$\boldsymbol{CV}_k(\Omega)$	space of curvature k-varifolds in Ω
$\lVert V \rVert$	weight of the varifold V
$\mathrm{reg}\, V$	set of regular points of the varifold V
$\mathrm{sing}\, V$	set of singular points of the varifold V

Function spaces

$\|x\|_G$	norm in a Banach space G
$L^p(\Omega)$	Lebesgue space of p-integrable functions on Ω
$W^{s,p}(\Omega)$	(s,p) Sobolev space
$H^k(\Omega)$	k-th Hilbert-Sobolev space ($W^{k,2}(\Omega)$)
$\mathcal{C}^k(\Omega)$	space of k-times continuously differentiable functions in Ω
$\mathcal{C}^k_c(\Omega)$	space of k-times continuously differentiable functions with compact support in Ω
$\delta^k F$	k-th variation of a functional F
$(f)^+$	positive part of the function f
$m(u)$	Morse index of u

Riemannian manifolds

$\Delta_g f$	Laplace-Beltrami operator
$\mathrm{d}\mu_g$	volume measure induced by the metric g
$\mathfrak{X}(M)$	space of smooth vector fields on a manifold M
TM	tangent bundle of M
NM	normal bundle of M
$\nabla^M f$	tangential gradient of f in M
$\operatorname{div}_M X$	tangential divergence of $X \in \mathfrak{X}(M)$
$\boldsymbol{H}$	mean curvature vector
$\boldsymbol{B}$	second fundamental form
$\operatorname{Hess} f$	Riemannian Hessian of f
$\exp_x$	exponential map at x
$\operatorname{inj}(x)$	injectivity radius at x

Allen–Cahn

W	double-well potential in the Allen–Cahn equation
σ	surface tension constant associated to the potential W
u_k, ε_k	sequence of critical points of the Allen–Cahn energy with parameter ε_k
C_0, E_0	uniform pointwise and energy bounds of the sequence
V_k	associated varifold sequence
ξ_k	discrepancy function associated to u_k

Contents

Chapter 1

Introduction

The study of minimal surfaces originates from the physical problem of determining the equilibrium shape of a soap film spanning a wire frame. Mathematically, minimal surfaces are defined as critical points of the area functional, serving as a natural generalization of area-minimizing surfaces. The classical problem of finding an area-minimizing surface with a prescribed boundary is Plateau's problem. Since minimizing sequences of smooth surfaces need not converge to smooth surfaces, the direct method of the calculus of variations fails as a means to show existence. This analytical obstruction led to the development of geometric measure theory, which generalizes smooth surfaces to currents and varifolds that possess the necessary compactness properties [Fed14; Sim84].

On closed manifolds (without boundary), strict minimization fails to produce non-trivial surfaces since unconstrained minimizing sequences collapse to a point. Global existence therefore requires finding unstable critical points of the area functional. This approach was first realized for one-dimensional minimal surfaces (geodesics) by Birkhoff [Bir17].

Theorem 1.1 (Birkhoff, 1917) *Every closed Riemannian 2-manifold contains at least one non-trivial closed geodesic.*

Almgren and Pitts generalized this min-max procedure to higher dimensions [Alm65; Alm62; Pit81]. Utilizing geometric measure theory to handle the singularities that arise in higher dimensional sweepouts, Pitts successfully carried out the min-max program. Schoen and Simon subsequently extended the regularity estimates to arbitrary dimensions [SS81].

Theorem 1.2 (Almgren, Pitts, Schoen–Simon) *Every closed Riemannian manifold of dimension $n \geq 3$ contains a smooth, closed, embedded minimal hypersurface.*

To circumvent the technical difficulties of the Almgren–Pitts construction, an alternative PDE-based approach uses the Allen–Cahn equation [AC79]. On a

Riemannian manifold (M, g), the stationary Allen–Cahn equation is

$$\varepsilon \Delta_g u = \frac{W'(u)}{\varepsilon},$$

where $\varepsilon > 0$ is a small parameter and W is a double-well potential. Solutions are critical points of the Allen–Cahn energy functional:

$$E_\varepsilon(u) = \int_M \left(\frac{\varepsilon}{2} |\nabla u|^2 + \frac{W(u)}{\varepsilon} \right) \, \mathrm{d}\mu_g.$$

As $\varepsilon \to 0^+$, solutions concentrate along increasingly sharp interfaces. Under appropriate assumptions, these interfaces converge to minimal hypersurfaces. This allows the replacement of the singular area functional with the analytically tractable Allen–Cahn energy. Applying min-max methods to E_ε yields unstable critical points whose transition layers converge to minimal hypersurfaces. This program was completed by Padilla, Hutchinson, Tonegawa, Wickramasekera, and Guaraco [PT98; HT00; Ton05; TW12; Gua18], providing a PDE-based proof of the existence theorem.

These notes develop the analytical and geometric tools leading to Guaraco's proof. Chapter 2 establishes the geometric preliminaries and introduces varifolds. Chapter 3 develops the Allen–Cahn model, including the Modica–Mortola theorem [MM77] on the Γ-convergence of the energy. Chapter 4 presents the varifold theory of Hutchinson and Tonegawa [HT00] for critical points of the Allen–Cahn energy. Chapter 5 covers Tonegawa's results on stable critical points [Ton05]. Chapter 6 develops the regularity theory of the limiting stable varifolds using the work of Wickramasekera [Wic14] and includes the generalization to closed Riemannian manifolds. Finally, Chapter 7 combines these ingredients to present Guaraco's existence proof [Gua18].

Throughout these notes, we assume familiarity with standard PDE and Sobolev space theory, Riemannian geometry, and basic geometric measure theory, including Hausdorff measures and Hausdorff dimension. Prior exposure to classical minimal surface theory and the theory of varifolds is helpful but not required, as the necessary definitions and fundamental properties will be reviewed in Chapter 2.

Chapter 2

Preliminaries

Minimal surfaces, which mathematically model the physical behavior of soap films in equilibrium, have been a central object of study in differential geometry and the calculus of variations since Lagrange introduced them in the 18th century. While the geometric definition of a minimal surface as a critical point of the area functional is very natural, the classical framework of smooth manifolds falls short when attempting to prove general existence results. The standard approach to finding minimizers or critical points, such as the direct method of the calculus of variations or min-max theory, relies heavily on compactness. However, the space of smooth submanifolds lacks the necessary compactness properties. Consequently, one must first enlarge the space of admissible surfaces to a class of generalized objects that enjoy strong compactness properties and solve the variational problem in a weaker sense.

This chapter establishes the geometric and analytic preliminaries required for the remainder of these notes. In the first section, we review the classical theory of minimal surfaces, focusing on the area functional, its first and second variations, and the concept of stability. In the second section, we introduce the theory of varifolds. Varifolds provide the measure-theoretic generalization of submanifolds that will serve as our main analytic tool for studying generalized minimal surfaces.

2.1 Some Background on Minimal Surfaces

In this section, we review some basic geometric aspects of minimal surface theory, focusing on the area functional and its first and second variations in the general setting of Riemannian manifolds. A good reference covering the geometric side of minimal surfaces is [CM11]; additionally the derivation of the formulas for the first and second variations in this section can be found in [Li12].

For the entirety of these notes, (M, g) will denote a closed Riemannian n-manifold, i.e., a $\mathcal{C}^\infty$ compact manifold M without boundary ($\partial M = \emptyset$) equipped with a $\mathcal{C}^\infty$ Riemannian metric g. It is assumed throughout that the reader is familiar with Riemannian manifolds, as presented for instance in [Lee18].

Definition 2.1 Let $f : \Sigma \hookrightarrow M$ be a closed embedded k-manifold. We define the *area* of Σ in M by

$$\mathrm{Area}_M(\Sigma) = \int_\Sigma \mathrm{d}\mathcal{H}^k,$$

where $\mathcal{H}^k$ is the k-dimensional Hausdorff measure on Σ pulled back from the one on M. The map which assigns to a submanifold its area is known as the *area functional*.

It is a common abuse of notation to identify the abstract manifold Σ with its image $f(\Sigma) \subset M$ under the embedding.

Remark 2.2 *The Hausdorff measure $\mathcal{H}^s$ on a Riemannian manifold is constructed in the same way as the Hausdorff measure on a metric space, with the distance being the geodesic distance induced by the Riemannian structure. This measure is equivalent to the measure induced by the volume form $\mathrm{d}V_g$ (or more generally the volume density if M is non-orientable) whenever s is an integer.*

The Plateau Problem is concerned with the existence of a k-submanifold Σ minimizing the area among all other submanifolds Σ' sharing the same fixed $(k-1)$-submanifold B as their boundary:

$$\mathrm{Area}_M(\Sigma) \leq \mathrm{Area}_M(\Sigma') \text{ for any } \Sigma' \text{ such that } \partial\Sigma' = B.$$

A first approach towards answering this question has us treating the area functional much like we would treat a real function, such that Σ can be seen as a critical point of $\mathrm{Area}_M : X \to [0, \infty)$ where X is the set of all suitable candidates. In these notes we are not concerned with the problem of minimizing area under a boundary constraint but rather the closely related notion of unconstrained critical points of Area_M. To correctly define these, we require a notion of the derivative of the area of a submanifold.

Definition 2.3 Let $f : \Sigma \hookrightarrow M$ be a closed embedded k-manifold and $X \in \mathfrak{X}(M)$ be a vector field. A *variation* of Σ in the direction of X is a smooth 1-parameter family of embeddings $F : (-\varepsilon, \varepsilon) \times \Sigma \to M$, denoted by $F_t(x) = F(t, x)$, such that $F_0 = f$ and $X(f(x)) = \partial_t|_{t=0} F_t(x)$.

The *first variation of area* of Σ in the direction of X is defined as

$$\delta\,\mathrm{Area}_M(\Sigma)(X) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathrm{Area}_M(\Sigma_t) := \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \int_\Sigma \mathrm{d}\mathcal{H}^k_t, \tag{2.1}$$

where $\mathcal{H}^k_t$ is the k-dimensional Hausdorff measure on Σ pulled back from the one on M with F_t.

Definition 2.4 A closed embedded k-manifold $f : \Sigma \hookrightarrow M$ is called a *minimal surface* if it is a critical point of the area functional, i.e.,

$$\delta \operatorname{Area}_M(\Sigma)(X) = 0$$

for all $X \in \mathfrak{X}(M)$.

It is clear that any area minimizing submanifold is a minimal surface; however, the converse is not necessarily true as a minimal surface could actually be a maximum or a saddle point of the area functional, much like in the case of a real function. Some basic examples of minimal surfaces include a plane or a catenoid in $\mathbb{R}^3$. Another familiar example is that of geodesics of M, as they are the critical points of the length functional which is exactly the area functional for 1-submanifolds.

Computing explicitly the RHS in (2.1) yields the very important formula for the first variation of area.

Theorem 2.5 (First Variation of Area) *Let $f : \Sigma \hookrightarrow M$ be a closed embedded k-manifold without boundary, and $X \in \mathfrak{X}(M)$. The first variation of area of Σ in the direction of X is given by*

$$\delta \operatorname{Area}_M(\Sigma)(X) = \int_\Sigma \operatorname{div}_\Sigma X \, \mathrm{d}\mathcal{H}^k,$$

where $\operatorname{div}_\Sigma$ is the divergence on $f(\Sigma)$ induced by the ambient Riemannian metric g. Furthermore, if $X = X^\top + X^\perp$ with $X^\top \in Tf(\Sigma)$ and $X^\perp \in Nf(\Sigma)$ then

$$\delta \operatorname{Area}_M(\Sigma)(X) = -\int_\Sigma g(\boldsymbol{H}, X^\perp) \, \mathrm{d}\mathcal{H}^k, \tag{2.2}$$

where $\boldsymbol{H}$ is the mean curvature vector of Σ in M.

Remark 2.6 *We recall that the second fundamental form $\boldsymbol{B}$ of the embedding $f(\Sigma)$ in M is the symmetric bilinear map $\boldsymbol{B} : Tf(\Sigma) \times Tf(\Sigma) \to Nf(\Sigma)$ such that for $X, Y \in Tf(\Sigma)$*

$$\boldsymbol{B}(X,Y) = \left(\nabla^M_{\widetilde{X}} \widetilde{Y}\right)^\perp,$$

where ∇^M is the Levi-Civita connection on (M, g), $\tilde{X}$ and $\tilde{Y}$ are local extensions of X and Y to TM and $(\cdot)^\perp$ denotes the orthogonal projection onto $Nf(\Sigma)$. The mean curvature vector $\boldsymbol{H}$ is then defined as the trace of the second fundamental form. Explicitly, if $\{e_1, \ldots, e_k\}$ is a local orthonormal frame for $T_{f(x)}f(\Sigma)$, we have

$$\boldsymbol{H} = \operatorname{tr} \boldsymbol{B} = \sum_{i=1}^k \boldsymbol{B}(e_i, e_i).$$

The expression for the first variation involving the mean curvature (2.2) has two immediate consequences. On one hand, since g is non-degenerate, we find that the condition $\boldsymbol{H} = 0$ gives an equivalent characterization of minimal surfaces. On the other hand, we see that only the normal part of the variation affects the area up to first order. This is to be expected as we can imagine that the tangential component only "slides" Σ on to itself without changing the area.

Another property that area minimizing surfaces enjoy is stability. Precisely, an area minimizing surface is not only a critical point of the area functional but it is also a minimizer, which indicates that its "second derivative" should be positive. In general this is a nice property to have which only requires critical points to be local minimizers.

Definition 2.7 Let $f : \Sigma \hookrightarrow M$ be a closed embedded k-manifold and F_t a variation in the direction of $X \in \mathfrak{X}(M)$. The *second variation of area* of Σ is defined as

$$\delta^2 \operatorname{Area}_M(\Sigma)(X) := \left.\frac{\mathrm{d}^2}{\mathrm{d}t^2}\right|_{t=0} \int_\Sigma \mathrm{d}\mathcal{H}_t^k,$$

where $\mathcal{H}_t^k$ is the k-dimensional Hausdorff measure on Σ pulled back from the one on M with F_t.

Definition 2.8 A minimal surface $f : \Sigma \hookrightarrow M$ is said to be *stable* if any small enough variation increases its area, i.e.,

$$\delta^2 \operatorname{Area}_M(\Sigma)(X) \geq 0$$

for all $X \in \mathfrak{X}(M)$.

Much like in the first variation case, an explicit formula for the second variation can be obtained. However, the second derivatives make the computations much more tedious and also require the involvement of an orthonormal frame for the normal bundle. By assuming that we deal only with normal variations a relatively simple formula can be written down.

Theorem 2.9 (Second Variation of Area) *Let $f : \Sigma \hookrightarrow M$ be a k-dimensional minimal surface and let F_t be a variation in the direction of a smooth normal vector field $X \in \Gamma(Nf(\Sigma))$. Then the second variation of area is given by*

$$\begin{aligned}\delta^2 \operatorname{Area}_M(\Sigma)(X) = \int_\Sigma \sum_{i=1}^{k} \sum_{j=k+1}^{n} \langle \nabla^M_{e_i} X, \nu_j \rangle^2 - \sum_{i,j=1}^{k} \langle X, \boldsymbol{B}(e_i, e_j) \rangle^2 \\ - \sum_{i=1}^{k} \langle R^M(e_i, X)X, e_i \rangle \, \mathrm{d}\mathcal{H}^k, \quad (2.3)\end{aligned}$$

where the terms in the integrand are defined as follows:

- *$\{e_1, \dots, e_k\}$ is a local orthonormal frame for the tangent bundle $Tf(\Sigma)$ and $\{\nu_{k+1}, \dots, \nu_n\}$ is a local orthonormal frame for the normal bundle $Nf(\Sigma)$;*
- *$\langle \cdot, \cdot \rangle$ denotes the ambient Riemannian metric g, R^M is the ambient Riemann curvature tensor and ∇^M is the ambient Levi-Civita connection.*

Particularly relevant for these notes is the case of $k = n-1$ for which the second variation formula is much simpler. Indeed, if Σ is a two-sided stable minimal hypersurface and ν is a globally defined unit normal vector field on $f(\Sigma)$ then Σ satisfies the following *stability inequality*

$$\int_\Sigma |\nabla \phi|^2 \, \mathrm{d}\mathcal{H}^{n-1} \geq \int_\Sigma \left(|\boldsymbol{B}|^2 + \mathrm{Ric}_M(\nu, \nu) \right) \phi^2 \, \mathrm{d}\mathcal{H}^{n-1} \tag{2.4}$$

for any $\phi \in C_c^\infty(\Sigma)$, where:

- $|\boldsymbol{B}|^2 = \sum_{i,j=1}^{n-1} \langle \nu, \boldsymbol{B}(e_i, e_j) \rangle^2$ is the squared norm of the second fundamental form;
- $\mathrm{Ric}_M(\nu, \nu) = \sum_{i=1}^{n-1} \langle R^M(e_i, \nu)\nu, e_i \rangle$ is the Ricci curvature of the ambient manifold M in the direction of the normal ν.

The previous inequality by itself carries some strong implications and an important consequence for us is found in the example of spheres. Indeed, the stability inequality implies that there are no stable minimal hypersurfaces if the ambient manifold is a sphere $M = \mathbb{S}^n$. This follows immediately from $\mathrm{Ric}_{\mathbb{S}^n}(\nu, \nu) = n - 1$ and testing against $\phi \equiv 1$ in (2.4). The intuition behind this result can be found by considering $n = 2$ such that that minimal surfaces correspond to geodesics and these are great circles. We can imagine the minimal surface as a rubber band located at the equator of the sphere so that if we slightly pull the rubber band in any direction its tension will make it shrink towards one of the poles, indicating that its initial configuration was unstable, see Fig. 2.1.

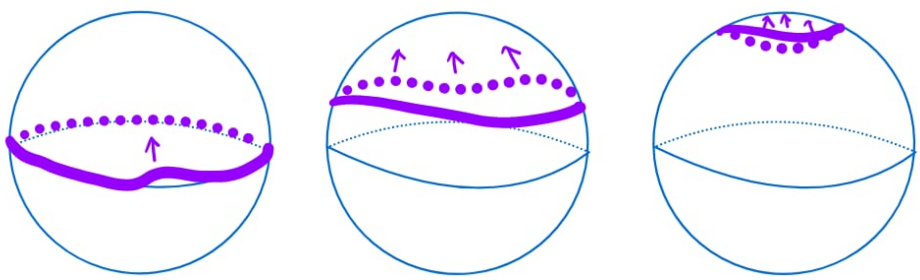

Figure 2.1: The geodesics of $\mathbb{S}^2$ are great spheres which are unstable with respect to variations of their length.

The previous argument clearly applies to any manifold with positive Ricci curvature, therefore we cannot expect to find minimal surfaces in arbitrary manifolds by simply minimizing the area functional. This geometric obstruction is one of the main reasons why the direct method of the calculus of variations is insufficient for proving the general existence of minimal hypersurfaces. Instead, one must search for unstable critical points of the area functional which leads to min-max theory (see Section 7.2 for an introduction to min-max). The foundational work in this direction is the Almgren-Pitts min-max theory, using advanced tools in geometric analysis to produce saddle-point critical points. Their main result guarantees the existence of at least one closed, embedded minimal hypersurface in any closed Riemannian manifold. More recently, an alternative method has emerged through the work of Padilla, Tonegawa, Hutchinson, Wickramasekera and Guaraco which is the main focus of these notes.

2.2 The Theory of Varifolds

Varifolds were introduced in 1965 by Almgren as a successor to Federer's theory of currents[1] in the context of the Plateau Problem, with the primary aim of resolving the cancellation issues inherent in the latter. The theory was subsequently expanded by Allard in the 1970s and has since been applied with great success in the study of minimal surfaces, the Plateau Problem, and mean curvature flow. In this section, we provide a brief introduction to the theory of varifolds, which will be needed throughout these notes. The lecture notes of Leon Simon [Sim84] serve as the standard textbook reference for the subject.

Throughout this section we assume that Ω is an open subset of $\mathbb{R}^n$ and k is an integer such that $0 \leq k \leq n$.

Definition 2.10 We define the (k,n)-*Grassmannian*, denoted by $G(k,n)$, as the collection of all (unoriented) k-dimensional subspaces of $\mathbb{R}^n$. We equip $G(k,n)$ with the distance $\rho(S,T) = |p_S - p_T|$ where p_S and p_T denote the orthogonal projections onto S and T and are identified as matrices inside $\mathbb{R}^{n^2}$ using the standard orthonormal basis.

It is very common to identify planes in the Grassmannian with the orthogonal projection onto them. Using this identification one obtains an atlas for $G(k,n)$ constituted by the charts coming from the coordinate expression of the projections in a fixed basis of $\mathbb{R}^n$. This gives $G(k,n)$ the structure of a smooth $k(n-k)$-dimensional compact manifold whose topology coincides with the one induced by the metric ρ. We note that the $k=1$ case coincides with the projective space $\mathbb{RP}^{n-1}$ and in this sense $G(k,n)$ can be thought as a

[1]See [Fed14].

generalization of projective space where instead of modding out by lines one mods out k-dimensional spaces. Moreover, by duality, $G(n-1,n)$ also coincides with $\mathbb{RP}^{n-1}$ and more generally $G(k,n) \cong G(n-k,n)$.

Definition 2.11 For a subset $\Omega \subset \mathbb{R}^n$, we define the *Grassmannian bundle* over $\Omega \subset \mathbb{R}^n$ as

$$G_k(\Omega) := \Omega \times G(k,n)$$

equipped with the product topology.

The Grassmannian bundle can be seen as a generalization of the tangent bundle as we have all possible choices of tangent planes at every point instead of a single one. We will commonly refer to points in the Grassmannian bundle as a pair $(x,S) \in G_k(\Omega)$ where $x \in \Omega$ and $S \in G(k,n)$.

Definition 2.12 A *k-varifold* V over Ω is a Radon measure on $G_k(\Omega)$. The *weight* of a varifold V is the Radon measure on Ω defined by

$$\|V\|(\phi) = \int_{\Omega\times G(k,n)} \phi(x)\, \mathrm{d}V(x,S), \text{ for } \phi \in \mathcal{C}_c(\Omega),$$

via the Riesz-Markov-Kakutani representation theorem[2]. The *support* of a varifold V is the support of its weight and is denoted by $\mathrm{spt}\|V\|$. We denote by $\mathbf{V}_k(\Omega)$ the set of all varifolds on Ω.

A smooth k-submanifold Σ of $\mathbb{R}^n$ corresponds naturally to the varifold

$$V(A) = \int_\Sigma \chi_A(x, T_x\Sigma)\, \mathrm{d}\mathcal{H}^k(x), \quad \text{for } A \subset G_k(\Omega) \text{ Borel},$$

often just written as $\mathcal{H}^k \llcorner \Sigma \otimes \delta_{T_x\Sigma}$. Thus, varifolds are a generalization of submanifolds in which we are (roughly) considering the submanifold to be the support of a measure of $\mathbb{R}^n$ which need not be smooth or even k-dimensional such that at each point we have a diffuse distribution of k dimensional planes.

Definition 2.13 We say a sequence $\{V_i\}_{i\geq 1}$ in $\mathbf{V}_k(\Omega)$ converges to $V \in \mathbf{V}_k(\Omega)$ if $V_i \overset{*}{\rightharpoonup} V$ as Radon measures.

One of the main advantages of working with such a weak notion of submanifold is that varifolds enjoy much better compactness properties than submanifolds. Indeed, any sequence of Radon measures V_i with a uniform mass bound ($\sup \|V_i\|(\Omega) < \infty$) has a subsequence that converges weakly to a limit Radon measure V thanks to the Banach-Alaoglu Theorem. In contrast, it is not hard to construct a sequence of smooth submanifolds converging (in the Hausdorff distance for example) to a very rough set.

[2]**Riesz-Markov-Kakutani Theorem:** Let X be a locally compact Hausdorff space. Then for any positive linear functional $\Lambda : \mathcal{C}_c(X) \to \mathbb{R}$, there exists a unique Radon measure μ on X such that $\Lambda(f) = \int_X f\, d\mu$ for all $f \in \mathcal{C}_c(X)$.

Definition 2.14 Let $f : \Omega \to \Omega'$ be a smooth diffeomorphism and $V \in \mathbf{V}_k(\Omega)$, we define the *pushforward* or *image varifold* $f_\# V \in \mathbf{V}_k(\Omega')$ by

$$f_\# V(A) := \int_{F^{-1}(A)} J_S f(x) \, \mathrm{d}V(x, S)$$

for $A \subset G_k(\Omega')$ Borel, with $F(x, S) = (f(x), \, \mathrm{d}f_x(S))$ and

$$J_S f(x) = \sqrt{\det\left((\, \mathrm{d}f_x|_S)^* \circ (\, \mathrm{d}f_x|_S) \right)}$$

for $(x, S) \in G_k(\Omega)$.

The pushforward of varifolds plays a very important role in defining their first variation and in taking blow-ups around a point. By taking blow-ups we mean rescaling the varifold around a given point to obtain a new varifold which captures the local behavior around said point. Precisely, we denote by

$$\eta_{x,r}(y) := \frac{y - x}{r}$$

such that

$$(\eta_{x,r})_\# V = r^{-k} V \left(\{ (x + ry, S) \mid (y, S) \in A \} \cap G_k(\Omega) \right). \tag{2.5}$$

Definition 2.15 Given $T \in G(k, n)$, $x \in \Omega$ and $\theta \in (0, \infty)$, we say that a k-varifold V on Ω has *tangent space T with multiplicity θ* at x if $(\eta_{x,r})_\# V \to \theta \mathcal{H}^k \otimes \delta_T$ as varifolds as $r \to 0^+$.

In contrast with submanifolds, the tangent space of a varifold need not exist which is one of the reasons that general varifolds are too weak of an object. Therefore, we turn our attention to a better behaved class of objects.

Definition 2.16 A $\mathcal{H}^k$-measurable set $\Sigma \subset \mathbb{R}^n$ is said to be *countably k-rectifiable* if it consists of the union of images of $\mathbb{R}^k$ under a countable number of Lipschitz maps $F_i : \mathbb{R}^k \to \mathbb{R}^n$ up to a set of $\mathcal{H}^k$-measure 0, i.e.

$$\mathcal{H}^k \left(\Sigma \setminus \bigcup_i F_i(\mathbb{R}^k) \right) = 0.$$

Theorem 2.17 *Let $\Sigma \subset \mathbb{R}^n$ a $\mathcal{H}^k$-measurable set with locally finite $\mathcal{H}^k$ measure. Then Σ is countably k-rectifiable if and only if for $\mathcal{H}^k$-a.e. $x \in \Sigma$ there exists a k-dimensional subspace $\pi_x \subset \mathbb{R}^n$ such that*

$$\lim_{r \to 0^+} r^{-k} \int_\Sigma \phi\left(\frac{y - x}{r} \right) \, \mathrm{d}\mathcal{H}^k(y) = \int_{\pi_x} \phi(y) \, \mathrm{d}\mathcal{H}^k(y)$$

for all $\phi \in \mathcal{C}_c(\mathbb{R}^n)$. The subspace π_x is known as the approximate tangent space *of Σ at x and is denoted by $T_x\Sigma$.*

The approximate tangent space is the tangent space in a measure theoretic sense as it carries the integration properties we expect from the tangent space of a submanifold of $\mathbb{R}^n$. Moreover, the fact that rectifiable sets have a tangent space almost everywhere means they naturally induce a varifold in the same way as smooth submanifolds.

Definition 2.18 Let $\Sigma \subset \mathbb{R}^n$ be countably k-rectifiable. We define the k-varifold induced by Σ by

$$v(\Sigma)(A) := \mathcal{H}^k(\{x \in \Omega \mid (x, T_x\Sigma) \in A\}) \tag{2.6}$$

for $A \subset G_k(\Omega)$ Borel.

Conversely, we make the following definition.

Definition 2.19 We say $V \in \mathbf{V}_k(\Omega)$ is a *k-rectifiable varifold* if there exist positive real numbers $\{c_j\}_{j\geq 1}$ and k-rectifiable sets $\{\Sigma_j\}_{j\geq 1}$ such that

$$V = \sum_{j=1}^{\infty} c_j v(\Sigma_j).$$

We denote the set of all k-rectifiable varifolds $\mathbf{RV}_k(\Omega)$

Clearly, any rectifiable varifold has a unique tangent space (in the sense of Definition 2.16) $\mathcal{H}^k$-a.e. and its multiplicity is given by

$$\theta(x) = \sum_{x \in \Sigma_j} c_j.$$

Moreover, $\|V\| = \theta \mathcal{H}^k \llcorner \Sigma$ with $\Sigma = \bigcup_j \Sigma_j$ and the support is given by

$$\operatorname{spt}\|V\| = \overline{\{x \in \Sigma \mid \theta(x) > 0\}}.$$

Definition 2.20 The *k-dimensional density* Θ^k of a k-varifold $V \in \mathbf{V}_k(\Omega)$ at $x \in \Omega$ is defined as the density of the measure $\|V\|$, i.e., the limit

$$\Theta^k(\|V\|, x) = \lim_{r \to 0} \frac{\|V\|(B_r(x))}{\omega_k r^k}$$

whenever it exists.

By definition, the previous limit exists whenever the upper and lower limits coincide $\Theta^{*,k}(\|V\|, x) = \Theta^k_*(\|V\|, x)$. If V is rectifiable, in view of (2.5) and Theorem 2.17, the density and the multiplicity coincide $\Theta(x) = \theta(x)$ for $\mathcal{H}^k$-a.e. $x \in \Sigma$.

When we are dealing with a sequence of smooth submanifolds converging towards a varifold, a common way in which the limit can develop singularities is if the submanifolds fold onto themselves. When this happens we expect the limit varifold to have multiplicity strictly larger than 1 but still be an integer.

Definition 2.21 A k-varifold $V \in \mathbf{V}_k(\Omega)$ is called *integer rectifiable* or *integral* if it has rectifiable support $\Sigma \subset \Omega$ and integer multiplicity, i.e., θ is locally $\mathcal{H}^k$-integrable and takes positive integer values almost everywhere. We denote the set of all integer rectifiable k-varifolds by $\mathbf{IV}_k(\Omega)$ and write $V = (\Sigma, \theta)$ for $V \in \mathbf{IV}_k(\Omega)$.

To use varifolds as substitute for submanifolds in the theory of minimal surfaces we require a notion minimality.

Definition 2.22 For $V \in \mathbf{V}_k(\Omega)$ we define its *first variation* by

$$\delta V(X) = \int_{G_k(\Omega)} \operatorname{div}_S X(x) \, \mathrm{d}V(x, S) \tag{2.7}$$

for any $X \in \mathcal{C}_c^1(\Omega, \mathbb{R}^n)$ where div_S denotes the tangential divergence with respect to the subspace S and is given by

$$\operatorname{div}_S X = \sum_{j=1}^{k} \langle e_j, \partial_{e_j} X \rangle = \sum_{i,j=1}^{n} S_{ij} \partial_j X_i$$

with $e_1, \ldots, e_k$ an orthonormal basis for S. If $\delta V(X) = 0$ for all $X \in \mathcal{C}_c^1(\Omega, \mathbb{R}^n)$ then V is called *stationary*.

Remark 2.23 *Similarly to the smooth case, the first variation of a varifold is usually first defined as the derivative at $t = 0$ of the mass of a variation. Precisely, if $F_t : \Omega \to \Omega$ is a family of diffeomorphisms such that $F_t|_{\Omega \setminus K} = \mathrm{Id}$ for $K \subset \Omega$ compact, $F_0 = \mathrm{Id}$ and $\partial_t|_{t=0} F_t = X$, then*

$$\delta V(X) := \left. \frac{\mathrm{d}}{\mathrm{d}t} \right|_{t=0} (F_t)_\# V(G_k(K)).$$

After some computations one finds the completely equivalent expression in (2.7).

A simple yet relevant example of a stationary varifolds is found in $\mathbb{R}^2$. It is easy to check that the union of any finite number of half-lines radiating from the origin, defined by unit direction vectors $v_j \in \mathbb{R}^2$ for $1 \leq j \leq N$, is a stationary 1-varifold if and only if

$$\sum_{j=1}^{N} v_j = 0,$$

meaning their directions balance out. Clearly, such varifolds do not correspond to smooth submanifolds, as they possess a singularity at the origin. These configurations are known as *junctions*.

By testing (2.7) against approximations of indicator functions of balls one can derive the following fundamental result for stationary varifolds.

Theorem 2.24 (Monotonicity Formula) *Let V be a stationary k-varifold on $\Omega \subset \mathbb{R}^n$, $x \in \Omega$ and $0 < r < R$ such that $B_R(x) \subset \Omega$, then*

$$\frac{\|V\|(B_R(x))}{R^k} - \frac{\|V\|(B_r(x))}{r^k} = \int_{G_k(B_R(x)\setminus B_r(x))} \frac{|S^\perp(y-x)|^2}{|y-x|^{k+2}} \,\mathrm{d}V(y,S), \quad (2.8)$$

where $S^\perp$ denotes the projection onto the orthogonal complement of S.

An immediate consequence of (2.8) is that any stationary varifold V has its density $\Theta(\|V\|,\cdot)$ well-defined everywhere. Moreover, if V is integral then $\Theta(\|V\|,\cdot) \geq 1$ everywhere on the support. The monotonicity formula is the starting point for the regularity theory of stationary varifolds (see Section 6.1) which aims to identify them with classical smooth minimal surfaces under suitable assumptions.

We have seen that a smooth submanifold induces a multiplicity 1 integral varifold and it is easy to check that if the submanifold is also minimal then the induced varifold is stationary. We can take this one step further by defining the mean curvature of varifolds. By definition, the first variation is a vector valued distribution of order 1. We say that δV is locally bounded if for every compact $K \subset \Omega$ there exists a constant C_K such that

$$|\delta V(X)| \leq C_K \sup_{x\in K} |X(x)|, \text{ for all } X \in \mathcal{C}^1_c(K,\mathbb{R}^n).$$

In this case, the Riesz representation theorem gives the existence of a vector valued Radon measure h characterized by

$$\delta V(X) = \int \langle X(x), \mathrm{d}h(x)\rangle.$$

If, additionally, the total variation $|\delta V|$ is absolutely continuous (as measures) with respect to $\|V\|$ then by the Radon-Nikodym Theorem there exists a $\|V\|$-measurable vector field $\boldsymbol{H}$ such that

$$\delta V(X) = -\int_\Omega \langle \boldsymbol{H}, X\rangle \,\mathrm{d}\|V\| \quad (2.9)$$

and in view of (2.2) we make the following definition.

Definition 2.25 Assume V is a varifold with a locally bounded first variation and such that $|\delta V|$ is absolutely continuous w.r.t. $\|V\|$, then we define the *generalized mean curvature* $\boldsymbol{H}$ as the unique $\|V\|$-measurable function satisfying (2.9).

Clearly, a smooth submanifold has a generalized mean curvature and it coincides with the classical one. On the other hand, a stationary varifold always admits a generalized mean curvature and it is identically 0. Therefore, stationary varifolds are characterized by $\boldsymbol{H} = 0$ much like classical minimal surfaces.

The first variation also carries information about the regularity of a varifold. The Theorem of Allard gives a very useful criterion for determining when a varifold is rectifiable.

Theorem 2.26 (Allard Rectifiability Theorem) *Let $V \in \mathbf{V}_k(\Omega)$ with locally bounded first variation in Ω and $\Theta^k(\|V\|, x) > 0$ for $\|V\|$-a.e. $x \in \Omega$, then $V \in \mathbf{RV}_k(\Omega)$.*

The second variation of a varifold is defined analogously to the first; precisely,

$$\delta^2 V(X) := \left.\frac{\mathrm{d}^2}{\mathrm{d}t^2}\right|_{t=0} (F_t)_\# V(G_k(K)),$$

where K, F_t, and X are defined as in Remark 2.23. However, unlike the first variation, obtaining an explicit formula for the second variation in the general case is significantly more challenging. Even when restricting our attention to rectifiable varifolds, the approximate tangent spaces need not vary smoothly enough from point to point to yield a meaningfully differentiable frame. Without such regularity, one cannot define a classical second fundamental form (cf. (2.3)). This issue was first addressed by Hutchinson [Hut86] and will be relevant to us later (see Section 5.1).

Chapter 3

The Allen–Cahn Theory

In this chapter, we introduce the mathematical theory behind the Allen–Cahn model. Our goal is twofold: first, to present the foundational theory and develop a geometric intuition for the shape of solutions to the Allen–Cahn equation and their behavior in the singular limit $\varepsilon \to 0^+$; second, to motivate the central problem of these notes by presenting the Modica–Mortola theorem on the convergence of Allen–Cahn minimizers to area-minimizing minimal surfaces. By doing so, we highlight the main obstructions to generalizing this minimization approach on closed manifolds.

The organization and contents of this chapter are heavily inspired by Chodosh's lecture notes [Cho19], which serve as the primary source for the relevant results presented herein.

3.1 The Allen–Cahn Model

The Allen–Cahn model was originally introduced by Sam Allen and John Cahn in 1979 [AC79] to describe the motion of boundaries in crystalline solids, such as iron-aluminum alloys. Since then, it has become a cornerstone of phase-field modeling for systems experiencing non-conserved order parameter dynamics. The time-dependent model is governed by the following heat-equation-like parabolic PDE

$$\partial_t u = \varepsilon \Delta u - \frac{1}{\varepsilon} W'(u), \tag{3.1}$$

where $\varepsilon > 0$ is a parameter controlling the width of the transition region separating the two distinct phases. The model assumes that W is a double-well potential, representing an energy density landscape with two stable local minima separated by an energy barrier, as illustrated in Fig. 3.1. Each minimum corresponds to a physical phase of the system. Consequently, regions where u takes values close to either minimum represent macroscopic domains occupied by that specific phase.

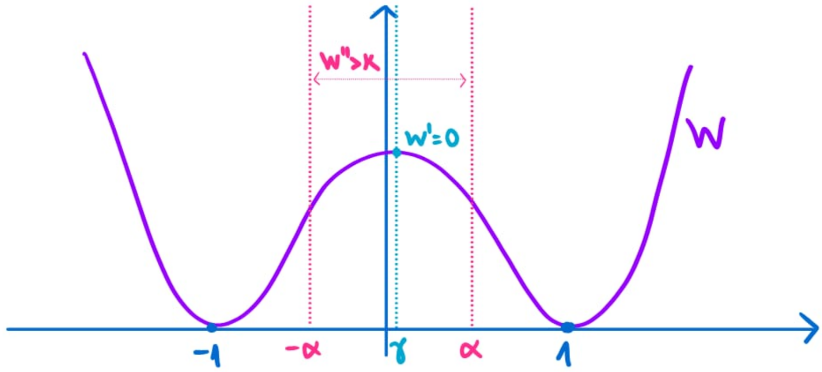

Figure 3.1: Sketch of the graph of a generic double-well potential W.

Definition 3.1 We say that a function $W : \mathbb{R} \to [0, \infty)$ is a *double-well potential* if it is at-least C^3, has two minima at 1 and -1, $W(\pm 1) = 0$, and

$$\begin{aligned} &\textit{there is } \gamma \in (-1,1),\ \alpha \in (0,1) \textit{ and } \kappa > 0 \textit{ such that } W' < 0 \textit{ on } (\gamma, 1) \\ &W' > 0 \textit{ on } (-1, \gamma) \textit{ and } W''(x) \geq \kappa \textit{ for all } |x| \geq \alpha. \end{aligned} \tag{A}$$

The *surface tension constant* associated to W is the positive real number defined by

$$\sigma := \int_{-1}^{1} \sqrt{2W(s)}\, \mathrm{d}s.$$

Remark 3.2 *The previous definition is not necessarily standard and the exact assumptions on W can vary depending on the use case. We choose these particular properties on W for these notes as they are the ones fitting the theory that is presented. For example, it is not necessary that the minima are located at ± 1 but it is arguably the simplest choice.*

Throughout the rest of the notes W will always denote a double-well potential. Physically, σ is the energy per unit area stored within the transition layer at equilibrium, which explains its name. This will become clearer when looking at higher dimensional solutions. The most common choice of W in the literature is the quartic potential $W(s) = \frac{1}{4}(1 - s^2)^2$ which has surface tension constant $\sigma = 2\sqrt{2}/3$.

The Allen–Cahn equation admits a natural variational formulation coming from the so-called Allen–Cahn energy.

Definition 3.3 Let $\Omega \subset \mathbb{R}^n$ open and $\varepsilon > 0$. The *Allen–Cahn energy* is the functional $E_\varepsilon : H^1(\Omega) \to [0, \infty]$ defined by

$$E_\varepsilon(u, \Omega) := \int_\Omega \frac{\varepsilon |\nabla u|^2}{2} + \frac{W(u)}{\varepsilon} \, \mathrm{d}^n x. \tag{ACE$_\varepsilon$}$$

We will just write $E_\varepsilon(u)$ when it is clear from context.

Remark 3.4 *As defined, E_ε can take on the value ∞ at some points due to the fact that $u \in H^1(\Omega)$ does not guarantee that the potential term of the energy is finite. For instance, if $W(u) \propto u^4$ away from 0 then $\int_\Omega W(u) < \infty$ is guaranteed only in dimension $n \leq 4$ by the Sobolev embedding. We choose for concreteness to keep the domain of the energy as $H^1(\Omega)$ and deal with any possible issues further down the road .*

The parabolic Allen–Cahn equation (3.1) corresponds precisely to the L^2-gradient flow of E_ε. Indeed, letting $\phi \in C_c^\infty(\Omega)$ a formal computation yields

$$\begin{aligned} \langle \nabla_{L^2} E_\varepsilon(u, \Omega), \phi \rangle &= \left. \frac{\mathrm{d}}{\mathrm{d}s} \right|_{s=0} E_\varepsilon(u + s\phi, \Omega) \\ &= \int_\Omega \varepsilon \langle \nabla u, \nabla \phi \rangle + \frac{W'(u)}{\varepsilon} \phi \, \mathrm{d}^n x = \int_\Omega \left(-\varepsilon \Delta u + \frac{W'(u)}{\varepsilon} \right) \phi \, \mathrm{d}^n x \end{aligned}$$

and from $\partial_t u = -\nabla_{L^2} E_\varepsilon(u, \Omega)$ we recover (3.1).

In these notes we are concerned with the time-independent case which corresponds to the following elliptic PDE

$$\varepsilon \Delta u = \frac{W'(u)}{\varepsilon} \quad \text{in } \Omega, \tag{AC$_\varepsilon$}$$

whose solutions are precisely the critical points of E_ε, i.e.,

$$\int_\Omega \left(\varepsilon \Delta u + \frac{W'(u)}{\varepsilon} \right) \phi \, \mathrm{d}^n x = 0$$

for any $\phi \in C_c^\infty(\Omega)$.

Proposition 3.5 (Regularity of Critical Points) *Let $u \in H^1(\Omega)$ be a weak solution to* (AC$_\varepsilon$) *for a double-well potential $W \in C^k(\mathbb{R})$, $k \geq 3$, then $u \in C^k(\Omega)$.*

Sketch of Proof $u \in H^1(\Omega)$ weakly solves (AC$_\varepsilon$) if for any $\phi \in H_0^1(\Omega)$

$$\int_\Omega \varepsilon \langle \nabla u, \nabla \phi \rangle + \frac{W'(u)}{\varepsilon} \phi \, \mathrm{d}^n x = 0. \tag{3.2}$$

If a weak solution is locally bounded we can use standard elliptic regularity theory to show that it is as regular as W. Precisely, if $u \in L^\infty_{\mathrm{loc}}(\Omega)$, then in particular $W'(u) \in L^p_{\mathrm{loc}}(\Omega)$ for any $1 < p < \infty$, which implies by interior elliptic regularity that $u \in W^{2,p}_{\mathrm{loc}}(\Omega)$. For $p > n$, the Morrey embedding implies that $u \in \mathcal{C}^{0,\alpha}_{\mathrm{loc}}(\Omega)$ for some $\alpha \in (0,1)$, and then using a Schauder bootstrapping argument, we find that if $W' \in \mathcal{C}^{k-1}$, then $u \in \mathcal{C}^{k,\alpha}_{\mathrm{loc}}(\Omega)$. For more details on elliptic regularity, see [GT01].

Therefore, we only need to show that u is locally bounded. This can be seen by testing (3.2) against $\phi = (u-k)^+\eta^2$, where $k > 0$, $(f)^+ = \max(0, f)$ and $\eta \in \mathcal{C}^\infty_c$ is a smooth cutoff and using the fact that there exist $a, b > 0$ such that $W(s) \geq as^2 - b$. Like this we obtain that $u \leq C$ for some $C > 0$ and an identical argument yields the lower bound. □

Inspecting E_ε we can develop a good understanding of how exactly the Allen–Cahn equation models two-phase systems. If u (locally) minimizes energy, the first term in (ACE$_\varepsilon$) (the *gradient* or *kinetic term*) penalizes large spatial variations $|\nabla u|$. This term encourages the solution to remain constant. Conversely, the second term (the *potential term*) forces u to remain close to the minima of W at ± 1.

Hence, any non-constant minimizer will have values close to 1 on some regions and close to -1 on others, separated by a narrow transition layer. The scaling of this layer can be deduced through a basic one-dimensional heuristic. Let D be the transition layer with width δ. The gradient scales along D as $|\nabla u| \sim \frac{2}{\delta}$; thus, evaluating the kinetic energy

$$\int_\Omega \frac{\varepsilon|\nabla u|^2}{2} \sim \int_D \frac{\varepsilon}{\delta^2} \sim \delta\frac{\varepsilon}{\delta^2} = \frac{\varepsilon}{\delta}.$$

Similarly, since $W(u) \sim 1$ inside D, we have that

$$\int_\Omega \frac{W(u)}{\varepsilon} \sim \int_D \frac{W(u)}{\varepsilon} \sim \frac{\delta}{\varepsilon}.$$

We can conclude by minimizing $E_\varepsilon(u) \sim \varepsilon/\delta + \delta/\varepsilon$ with respect to δ, which gives $\delta = \varepsilon$. Thus, the physical width of the phase interface is of order $\mathcal{O}(\varepsilon)$. As we will see shortly, the Allen–Cahn theory is centered around studying the phenomena which arise at the transition region as $\varepsilon \to 0^+$.

3.2 Solutions & Geometric Aspects of Allen–Cahn

The connection between Allen–Cahn solutions and the theory of minimal surfaces is made by looking at the level sets around the transition regions of the solutions as $\varepsilon \to 0^+$. For this reason, it is very helpful to develop an intuition for what these solutions actually look like. In this section we explore this, focusing on the $n = 1$ case, which is the most relevant for our purposes. We will set aside the variational formulation and focus strictly on the PDE perspective; consequently, we will consider solutions that do not lie in H^1. The study of these phase transitions falls under the topic commonly referred to as the geometric aspects of the Allen–Cahn equation.

We begin by pointing out that the constant functions $u \equiv 0$ and $u \equiv \pm 1$ are solutions of (AC_ε), with the former having infinite energy on unbounded domains[1] and the latter two having 0 energy as they are global minima. However, since these trivial solutions do not exhibit phase-transition behavior we are forced to look for non-trivial ones.

First, we note that if u is a solution to (AC_ε) then by defining the rescaled function $\hat{u}(x) := u(\varepsilon x)$ we have that

$$\Delta \hat{u} = W'(\hat{u}), \tag{3.3}$$

which allows us to reduce to the case $\varepsilon = 1$. It turns out that we can classify all entire solutions for $n = 1$ that have finite energy, i.e., those solving

$$u''(x) = W'(u(x)) \tag{3.4}$$

for $x \in \mathbb{R}$ such that

$$\int_{\mathbb{R}} \frac{(u'(x))^2}{2} + W(u(x))\, \mathrm{d}x < \infty. \tag{3.5}$$

Proposition 3.6 *Let W be a double-well potential, then for each $u_0 \in \mathbb{R}$ the equation* (3.4) *admits a unique solution u defined on all of $\mathbb{R}$ with finite energy and such that $u(0) = u_0$ if and only if $|u_0| \leq 1$. In addition, any such solution is necessarily monotonic, and if u is non-constant then its limits at $\pm\infty$ are ± 1 if u is non-decreasing, or ∓ 1 if u is non-increasing.*

Proof We begin by noticing that

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[(u'(x))^2 - 2W(u(x))\right] = 2u'(x)u''(x) - 2W'(u(x))u'(x) = 0$$

in view of (3.4), so we must have that

$$(u'(x))^2 - 2W(u(x)) = C$$

where $C \in \mathbb{R}$ is an integration constant. Now, since each term in the finite integral (3.5) is positive and the domain is all of $\mathbb{R}$ it must be that there exists a sequence $\{x_k\}_{k \geq 0}$ such that both $u'(x_k) \to 0$ and $W(u(x_k)) \to 0$ as $k \to \infty$. This implies that $C = 0$ in the previous equation, meaning any solution with finite energy satisfies

$$(u'(x))^2 = 2W(u(x)).$$

[1] It can easily be shown that the solution $u \equiv 0$ has the absolute maximum energy on bounded domains.

We show that the solution is necessarily monotonic. Assume that u solves (3.4) with $u(0) = u_0$. If u were not monotonic there would exist $x_c \in \mathbb{R}$ such that $u'(x_c) = 0$. Now, if $|u(x_c)| < 1$ then

$$(u'(x_c))^2 - 2W(u(x_c)) = 0$$

implies $W(u(x_c)) = 0$ thus $u(x_c) = \pm 1$, a contradiction. On the other hand, if $u(x_c) = \pm 1$ then by the Picard-Lindelöf Theorem applied to the second order ODE (3.4) it must be that $u \equiv \pm 1$, which is monotonic.

Now, if u is non-decreasing, we can take the positive square root to obtain the initial value problem

$$u'(x) = \sqrt{2W(u(x))} \quad \text{with } u(0) = u_0 \tag{3.6}$$

We claim that the function $\sqrt{2W(\cdot)}$ is locally Lipschitz on $\mathbb{R}$. Indeed, by the definition of W, the only points around which $\sqrt{2W(\cdot)}$ can fail to be locally Lipschitz are those where $W(y) = 0$, i.e., $y = \pm 1$. However, since these are critical points of W, a second order Taylor expansion shows that this is not the case.

Thus, we can apply the Picard–Lindelöf Theorem for first order ODEs to guarantee the uniqueness of local solutions around 0 to the initial value problem (3.6). If $u_0 = \pm 1$ we already know that $u \equiv \pm 1$, and if $|u_0| < 1$ by uniqueness we find that $-1 < u(x) < 1$ which implies the solution is defined for all $x \in \mathbb{R}$ as it does not blow-up. If $|u_0| > 1$ then necessarily $|u(x)| > 1$ for any x in the solution domain which cannot be the entirety of $\mathbb{R}$ as this would contradict the finite energy assumption. Finally, since u is bounded and strictly increasing on $\mathbb{R}$, the limits $\lim_{x\to\pm\infty} u(x)$ must exist and since the energy is finite, these limits must be zeroes of W, so $\lim_{x\to\infty} u(x) = 1$ and $\lim_{x\to-\infty} u(x) = -1$.

If u is non-increasing, the same argument holds for $u'(x) = -\sqrt{2W(u(x))}$.□

Remark 3.7 *The Allen–Cahn equation admits many solutions over $\mathbb{R}$ with infinite energy. A way of constructing them is by periodically extending solutions on a bounded interval. See [LMN20] for examples.*

An immediate consequence of Proposition 3.6 is that the only solutions to (AC_ε) in 1 dimension with finite energy are the constant solutions $u \equiv \pm 1$ and the monotonic solutions $u_\varepsilon(x) = \pm\varphi\left(\frac{x-x_0}{\varepsilon}\right)$ interpolating between -1 and $+1$. Here, φ is the unique solution to (3.6) satisfying $\varphi(0) = 0$, and $x_0 \in \mathbb{R}$ is a translation parameter. Moreover, the energy of the former is 0 and the energy of the latter is σ. Indeed,

$$E_\varepsilon(u_\varepsilon) = \int_{\mathbb{R}} \frac{\varepsilon(u'(x))^2}{2} + \frac{W(u(x))}{\varepsilon}\,\mathrm{d}x = \int_{\mathbb{R}} \frac{1}{2}(\varphi'(y))^2 + W(\varphi(y))\,\mathrm{d}y$$
$$= \int_{\mathbb{R}} 2W(\varphi(y))\,\mathrm{d}y = \int_{-1}^{1} \sqrt{2W(z)}\,\mathrm{d}z = \sigma$$

where the second equality follows from the change of variables $y = \frac{x-x_0}{\varepsilon}$, the third from (3.6) and fourth from the change $z = \varphi(y)$. The previous computation also shows that in dimension 1 the energy does not change under the rescaling of solutions $x \mapsto x/\varepsilon$.

Definition 3.8 The non-trivial, monotonic, 1-dimensional solutions to (AC_ε) with finite energy are referred to as *heteroclinic* or *standing wave* solutions.

Remark 3.9 *Since all heteroclinics in 1-dimension have the same energy they cannot be strictly energy minimizing. This is because any perturbation which effectively acts as a translation along the x-axis will yield a function with the same energy. However, they are stable with respect to compact variations with the heuristic being that any perturbation attempting to flatten them out towards $+1$ or -1 would require infinite energy to move an entire half-line from one minima to the other.*

If we take $W(s) = \frac{1}{4}(1-s^2)^2$, we can actually obtain an explicit expression for the heteroclinic solutions which showcase the behavior we have described. Indeed, it easy to check that the function $\mathbb{H}(x) := \tanh(x/\sqrt{2})$ solves the corresponding initial value problem

$$u' = \frac{1}{\sqrt{2}}(1-u^2) \text{ with } u(0) = 0.$$

Hence, the family of solutions $\mathbb{H}_\varepsilon(x) := \pm\tanh(\frac{x-x_0}{\varepsilon\sqrt{2}})$ for $x_0 \in \mathbb{R}$ are the heteroclinic solutions if $n = 1$ and W is the standard quartic potential. An important observation is that any heteroclinic solution converges as $\varepsilon \to 0^+$ to a step function with a discontinuity at x_0, see Fig 3.2.

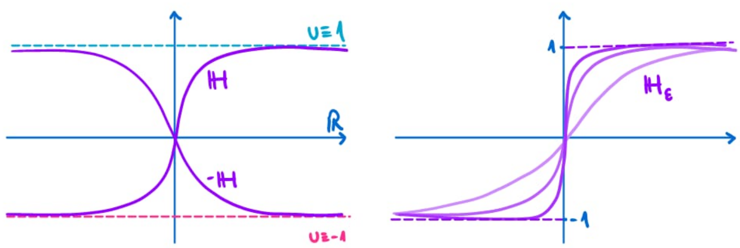

Figure 3.2: Sketch of the finite energy solutions of the 1-dimensional Allen–Cahn equation (left) and convergence of the heteroclinic solutions towards a step function as $\varepsilon \to 0^+$ (right).

We can also use the heteroclinics in 1 dimension to construct solutions in higher dimensions. Indeed, it is easy to check that $u(x) := \varphi(\langle a, x\rangle - b)$ is a solution to (3.3) for any $a \in \mathbb{S}^{n-1}$ and $b \in \mathbb{R}$. In this case, the solutions no longer possess finite energy but only local finite energy. This is due to the fact that now the energy is located along an unbounded region, so it makes more sense to talk about energy density. By rotating our coordinate system such that $a = e_1$, we can consider the cylinder $\Omega = (-\infty, +\infty) \times B_{n-1}$ and then $E_1(u, \Omega) = \sigma\omega_{n-1}$. For the same reason, the energy of the rescaled solutions $u_\varepsilon(x) = u(x/\varepsilon)$ scales as $E_\varepsilon(u_\varepsilon) = \varepsilon^{n-1}E_1(u)$. In this case we also find that

$$u_\infty(x) := \lim_{\varepsilon\to 0^+} u_\varepsilon(x) = \begin{cases} 1 & \langle a, x\rangle > b, \\ -1 & \langle a, x\rangle < b, \end{cases}$$

and the limiting interface $\partial\{u_\infty = 1\} = \{\langle a, x\rangle = b\}$ is an affine hyperplane, which we know is a minimal surface.

It turns out that for $n \geq 2$ there are many more solutions of (AC_ε) compared to $n = 1$. For instance, a large family of examples is given by the following result.

Theorem 3.10 (Kowalczyk–Liu–Pacard [KLP12]) *Given any two lines $\ell_1, \ell_2 \subset \mathbb{R}^2$ intersecting at the origin, there is a solution u on $\mathbb{R}^2$ to* (3.3) *whose nodal set $\{u = 0\}$ is asymptotic at infinity to $\ell_1 \cup \ell_2$.*

As before, if we take the limit as $\varepsilon \to 0^+$ of the solutions to (AC_ε) obtained by rescaling the previous solution u we find that the limit interface is precisely $\ell_1 \cup \ell_2$. We note that this interface corresponds to a junction, i.e., a stationary varifold which is unstable and unlike the previous cases it cannot be identified with a regular submanifold of $\mathbb{R}^2$.

A central question in the Allen–Cahn theory regarding the classification of solutions in $\mathbb{R}^n$ was posed by De Giorgi in 1978.

Conjecture 3.11 (De Giorgi) *Consider $u \in \mathcal{C}^2(\mathbb{R}^n)$ solving the Allen–Cahn equation for the standard quartic potential*

$$\Delta u = W'(u) = u^3 - u$$

so that $|u| \leq 1$ and $\partial_n u > 0$. Is it true that necessarily $u(x) = \mathbb{H}(\langle a, x\rangle - b)$?

This conjecture is the direct Allen–Cahn analog of the Bernstein conjecture[2] for entire minimal surfaces. It is known that for $n = 2, 3$ the conjecture holds true, and that for $n \geq 9$ there are solutions other than the 1-dimensional ones; however, the conjecture remains open for $4 \leq n \leq 8$. For more details, see Section 5 in [Cho19].

[2]The Bernstein conjecture states that any entire minimal graph in $\mathbb{R}^n$ must be an affine hyperplane. This is known to be true for dimensions $n \leq 8$, but fails for $n \geq 9$.

3.3 Convergence of Minimizers

Our goal in this section is to motivate the coming chapters by covering the theorem of Modica and Mortola [MM77], which established the first connection between the Allen–Cahn theory and minimal surfaces. The theorem says that a sequence of minimizers of the Allen–Cahn energy converges (in an appropriate sense) as $\varepsilon \to 0^+$ to a minimizer of $(n-1)$-dimensional area (in another appropriate sense). The machinery behind this result is that of functions of bounded variation and the notion of Γ-convergence of functionals, for which we will cover the necessary basics.

The Γ-convergence of functionals is a form of convergence which ensures that sequences of minimizers converge to minimizers. It was first introduced by De Giorgi and it allows the study of singular functionals by approximating them with better behaved ones.

Definition 3.12 Let (X, τ) be a first countable topological space and $F_n : X \to \overline{\mathbb{R}}$ be a sequence of functionals, where $\overline{\mathbb{R}} = \mathbb{R} \cup \{\pm\infty\}$. We say that the sequence F_n Γ-*converges* to a limit functional $F_\infty : X \to \overline{\mathbb{R}}$ with respect to the τ-topology as $n \to \infty$ if, for every $x \in X$, the following two conditions hold:

(a) For every sequence such that $x_n \to x$ in τ,

$$F_\infty(x) \leq \liminf_{n\to\infty} F_n(x_n).$$

(b) There exists a sequence such that $x_n \to x$ in τ and

$$F_\infty(x) \geq \limsup_{n\to\infty} F_n(x_n).$$

Theorem 3.13 *Let (X, τ) be a first countable topological space and suppose that F_n is a sequence of functionals that Γ-converges to a functional F_∞. Suppose that $\{x_n\}_{n\geq 1}$ is a sequence of minimizers for F_n, i.e.,*

$$F_n(x_n) = \inf_{y\in X} F_n(y).$$

If the sequence converges to some $x \in X$, $x_n \to x$ in τ, then:

(i) $F_\infty(x) = \inf_{y\in X} F_\infty(y)$ *and*

(ii) $\lim_{n\to\infty} F_n(x_n) = F_\infty(x)$.

Now, we precise the functional towards which E_ε will converge and the space on which it is defined on.

De Giorgi initially developed the theory of functions of bounded with the Plateau Problem for hypersurfaces in mind. Similarly to the theory of varifolds or currents, the starting point is a generalized notion of hypersurfaces enjoying good compactness properties. For a comprehensive treatment, see [Mag12].

Definition 3.14 Let $u \in L^1(\Omega)$ and define

$$\int_\Omega |Du| := \sup \left\{ \int_\Omega u \operatorname{div} \phi \, \mathrm{d}^n x \mid \phi \in \mathcal{C}_c^1(\mathbb{R}^n) \right\}.$$

Then u is said to have *bounded variation* in Ω if $\int_\Omega |Du| < \infty$. We define $BV(\Omega)$ as the space of all functions in $L^1(\Omega)$ with bounded variation.

Remark 3.15 *$u \in BV(\Omega)$ if and only if its distributional gradient Du can be represented as a vector valued Radon measure with finite total variation.*

From the definition, it is clear that $W^{1,1}(\Omega) \subset BV(\Omega)$; however, the converse does not hold. A relevant counterexample is the indicator function $\chi_E \in L^1(\Omega)$, where $E \Subset \Omega$ is a bounded open set with a $\mathcal{C}^2$ boundary. It can be shown that $\chi_E \in BV(\Omega) \setminus W^{1,1}(\Omega)$. This is because its distributional derivative is not in $L^1(\Omega)$, but rather a Radon measure concentrated entirely on the boundary ∂E, representing the surface area of the set.

Definition 3.16 Let E be a Borel of $\mathbb{R}^n$, we define the *perimeter of E in Ω* as

$$P(E, \Omega) = \int_\Omega |D\chi_E|.$$

E is called a *Caccioppoli set* if it has finite perimeter, i.e., $\chi_E \in BV(\Omega)$.

In the BV theory, Caccioppoli sets play the role of hypersurfaces and the perimeter corresponds to the area functional. Naturally, minimal hypersurfaces will correspond to the critical points of P.

Definition 3.17 We say a Borel set $E \subset \mathbb{R}^n$ of locally finite perimeter in Ω *minimizes perimeter* in Ω if, for any Borel set $E' \subset \mathbb{R}^n$ such that $E \triangle E' \Subset \Omega$, we have

$$P(E, \Omega) \leq P(E', \Omega).$$

The existence of minimizers of perimeter is guaranteed mainly by the compactness of BV functions.

Theorem 3.18 (BV compactness) *Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with Lipschitz boundary. Then sets of functions uniformly bounded in the BV-norm, defined by*

$$\|u\|_{BV(\Omega)} := \|u\|_{L^1(\Omega)} + \int_\Omega |Du|,$$

are relatively compact in $L^1(\Omega)$.

We can now look at the ideas behind the Γ-convergence of the Allen–Cahn functional towards the perimeter functional. Precisely, we will see that E_ε converges towards σP, which is to be expected given that the interfaces separating the two phases will store energy equal to their size times their energy density.

The first of the properties of Γ-convergence follows from the following computation and BV compactness. Define

$$\Phi(t) := \int_0^t \sqrt{2W(s)}\,\mathrm{d}s,$$

then for any $u \in H^1(\Omega)$

$$\int_\Omega |\nabla\Phi(u)| = \int_\Omega \sqrt{2W(u)}|\nabla u| \leq \int_\Omega \frac{\varepsilon|\nabla u|^2}{2} + \frac{W(u)}{\varepsilon},$$

where the inequality comes from $2ab \leq a^2 + b^2$. Thus, if u_ε is a sequence of critical points of (ACE$_\varepsilon$) with uniformly bounded energy $E_\varepsilon(u_\varepsilon) \leq E_0$, we can use BV compactness to extract a converging subsequence of $\{\Phi(u_\varepsilon)\}_\varepsilon$ to some $w_\infty \in BV(\Omega)$. Defining $u_\infty := \Phi^{-1}(w_\infty)$ and using that

$$\int_\Omega W(u_\varepsilon) \leq E_0\varepsilon,$$

we have the following result.

Proposition 3.19 *Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with Lipschitz boundary and $\{u_\varepsilon\}_\varepsilon$ a sequence of solutions such that $E_\varepsilon(u_\varepsilon) \leq E_0$ for some $E_0 > 0$. Then there exists a subsequence $\varepsilon_k \to 0$ and $u_\infty \in BV(\Omega)$ with $u_\infty \in \{\pm 1\}$ a.e. and $u_{\varepsilon_k} \to u_\infty$ in $L^1(\Omega)$ as $k \to \infty$. Moreover,*

$$\sigma P(\{u_\infty = 1\}, \Omega) \leq \liminf_{k\to\infty} E_{\varepsilon_k}(u_{\varepsilon_k}, \Omega).$$

The second property in the definition of Γ-convergence (known as the *recovery property*) can be obtained by placing a heteroclinic profile over the boundary of any set of finite perimeter. A rigorous version of this idea gives the following.

Proposition 3.20 *If $E \subset \Omega$ has finite perimeter then there exists a sequence $\{u_\varepsilon\}_\varepsilon$ in $H^1(\Omega) \cap L^4(\Omega)$ such that*

$$\sigma P(E, \Omega) = \lim_{\varepsilon\to 0^+} E_\varepsilon(u_\varepsilon, \Omega)$$

and $u_\varepsilon \to \chi_E - \chi_{\Omega\setminus E}$ in $L^1(\Omega)$.

The combination of the previous two results gives the Γ-convergence of the Allen–Cahn functional towards the perimeter functional.

Theorem 3.21 (Modica–Mortola) *Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with Lipschitz boundary. Define the extended Allen–Cahn energy $F_\varepsilon : L^1(\Omega) \to \overline{\mathbb{R}}$ by*

$$F_\varepsilon(u) = \begin{cases} E_\varepsilon(u,\Omega) & \text{if } u \in H^1(\Omega), \\ +\infty & \text{otherwise,} \end{cases}$$

and $F_\infty : L^1(\Omega) \to \overline{\mathbb{R}}$ by

$$F_\infty(u) = \begin{cases} \sigma P(\{u=1\},\Omega) & \text{if } u \in BV(\Omega;\{-1,1\}), \\ +\infty & \text{otherwise.} \end{cases}$$

Then, F_ε Γ-converges to F_∞ with respect to the strong $L^1(\Omega)$ topology as $\varepsilon \to 0^+$.

We can apply the previous result to sequences of minimizers of the Allen–Cahn energy to find minimizers of perimeter with a small caveat. Since the only global minimizers of E_ε are constant functions, the corresponding perimeter minimizing set will be empty. To fix this, we have to consider instead local minimizers.

Definition 3.22 Fix $\delta > 0$. A function $u \in H^1(\Omega)$ with finite energy $E_\varepsilon(u,\Omega) < \infty$ is a *strict δ-local minimizer* of E_ε if u minimizes the energy in a ball of radius δ in $L^1(\Omega)$, i.e., $E_\varepsilon(u) < E_\varepsilon(v)$ for any $v \in L^1(\Omega)$ with $0 < \|u-v\|_{L^1(\Omega)} \leq \delta$.

Definition 3.23 We say a Borel set $E \subset \mathbb{R}^n$ of locally finite perimeter in Ω *(strictly) locally minimizes perimeter* in Ω if there exists $\delta > 0$ such that for any $E' \subset \Omega$ such that $0 < \|\chi_E - \chi_{E'}\|_{L^1(\Omega)} \leq \delta$ and $E \triangle E' \Subset \Omega$ then $P(E,\Omega) \leq (<) P(E',\Omega)$.

Combining Theorem 3.21 and Theorem 3.13 with X a ball of radius δ in $L^1(\Omega)$ we have the following.

Proposition 3.24 *Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with Lipschitz boundary and $\{u_\varepsilon\}_\varepsilon$ a sequence of δ-local minimizers of $E_\varepsilon(\cdot,\Omega)$. For $u_\infty \in BV(\Omega)$ such that (up to a subsequence) $u_\varepsilon \to u_\infty$ in $L^1(\Omega)$ and $u_\infty \in \{\pm 1\}$ a.e.; the set $E := \{u_\infty = 1\}$ locally minimizes perimeter.*

In combination with the following well-known regularity theory for area-minimizing hypersurfaces (see [Fed14]), we conclude that the level sets of a sequence of minimizers of the Allen–Cahn converge towards smooth (locally) area-minimizing minimal hypersurfaces.

Theorem 3.25 *If $E \subset \Omega$ is a local minimizer of perimeter for $2 \leq n \leq 7$, then after changing E by a set of measure zero, the topological boundary of E, ∂E , is a smooth hypersurface.*

All of the previous results are also true if we replace Ω with a closed Riemannian manifold (M,g) (and make the necessary adaptations, see Section

6.3). By the observation made at the end of Section 2.1, we see that not every closed manifold will have a local minimizer of the Allen–Cahn energy as otherwise this would mean that it also has a stable minimal hypersurface.

In view of this, if we hope to use critical points of the Allen–Cahn energy to find minimal hypersurfaces in arbitrary closed Riemannian manifolds, we are forced to look for unstable critical points. This introduces two major challenges. First, we cannot use the direct method of the calculus of variations; instead, we must employ a min-max scheme, which is generally more involved. Second, the regularity theory for non-area-minimizing minimal surfaces is much harder than that of area-minimizing ones. In Chapter 7 we present the solution of Guaraco [Gua18] to these two problems.

To finish, we present an example demonstrating that all hope is not lost before delving into the full solution. At the end of Section 2.1, we saw that the equator of a sphere is an unstable minimal hypersurface. We now claim that it can be obtained as the limit of a sequence of critical points of E_ε. Indeed, it is possible to construct a family of solutions $u_\varepsilon : \mathbb{S}^2 \to [-1, 1]$ to the Allen–Cahn equation such that the nodal set $\{u_\varepsilon = 0\}$ lies exactly at the equator, with $\{u_\varepsilon > 0\}$ and $\{u_\varepsilon < 0\}$ occupying opposite hemispheres[3] (see Fig. 3.3). As $\varepsilon \to 0^+$, the transition layers $\{|u_\varepsilon| \leq 1 - b\}$ converge strictly to the equator for any $b \in (0, 1)$. Hence, our goal for the remainder of these notes is to guarantee the existence of such a sequence of critical points without a priori knowledge of the underlying minimal surface or the ambient manifold.

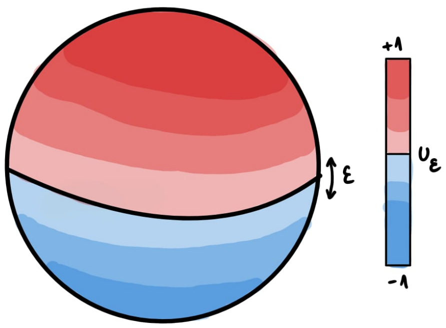

Figure 3.3: Sketch of the function u_ε solving the Allen–Cahn equation on $\mathbb{S}^2$ and nodal set $\{u_\varepsilon = 0\}$ at the equator.

[3] For example, one can find a minimizer of the Allen–Cahn energy on one of the hemispheres subject to the Dirichlet boundary condition $u = 0$ on the equator. The solution can be shown to be non-zero in the interior, and a full, smooth solution on $\mathbb{S}^2$ can then be constructed via odd reflection across the equator. See, for instance, [LMN20] for more details.

Chapter 4

Stable Stationary Varifolds from Allen–Cahn

In this first chapter we begin our look into the theory of general critical points of the Allen–Cahn energy (ACE_ε). Unlike in the previous chapter, we don't focus only on minimizers of $E_\varepsilon(\cdot, \Omega)$ and instead look at general critical points, i.e., those solving the Allen–Cahn equation (AC_ε). We will follow the remarkable paper of Hutchinson and Tonegawa [HT00] which studied the varifold theory of the Allen–Cahn energy first proposed in [PT98] refining the ideas in the 1993 paper of Illmanen for the parabolic Allen–Cahn equation [Ilm93].

The chapter is organized into three parts. First, we look at how to associate a varifold with a solution of the Allen–Cahn equation. The starting point is a family of solutions to (AC_ε), one for each value of ε, which are critical points of E_ε with uniformly bounded energy and L^∞-norm. A varifold can be associated with each of these functions by averaging over $\mathbb{R}$ the hypersurfaces defined by their level sets. Using arguments analogous to those in Section 3.3, a converging subsequence of varifolds can be extracted. Following this construction, we begin our study of the limit varifold by deriving a local monotonicity type formula for the energy. By utilizing this monotonicity we obtain important properties of the related limiting energy and discrepancy measures, which we then use to prove that the limiting varifold is rectifiable, stationary and integral.

In addition to the assumption (**A**) on the potential W, we will assume throughout the following hypothesis:

> *$\Omega \subset \mathbb{R}^n$ is a bounded open set with Lipschitz boundary and $\{u_k\}_{k\geq 1}$ is a sequence of $C^3(\Omega)$ functions satisfying*
>
> $$\varepsilon_k \Delta u_k = \varepsilon_k^{-1} W'(u_k)$$
>
> *on Ω, with $\varepsilon_k \to 0$ as $k \to \infty$. Moreover, the functions and their energy are uniformly bounded* **(B)**
>
> $$\sup_{\Omega} |u^k| \leq C_0; \quad \int_{\Omega} \varepsilon_k \frac{|\nabla u_k|^2}{2} + \frac{W(u_k)}{\varepsilon_k} \leq E_0$$
>
> *for C_0, $E_0 \geq 0$ constants independent of k.*

The existence of such a sequence can be proven in $H^1(\Omega)$ using standard tools in functional analysis[1] and Proposition 3.5 guarantees the regularity properties.

Remark 4.1 *The original paper [HT00] includes the case in which the critical points are taken with the additional mass constraint $\int_{\Omega} u = m$. This corresponds to solving*

$$\varepsilon_k \Delta u_k = \varepsilon_k^{-1} W'(u_k) - \lambda_k$$

with λ_k the Lagrange multiplier of the constraint. This is modeling, for example, the case of a mixture of two different components that do not mix as opposed to the unconstrained case which models phase transitions of a single two-phase component. For the sake of simplicity we deal here with the unconstrained case (corresponding to $\lambda_k = 0$) which is all we will need in the rest of the work.

4.1 Defining the Varifolds

Up to now, we have only utilized the energy measure associated with the critical points u_k of E_{ε_k},

$$\mathrm{d}\mu_k = \left(\varepsilon_k \frac{|\nabla u_k|^2}{2} + \frac{W(u_k)}{\varepsilon_k} \right) \mathrm{d}^n x,$$

to deduce properties of the limit. However, the geometric information captured solely by the measure μ_k is limited. A much finer analysis requires an object that records not only the energy density but also the infinitesimal geometry of the level sets. This is naturally achieved using the framework of varifolds.

[1] For example, in Section 7.2 this is done on a closed manifold with the additional property of a uniformly bounded Morse index.

To construct the varifolds we use the same ideas as in Section 3.3. Define $w_k = \Phi \circ u_k$ with

$$\Phi(s) := \int_0^t \sqrt{W(s)/2}\,\mathrm{d}s. \tag{4.1}$$

Since $|\nabla w_k| = \sqrt{W(u_k)/2}|\nabla u_k|$ we have $\int_\Omega |\nabla w_k| \leq E_0/2$ and $\Phi(-C_0) \leq w_k \leq \Phi(C_0)$. By compactness of BV functions $w_k \to w_\infty$ a.e. and in L^1 up to a subsequence, therefore $u_k \to u_\infty := \Phi^{-1}(w_\infty)$ a.e. and in L^1. Also, $u_\infty \in \{\pm 1\}$ a.e. on Ω and the sets $\{u_\infty = \pm 1\}$ are of finite perimeter.

Remark 4.2 (Warning on Φ and σ) *A closer look at* (4.1) *reveals that it is not exactly the same function as the one defined in Section 3.3, as it differs by a factor of* 2. *Consequently, the constant σ also differs from the one in Section 3.3 by this same factor. This convention is maintained throughout the rest of the notes to simplify computations by avoiding an extra factor of* 2.

Since w_k are smooth, by Sard's theorem[2] the set $\{w_k = t\} \subset \Omega$ is $(n-1)$-rectifiable for $\mathcal{L}$-a.e. $t \in \mathbb{R}$. We then define the $(n-1)$-varifold associated to u_k via w_k as

$$V_k(A) := \int_{\mathbb{R}} v(\{w_k = t\})(A)\,\mathrm{d}t \tag{4.2}$$

for $A \subset G_{n-1}(\Omega)$ Borel (see Definition 2.18). The varifolds V_k can be interpreted as the averaging the varifolds associated to the hypersurfaces coming from the level sets, leading to a diffuse object with mass concentrated around the phase transition.

Using the coarea formula (see Appendix (A)) we find the weight of V_k is given by

$$\|V_k\|(B) = \int_{\mathbb{R}} \mathcal{H}^{n-1}(\{w_k = t\} \cap B)\,\mathrm{d}t = \int_B |\nabla w_k| = \int_B \sqrt{W(u_k)/2}|\nabla u_k| \tag{4.3}$$

for $B \subset \Omega$ Borel. As in Section 3.3, using $2ab \leq a^2 + b^2$ and the fact that the u_k's have uniformly bounded energy we find that $\|V_k\|(\Omega)$ is uniformly bounded and therefore, by the theory of varifolds, has a convergent subsequence to some $V \in \mathbf{V}_{n-1}(\Omega)$ in the varifold sense.

With this, the following theorem summarizes the results we are after in this section.

Theorem 4.3 (Limiting interface) *Let V_k be the varifolds associated to u_k via w_k in* (4.2) *under hypothesis* (**A**) *and* (**B**). *Then, up to a subsequence $u_k \to u_\infty$ a.e. and there exists a stationary rectifiable varifold V such that $V_k \to V$ as varifolds and $\sigma^{-1}V$ is integral. Furthermore*

[2] Precisely, the version of Federer: if $f : \mathbb{R}^m \to \mathbb{R}^n$ is Lipschitz and $m > n$ then $f^{-1}(y)$ is $(m-n)$-rectifiable for $\mathcal{L}^n$ almost all y (see Theorem 3.2.15 in [Fed14]). We note that in [PT98] and [HT00] Sard's theorem is mentioned to justify rectifiability however this is a slight abuse as Sard's theorem would require w_k to be $\mathcal{C}^n$.

(i) for $\phi \in \mathcal{C}_c(\Omega)$

$$\|V\|(\phi) = \lim_{k\to\infty} \int \phi \frac{\varepsilon_k |\nabla u_k|^2}{2} = \lim_{k\to\infty} \int \phi \frac{W(u_k)}{\varepsilon_k};$$

(ii) $\operatorname{spt}\|\partial\{u_\infty = 1\}\| \subset \operatorname{spt}\|V\|$ *and u_k converges locally uniformly to ± 1 on $\Omega \setminus \operatorname{spt}\|V\|$;*

(iii) for each open subset $\Omega' \Subset \Omega$ and $0 < b < 1$, $\{|u_k| \leq 1 - b\} \cap \overline{\Omega'} \to$ $\operatorname{spt}\|V\| \cap \overline{\Omega'}$ *in the Hausdorff distance.*

The interpretation of the results in Theorem 4.3 is quite clear. Notably, since $|\nabla w_k|\, \mathrm{d}^n x = \|V_k\|$, point (i) says that the weight of the limit varifold $\|V\|$ is precisely half of the limiting energy measure of $\mathrm{d}\mu_{\varepsilon_k}$, in particular the energy in the limit is *equipartitioned* between the kinetic and potential energy. We also see how the V_k's carry at least all the information that the $\mathrm{d}\mu_k$'s do.

4.2 The Monotonicity Type Formula

The first step towards proving Theorem 4.3 is to derive a local monotonicity type formula for the energy. After the computation in Lemma 4.6 we will see that this amounts to studying the discrepancy function

$$\xi_\varepsilon(u) := \varepsilon \frac{|\nabla u|^2}{2} - \frac{W(u)}{\varepsilon}.$$

Using a modified discrepancy function to prove a technically involved lemma leads to a lower bound for ξ_ε and the main monotonicity result in Proposition 4.8.

Remark 4.4 *If u_ε solves* (AC_ε) *then the discrepancy function $\xi_\varepsilon(u)$ has an important geometric interpretation tied directly to the curvature of the level sets of u. At any point where $|\nabla u_\varepsilon| > 0$, let $\boldsymbol{\nu}$ denote the unit normal vector to the level set, and let $\boldsymbol{H}$ be its mean curvature. Given that $\boldsymbol{H} = \operatorname{div} \boldsymbol{\nu}$ it can shown that*

$$\langle \nabla \xi_\varepsilon, \boldsymbol{\nu} \rangle = -\varepsilon \boldsymbol{H} |\nabla u_\varepsilon|^2.$$

With this, we see that if $\xi_\varepsilon \to 0$ as $\varepsilon \to 0^+$ we can expect the level sets $\{u_\varepsilon = t\}$ to converge towards minimal surfaces. It is very useful to keep this relationship in mind in everything that follows.

In the following, we chose an arbitrary element of the sequence in (**B**) and drop the subindex for ease of notation $u = u_k$. Some results will require a small choice of $\varepsilon = \varepsilon_k$ amounting to choosing u further down the sequence. We also fix $\Omega' \Subset \Omega$ and define for $B_r(x) \Subset \Omega$ the following quantity

$$E_\varepsilon(r, x) := \frac{1}{r^{n-1}} \int_{B_r(x)} \varepsilon \frac{|\nabla u|^2}{2} + \frac{W(u)}{\varepsilon} \tag{4.4}$$

which will appear in the LHS of the monotonicity formula.

Remark 4.5 *The constrained case requires changing the potential in $E(r,x)$ for a modified potential whose role is to absorb the terms coming from the constraint. As we are not dealing with this case some of the arguments and notation presented here will be simpler compared to those in [HT00].*

Lemma 4.6 *If $B_R(x) \Subset \Omega$ and $R > r > 0$, then*

$$E_\varepsilon(R,x) - E_\varepsilon(r,x) = \int_r^R \left(\frac{1}{\rho^n} \int_{B_\rho(x)} \frac{W(u)}{\varepsilon} - \varepsilon \frac{|\nabla u|^2}{2} \right) \mathrm{d}\rho \\ + \varepsilon \int_{B_R(x) \setminus B_r(x)} \frac{\langle (y-x), \nabla u \rangle^2}{|y-x|^{n+1}} \mathrm{d}^n y. \quad (4.5)$$

Proof Let $\phi = (\phi^1, \cdots, \phi^n) \in \mathcal{C}_c^1(\Omega, \mathbb{R}^n)$. Multiplying both sides of the Allen–Cahn equation by $\langle \nabla u, \phi \rangle$ and integrating by parts we find

$$0 = \int_\Omega \varepsilon \operatorname{div}(\nabla u) \langle \nabla u, \phi \rangle - \varepsilon^{-1} \langle W'(u) \nabla u, \phi \rangle \\ = - \int_\Omega \varepsilon \nabla u \nabla(\langle \nabla u, \phi \rangle) + W(u) \operatorname{div} \phi.$$

Writing out the first term using Einstein notation

$$\nabla u \nabla(\langle \nabla u, \phi \rangle) = \partial_i u (\partial_i \partial_j u) \phi^j + \partial_i u \partial_j u \partial_i \phi^j = \frac{1}{2} \partial_i (\partial_j u)^2 \phi^j + \partial_i u \partial_j u \partial_i \phi^j$$

and integrating by parts once more

$$\int_\Omega \left(\varepsilon \frac{|\nabla u|^2}{2} + \frac{W(u)}{\varepsilon} \right) \operatorname{div} \phi - \varepsilon \sum_{i,j} \partial_i u \partial_j u \partial_i \phi^j \, \mathrm{d}^n y = 0. \quad (4.6)$$

Now, let $z = y - x$ and choose $\phi(z) = z\psi(\rho)$ with $\rho = |z|$ and ψ a radial smooth approximation of $\chi_{B_\rho(0)}$, then

$$\int_\Omega \left(\varepsilon \frac{|\nabla u|^2}{2} + \frac{W(u)}{\varepsilon} \right) (\rho \psi' + n\psi) - \varepsilon \left(\frac{\psi'}{\rho} \langle z, \nabla u \rangle^2 + |\nabla u|^2 \psi \right) \mathrm{d}^n z = 0.$$

Letting $\psi \to \chi_{B_\rho(0)}$ in L^1 we have that $\nabla \psi \to \nabla \chi_{B_\rho} = -\nu \mathcal{H}^{n-1} \llcorner \partial B_\rho$ as distributions/measures, with $\nu = z/|z|$, therefore

$$-(n-1) \int_{B_\rho} \left(\varepsilon \frac{|\nabla u|^2}{2} + \frac{W(u)}{\varepsilon} \right) + \rho \int_{\partial B_\rho} \left(\varepsilon \frac{|\nabla u|^2}{2} + \frac{W(u)}{\varepsilon} \right) \\ = \int_{B_\rho} \left(\frac{W(u)}{\varepsilon} - \varepsilon \frac{|\nabla u|^2}{2} \right) + \varepsilon \int_{\partial B_\rho} \frac{\langle z, \nabla u \rangle^2}{\rho}.$$

Multiplying by ρ^n, integrating between r and R and substituting $z = y - x$ we immediately find the RHS is the one in (4.5). For the LHS we also have to integrate by parts the first term and use once again that $\nabla \chi_{B_\rho} = -\nu \mathcal{H}^{n-1} \llcorner \partial B_\rho$ to conclude. □

We note that only the first term in the RHS of (4.5) can be negative and it contains the discrepancy function. Thus, to obtain a monotonicity formula we just need to control ξ_ε from below.

Proposition 4.7 *There exists constants c_2 and $\overline{\varepsilon_2}$ depending only on C_0, $d(\Omega', \partial\Omega)$ and W such that*

$$\inf_{\Omega'} \xi_\varepsilon \geq -c_2$$

whenever $\varepsilon < \overline{\varepsilon_2}$.

As an immediate corollary of the previous estimate we have the monotonicity type formula, therefore the remainder of the proof consists on showing Proposition 4.7.

Proposition 4.8 (Monotonicity Type Formula) *Suppose $B_R(x) \subset \Omega'$, $R > r > 0$ and $\varepsilon \leq \overline{\varepsilon_2}$. Then*

$$\begin{aligned} E_\varepsilon(R,x) - E_\varepsilon(r,x) \geq \int_r^R &\left[\frac{1}{\rho^n} \int_{B_\rho(x)} \left(\frac{W(u)}{\varepsilon} - \varepsilon \frac{|\nabla u|^2}{2} \right)^+ \right] \mathrm{d}\rho \\ &- c_3 R + \varepsilon \int_{B_R(x) \setminus B_r(x)} \frac{\langle (y-x), \nabla u \rangle^2}{|y-x|^{n+1}} \, \mathrm{d}^n y. \quad (4.7) \end{aligned}$$

where $(f)^+ = \max(f, 0)$ and $c_3 = c_2 \omega_n$.

Before discussing the discrepancy function, we prove a small but useful maximum principle type lemma.

Lemma 4.9 *There exists constants c_1 and $\overline{\varepsilon_1}$ depending only on C_0, $d(\Omega', \partial\Omega)$ and W such that*

$$\sup_{\Omega'} |u| \leq 1 + c_1 \varepsilon$$

whenever $\varepsilon < \overline{\varepsilon_1}$.

Proof Let $d > 0$ such that $B_{3d} \subset \Omega$ and $\zeta \in \mathcal{C}^\infty(B_{3d})$ such that

1. $1 + c_1\varepsilon/2 \leq \zeta \leq 1 + c_0$ on B_{3d},
2. $\zeta \equiv 1 + c_1\varepsilon/2$ on B_d,
3. $\zeta \equiv 1 + c_0$ on $B_{3d} \setminus B_{2d}$,

where $c_1 \geq 0$ is to be fixed later. For the sake of contradiction assume that $\sup_{B_d} u \geq 1 + c_1\varepsilon$ and define $g = u - \zeta$. Then $g \leq -1$ on ∂B_{3d} and $\sup_{B_{3d}} g \geq c_1\varepsilon/2$ therefore it has an interior maximum point x_0 such that $g(x_0) \geq c_1\varepsilon/2$. Furthermore, at x_0

$$\begin{aligned} 0 \geq \varepsilon\Delta g &= W'(u)/\varepsilon - \varepsilon\Delta\zeta \\ &= \frac{g}{\varepsilon} \int_0^1 W''(tu + (1-t)\zeta) \, \mathrm{d}t + \frac{W'(\zeta)}{\varepsilon} - \varepsilon\Delta\zeta \geq \frac{\kappa c_1}{2} - \varepsilon \max|\Delta\zeta| \end{aligned}$$

where the last inequality follows from the assumptions (**A**) on W, in particular since $u(x_0), \zeta(x_0) \geq 1$ we have $W'' > \kappa$ on the line joining them and $W' \geq 0$. Thus, by choosing c_1 large enough, we arrive at a contradiction due to how ζ was chosen. Showing $\inf u \geq 1 + c_1\varepsilon$ is completely analogous. $\square$

We look now at the discrepancy function to prove Proposition 4.7. Without loss of generality assume $0 \in \Omega'$, let $B_{3d} \subset \Omega'$ and consider the rescaling $x \mapsto x/d\varepsilon$ yielding the equivalent problem

$$\Delta u = W'(u) \quad \text{on } B_{3/\varepsilon}(0).$$

For simplicity, we work with the rescaled problem during the rest of the proof as well as the rescaled discrepancy function

$$\xi = \frac{|\nabla u|^2}{2} - W(u).$$

We also introduce the modified discrepancy function

$$\xi_G := \xi - G(u) = \frac{|\nabla u|^2}{2} - W(u) - G(u),$$

where $G : \mathbb{R} \to \mathbb{R}$ is some function to be chosen later.

Lemma 4.10 *On the set $\{|\nabla u| > 0\}$ it holds*

$$\Delta\xi_G - \frac{2(W' + G')}{|\nabla u|^2}\langle \nabla u, \nabla \xi_G\rangle + 2G''\xi_G \geq (G')^2 + G'W' - 2G''(W + G).$$

Proof Using the definition of ξ_G and the fact that u solves $\Delta u = W'(u)$ we find

$$\Delta\xi_G = \sum_{i,j}(\partial_i\partial_j u)^2 - W'(W' + G') - 2G''(G + W + \xi_G).$$

The result then follows from the following inequality applied to the first term

$$\begin{aligned} |\nabla u|^2 \sum(\partial_i\partial_j u)^2 &\geq \sum_j \left(\sum_i \partial_i u_{x_i}\partial_i\partial_j u\right)^2 \\ &= \sum_j \left(\partial_j\xi_G + (W' + G')\partial_j u\right)^2 \\ &\geq 2(W' + G')\langle \nabla u, \nabla\xi_G\rangle + (W' + G')^2|\nabla u|^2. \end{aligned}$$

where the first inequality follows from Cauchy-Schwarz. $\square$

The heart of the proof of Proposition 4.7 is in the next result for the rescaled discrepancy function. Indeed, assuming Lemma 4.11 it is enough to undo the rescaling and apply the bound on a finite number of balls covering Ω' and contained in Ω.

The proof of Lemma 4.11 is technical and not very enlightening so we only provide a sketch. The main ideas to keep in mind are the choice of an adequate G and to exploit the general shape of W described in (**A**). For the full proof see Proposition 3.6 in [HT00].

Lemma 4.11 *There exists constants c_4 and $\overline{\varepsilon_4}$ depending only on C_0 and W such that*

$$\sup_{B_{1/\varepsilon}(0)} \xi \leq c_4 \varepsilon$$

whenever $\varepsilon < \overline{\varepsilon_4}$.

Sketch of Proof Since $\sup_{\Omega'}|u| \leq C_0$, by standard elliptic [GT01] estimates there exists a constant c_5 such that $\sup_{B_{3/\varepsilon-1}}|\xi| \leq c_5$. Choosing first

$$G(u) = \varepsilon^{1/2}(1-(u-\gamma)^2/8)$$

and ε small enough it holds that $\sup_{B_{2/\varepsilon}} \xi_G \leq \varepsilon^{1/2}$ which implies that

$$\sup_{B_{2/\varepsilon}} \xi \leq \varepsilon^{1/2}.$$

Next choose

$$G(u) = L\varepsilon(1-(u-\gamma)^2/8)$$

then it holds that $\sup_{B_{1/\varepsilon}} \xi_G \leq L\varepsilon$ for L large enough. Since $G(u) \leq L\varepsilon$ the result follows from choosing $c_4 = 2L$. □

4.3 Stationarity & Rectifiability

With the monotonicity formula at our disposal we set out to prove the rectifiability and stationarity of the limiting varifold V. For this we will study the energy and discrepancy measures on $\mathbb{R}^n$ which will give us information at the limit. We will then use the results to control the first variation of V with the goal of applying the Rectifiability Theorem of Allard. Along the way we will also obtain the proofs of points (i) and (ii) in Theorem 4.3 and at the end we prove (iii).

By the Banach-Alaoglu theorem on the space of Radon measures of $\overline{\Omega}$ there exists a Radon measure μ_∞ such that up to a subsequence $\mu_k \xrightarrow{*} \mu_\infty$, i.e.,

$$\mu_\infty(\phi) := \lim_{k\to\infty} \int_\Omega \left(\varepsilon_k \frac{|\nabla u_k|^2}{2} + \frac{W(u_k)}{\varepsilon_k} \right) \phi.$$

We now study the properties of μ_∞ to see how it is related to the varifold V and what it allows us to say about it. The following result tells us that on small enough scales μ_∞ behaves like the weight of an $(n-1)$-submanifold.

Proposition 4.12 *There exists constants $0 < D_1 \leq D_2 < \infty$ and $r_0 > 0$ depending only on $C_0, E_0, d(\Omega', \partial\Omega)$ and W such that*

$$D_1 r^{n-1} \leq \mu_\infty(B_r(x)) \leq D_2 r^{n-1}$$

for all $0 < r < r_0$, $x \in \operatorname{spt}(\mu_\infty) \cap \Omega'$ and $B_r(x) \subset \Omega'$.

The proof of this proposition relies on the following lemma, which simply says that any point in the support of μ_∞ can be approximated by a sequence of high energy points of u_k, see Fig. 4.1.

Lemma 4.13 *Let $x \in \operatorname{spt}\mu_\infty \cap \Omega'$, then there exists a sequence $x_k \in \Omega'$ such that $u_k(x_k) \in [-\alpha, \alpha]$ and $x_k \to x$, with α as in* (**A**).

Proof For the sake of contradiction, assume that there exists $d > 0$ such that $B_d(x) \subset \Omega'$ and $B_d(x) \cap \{|u_k| \leq \alpha\} = \emptyset$ for all k larger than some N. Without loss of generality, suppose that $u_k > \alpha$ on $B_d(x)$, then by an analogous argument to those in Lemma 4.9 we find $u_k \in [1 - c\varepsilon_k, 1 + c\varepsilon_k]$ on $B_{d/2}(x)$, for a possibly larger N. Thus $W(u_k) = \mathcal{O}(\varepsilon_k^2)$ and then $W(u_k)/\varepsilon_k \to 0$ uniformly on $B_{d/2}(x)$ as $k \to \infty$.

On the other hand, using the Bochner identity[3] and the fact that u_k solves (AC_ε) we find

$$\varepsilon_k \Delta \left(\frac{1}{2} |\nabla u_k|^2 \right) = \frac{W''(u_k)}{\varepsilon} |\nabla u_k|^2 + \varepsilon_k |D^2 u_k|^2 \geq \frac{\kappa}{\varepsilon_k} |\nabla u_k|^2,$$

in particular for any non-negative $\phi \in \mathcal{C}_c^2(B_{d/2}(x))$

$$\int \varepsilon_k |\nabla u_k|^2 \phi \leq \frac{\varepsilon_k^2}{2\kappa} \int \Delta(\varepsilon_k |\nabla u_k|^2)\phi = \frac{\varepsilon_k^2}{2\kappa} \int \varepsilon_k |\nabla u_k|^2 \Delta\phi \xrightarrow{k\to\infty} 0$$

since $E_{\varepsilon_k}(u_k) \leq E_0$ and $\varepsilon_k \to 0$.

We have found that both the kinetic and potential energy terms go uniformly to 0 on $B_{d/2}(x)$ hence $\mu_\infty(B_{d/2}(x)) = 0$ contradicting the fact that $x \in \operatorname{spt}\mu_\infty$. $\square$

An immediate consequence of the previous proof is that on any compact subset of $\Omega \setminus \operatorname{spt}\mu_\infty$

$$\varepsilon_k \frac{|\nabla u_k|^2}{2} \xrightarrow{k\to\infty} 0 \quad \text{and} \quad \frac{W(u_k)}{\varepsilon_k} \xrightarrow{k\to\infty} 0 \tag{4.8}$$

uniformly and in particular either $u_k \to 1$ or $u_k \to -1$ on connected subsets. This also means $\operatorname{spt}\|\partial\{u_\infty = 1\}\| \subset \operatorname{spt}\mu_\infty$ and proves point (ii) of Theorem 4.3 once it has been established that $\mu_\infty = 2\|V\|$.

[3] $\Delta \frac{1}{2}|\nabla u|^2 = |D^2 u|^2 + \langle \nabla u, \nabla(\Delta u)\rangle$, where $|D^2 u|^2 = \sum_{i,j}(\partial_i \partial_j u)^2$

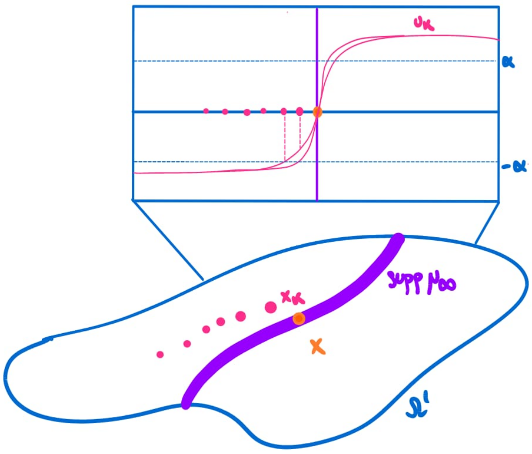

Figure 4.1: Sequence of high energy points converging towards a point in the support of μ_∞.

Proof (of Prop. 4.12) The upper bound is a direct consequence of the monotonicity formula. Indeed, inequality (4.7) implies

$$E_\varepsilon(x, R) + c_2 R \geq E_\varepsilon(x, r),$$

and the result follows from $r^{n-1}E_{\varepsilon_k}(x, r) = \mu_{\varepsilon_k}(B_r(x))$ and taking the limit $k \to \infty$.

For the lower bound let $\{x_k\}_k$ be a sequence as in Lemma 4.13 and $B_R(x) \subset \Omega'$, then

$$\begin{aligned} \frac{1}{R^{n-1}}\mu_\infty(B_R(x)) &\geq \lim_{k\to\infty} \frac{1}{R^{n-1}} \int_{B_{R/2}(x_k)} \varepsilon_k \frac{|\nabla u_k|^2}{2} + \frac{W(u_k)}{\varepsilon_k} \\ &\geq -\frac{c_3 R}{2^{n-1}} + \lim_{k\to\infty} \frac{1}{(2\varepsilon_k)^{n-1}} \int_{B_{\varepsilon_k}(x_k)} \varepsilon_k \frac{|\nabla u_k|^2}{2} + \frac{W(u_k)}{\varepsilon_k} \\ &\geq -\frac{c_3 R}{2^{n-1}} + \lim_{k\to\infty} \frac{1}{(2\varepsilon_k)^{n-1}} \int_{B_{\varepsilon_k}(x_k)} \frac{W(u_k)}{\varepsilon_k} \end{aligned}$$

where the first inequality comes from $B_{R/2}(x_k) \subset B_R(x)$ for large enough k and μ_k, μ_∞ being non-negative and the second inequality comes from the monotonicity (4.7) applied to $0 < \varepsilon_k < R/2$ for possibly larger k. Finally, we claim that the last integral is uniformly bounded below by a positive constant which concludes the proof after taking R small enough.

The idea behind the claim is the following: using the change of variables

$y = (x - x_k)/\varepsilon_k$ and a first order Taylor expansion around 0

$$\frac{1}{\varepsilon_k^{n-1}} \int_{B_{\varepsilon_k}(x_k)} \frac{W(u_k)}{\varepsilon_k} \, \mathrm{d}^n x = \int_{B_1(0)} W(u_k(y\varepsilon_k + x_k)) \, \mathrm{d}^n y$$
$$= \int_{B_1(0)} W(u_k(x_k)) + W'(u_k(x_k)) \langle \nabla u_k(x_k), y \rangle \varepsilon_k + \mathcal{O}(\varepsilon_k^2) \, \mathrm{d}^n y.$$

Since $W(u_k(x_k)) \in [-\alpha, \alpha]$ the first term is positive and uniformly lower-bounded. Moreover, higher order terms go to 0 as $\varepsilon_k \to 0$ due to $\widetilde{u}_k(y) := u_k(y\varepsilon_k + x_k)$ solving $\Delta \widetilde{u}_k = W'(\widetilde{u}_k)$ which implies derivatives are bounded on a compact subset due to elliptic regularity [GT01]. □

Now, for the same reason as in the case of the energy measure, we can consider the limiting absolute discrepancy measure $|\xi_\infty|$ defined by

$$|\xi_\infty| = \lim_{k \to \infty} \int_\Omega \left| \varepsilon_k \frac{|\nabla u_k|^2}{2} - \frac{W(u_k)}{\varepsilon_k} \right| \phi.$$

By the triangle inequality $|\xi_\infty| \leq \mu_\infty$ therefore $\operatorname{spt}|\xi_\infty| \subset \operatorname{spt} \mu_\infty$ and we actually have the following.

Proposition 4.14 *$|\xi_\infty|$ is the zero measure and so $\xi_k \to 0$ in $L^1_{loc}(\Omega)$. Moreover, both $\frac{\varepsilon_k}{2}|\nabla u_k|^2 - |\nabla w_k|$ and $\varepsilon_k^{-1} W(u_k) - |\nabla w_k|$ also converge to 0 in $L^1_{loc}(\Omega)$.*

Proof It is a standard result in measure theory that if $\operatorname{spt}|\xi_\infty| \subset \operatorname{spt} \mu_\infty$ and

$$\liminf_{r \to 0} \frac{|\xi_\infty|(B_r(x))}{\mu_\infty(B_r(x))} = 0 \quad \text{for all } x \in \operatorname{spt}|\xi_\infty|$$

then $|\xi_\infty| \equiv 0$; given that both measures are Radon (see Lemma 1.2 in [EG15]). In view of Proposition 4.12, it is then enough to show that

$$\liminf_{r \to 0} \frac{|\xi_\infty|(B_r(x))}{r^{n-1}} = 0$$

for all $x \in \operatorname{spt}|\xi_\infty| \cap \Omega'$ with $\Omega' \Subset \Omega$. For the sake of contradiction, assume that there is $x \in \operatorname{spt}|\xi_\infty| \cap \Omega'$, $0 < R \leq r_0$ and $b > 0$ such that $|\xi_\infty|(B_r(x)) \geq br^{n-1}$ for all $0 < r \leq R$. The contradiction will come from the monotonicity formula as otherwise the main term in the RHS involving $(\xi)^+$ will blow-up logarithmically at small scales if $|\xi_\infty|(B_r(x)) \geq br^{n-1}$. Precisely, choose

$$r_1 = \min\{b/4c_2\omega_n, \ R\} \quad \text{and} \quad r_2 = r_1 \min\{\exp(-4(4D_2 + c_3 r_1)/b), \ 1/2\}$$

and k large enough such that

$$\frac{1}{r_1^{n-1}} \int_{B_{r_1}(x)} \frac{\varepsilon_k |\nabla u_k|^2}{2} + \frac{W(u_k)}{\varepsilon_k} \leq 2D_2 \quad \text{and} \quad \frac{1}{\tau^{n-1}} \int_{B_\tau(x)} |\xi_k| \geq \frac{b}{2}$$

for all $r_2 \leq \tau \leq r_1$, which is possible thanks to the convergence of the respective measures. Then by Proposition 4.7 and choice of r_1

$$\begin{aligned}\frac{1}{\tau^{n-1}} \int_{B_\tau(x)} \left(\frac{W(u_k)}{\varepsilon_k} - \frac{\varepsilon_k |\nabla u_k|^2}{2} \right)^+ \\ \geq \frac{1}{\tau^{n-1}} \int_{B_\tau(x)} |\xi_k| - c_2 \omega_n \tau \geq \frac{b}{2} - c_2 \omega_n r_1 \geq \frac{b}{4}\end{aligned}$$

for all $r_2 \leq \tau \leq r_1$. By the monotonicity formula (4.7) and choice of r_2

$$\begin{aligned}2D_2 \geq \frac{1}{r_1^{n-1}} \int_{B_{r_1}(x)} \frac{\varepsilon_k |\nabla u_k|^2}{2} + \frac{W(u_k)}{\varepsilon_k} \\ \geq \frac{b}{4} \int_{r_2}^{r_1} \frac{\mathrm{d}\tau}{\tau} = \frac{b}{4} \ln(r_1/r_2) - c_3 r_1 \geq 4D_2\end{aligned}$$

which is a contradiction, hence $|\xi_\infty| \equiv 0$. It immediately follows that $|\xi_k| \to 0$ in $L^1_{\mathrm{loc}}(\Omega)$. Finally, given that $2|\nabla w_k| = \sqrt{2W(u_k)}|\nabla u_k|$ we find

$$\begin{aligned}\left| \frac{\varepsilon_k |\nabla u_k|^2}{2} + \frac{W(u_k)}{\varepsilon_k} - 2|\nabla w_k| \right| = \left(\sqrt{\varepsilon_k/2}|\nabla u_k| - \sqrt{W(u_k)/\varepsilon_k} \right)^2 \\ \leq \left| \frac{\varepsilon_k |\nabla u_k|^2}{2} - \frac{W(u_k)}{\varepsilon_k} \right| = |\xi_k| \qquad (4.9)\end{aligned}$$

which, together with $|\xi_k| \to 0$, proves the rest of the claims. □

We have just obtained the anticipated result of the equipartitioned energy in (i) of Theorem 4.3. As mentioned previously, what this means is that as the energy levels accumulate towards the interface corresponding to μ_∞, they do so while maintaining the kinetic and potential energies balanced. In the case of a sequence of energy minimizing critical points this is to be expected as both the kinetic and potential energy term are contributing the most energy around $u \approx 0$. It is then remarkable that the same happens for general critical points.

Now, using the tools we have developed up to this point, we can prove the rectifiability of V and in order to do this we will need the first variation of the varifolds V_k.

Lemma 4.15 (First variation) *The first variation of the varifolds V_k as defined in* (4.2) *is given by*

$$\delta V_k(\phi) = \int_\Omega \left(\operatorname{div} \phi - \sum_{i,j=1}^{n} \frac{\partial_i w_k \, \partial_j w_k}{|\nabla w_k|^2} \, \partial_i \phi^j \right) |\nabla w_k| \qquad (4.10)$$

for $\phi \in C^1_c(\Omega, \mathbb{R}^n)$.

Remark 4.16 *We note that the the first variation has the same form for any varifold constructed from the level sets of a function such as the present case. This expression can be obtained from the definition and an application of the coarea formula, for more details see Section 2.1 in [PT98]. Moreover, we have that*

$$\operatorname{div}\phi - \sum_{i,j=1}^{n} \frac{\partial_i w_k \, \partial_j w_k}{|\nabla w_k|^2} \, \partial_i \phi^j$$

is exactly the tangential divergence of the hypersurface coming from the level sets of w_k.

Theorem 4.17 (Rectifiability and Stationarity) *The limit varifold V satisfies $2\|V\| = \mu_\infty$, is rectifiable in $\operatorname{spt}\|V\|$ and stationary.*

Proof Since the V_k converge as varifolds towards V we have $\|V_k\| \to \|V\|$ and from $|V_k| = |\nabla w_k| \, \mathrm{d}^n x$ and the estimate (4.9) in the previous proposition we have $\mu_\infty = 2\|V\|$, which proves the first claim.

The rectifiability will follow from Allard's Rectifiability Theorem so we first have to determine δV. For this, note that $\partial_i u_k / |\nabla u_k| = \partial_i w_k / |\nabla w_k|$ which in combination with equation (4.6) gives

$$\int_\Omega \left(\operatorname{div}\phi - \sum_{i,j=1}^{n} \frac{\partial_i w_k \, \partial_j w_k}{|\nabla w_k|^2} \, \partial_i \phi^j \right) \varepsilon_k |\nabla u_k|^2 - \xi_k \operatorname{div}\phi = 0,$$

for $\phi \in C^1_c(\Omega, \mathbb{R}^n)$. Using this we can estimate (twice) the first variation in (4.10)

$$\begin{aligned} |2\delta V_k(\phi)| &= \left| \int_\Omega \left(\operatorname{div}\phi - \sum_{i,j=1}^{n} \frac{\partial_i w_k \, \partial_j w_k}{|\nabla w_k|^2} \, \partial_i \phi^j \right) (2|\nabla w_k| - \varepsilon_k |\nabla u_k|^2) + \xi_k \operatorname{div}\phi \right| \\ &\leq (2(n+n^2)+n) \sup_{\Omega'} |\nabla \phi| \int_{\Omega'} |\xi_k| \end{aligned}$$

with $\operatorname{spt}\phi \subset \Omega'$, where the bound follows from the triangle inequality combined with the estimates $|\operatorname{div}\phi| \leq n \sup|\nabla\phi|$, $|\partial_i w_k / |\nabla w_k|| \leq 1$ and

$$\begin{aligned} &\left| 2|\nabla w_k| - \varepsilon_k |\nabla u_k|^2 \right| \\ &\quad \leq \left| \frac{\varepsilon_k |\nabla u_k|^2}{2} + \frac{W(u_k)}{\varepsilon_k} - 2|\nabla w_k| \right| + \left| \frac{\varepsilon_k |\nabla u_k|^2}{2} - \frac{W(u_k)}{\varepsilon_k} \right| \leq 2|\xi_k| \end{aligned}$$

which again follows from inequality (4.9). Thus, we have found that $\delta V = 0$ which is clearly locally bounded and implies that V is stationary. Finally, using the lower bound in Proposition 4.12 and the fact that $\mu_\infty = 2\|V\|$ we have

$$\Theta^{*,n-1}(\|V\|, x) = \limsup_{r \to 0} \frac{\mu_\infty(B_r(x))}{\omega_{n-1} r^{n-1}} \geq \frac{D_1}{\omega_{n-1}} > 0$$

for $x \in \operatorname{spt} \mu_\infty$. Therefore V is rectifiable in the underlying set $\operatorname{spt}\|V\| = \operatorname{spt} \mu_\infty$. $\square$

We finish this section by proving point (iii) of Theorem 4.3. We recall that the Hausdorff convergence of a sequence of compact sets K_j to a compact set K is defined by the condition that for every $\delta > 0$, there exists $N \in \mathbb{N}$ such that for all $j \geq N$,

$$K_j \subset \{x \in \mathbb{R}^n : d(x, K) \leq \delta\} \quad \text{and} \quad K \subset \{x \in \mathbb{R}^n : d(x, K_j) \leq \delta\}.$$

Hence, what the Hausdorff convergence $\{|u_k| \leq 1 - b\} \cap \Omega' \to \operatorname{spt}\|V\| \cap \Omega'$ says is that the sequence of regions with high energy thin out uniformly towards the support of V as $k \to \infty$.

Proposition 4.18 *For each open subset $\Omega' \Subset \Omega$ and $0 < b < 1$, $\{|u_k| \leq 1 - b\} \cap \overline{\Omega'} \to \operatorname{spt}\|V\| \cap \overline{\Omega'}$ in the Hausdorff distance.*

Proof For ease of notation let $X_k = \{|u_k| \leq 1 - b\} \cap \overline{\Omega'}$ and $Y = \operatorname{spt}\|V\| \cap \overline{\Omega'}$. Assume for contradiction that the statement is not true, then there are 2 possible cases:

Case 1: There exists a (possibly sub-) sequence $x_k \in X_k$ and $d > 0$ such that $d(x_k, Y) \geq d$. Due to compactness we may assume that $x_k \to x_\infty$ for some $x_\infty \in \overline{\Omega'}$. Then $B_{d/2}(x_\infty) \cap Y = \emptyset$ and by the observation in (4.8) and $\operatorname{spt}\|V\| = \operatorname{spt} \mu_\infty$ it must be that either $u_k(x_k)$ converges to either 1 or -1. However this is a contradiction since $|u_k(x_k)| \leq 1 - b$.

Case 2: There exists $y \in Y$ and $d > 0$ such that $d(y, X_k) > d$, up to possibly a subsequence. In particular, $B_{d/2}(y) \cap X_k = \emptyset$ hence $|u_k(y)| > 1 - b$ for all k. A contradiction is then obtained with the same argument as in the proof of Lemma 4.13. $\square$

4.4 Integrality

This section covers the final part of [HT00], in which the integrality of the limiting varifold is established. The presentation here is structured to highlight essential steps and ideas, with some technical results omitted to preserve the flow of the proof. Additionally, the organization differs from the original text in that tools are introduced as they are needed, rather than being proved all at once at the beginning.

We begin by observing that the monotonicity formula in Proposition 4.8 implies that $E_{\varepsilon_k}(R, x) + c_3 R \geq E_{\varepsilon_k}(r, x)$ for $x \in \Omega$ and $0 < r < R$ small enough. Given that $\mu_k \to \mu_\infty$ we then have that the same monotonicity holds for the ratios of the energy measure in the limit

$$\frac{\mu_\infty(B(R, x))}{R^{n-1}} + c_3 R \geq \frac{\mu_\infty(B(r, x))}{r^{n-1}}.$$

From the fact that $\mu_\infty = 2\|V\|$, the previous inequality implies that the density of V is well defined[4] for any $x \in \Omega$ and

$$\theta(x) = \lim_{r\to 0} \frac{\|V\|(B_r(x))}{\omega_{n-1} r^{n-1}}.$$

Moreover, thanks to Proposition 4.12, it satisfies $\frac{D_1}{2\omega_{n-1}} \le \theta \le \frac{D_2}{2\omega_{n-1}}$ on $\operatorname{spt}\|V\|$. Therefore, to show that $\sigma^{-1}V$ is integral we have to show that θ/σ takes on integer values a.e. on $\operatorname{spt}\|V\|$. Precisely, we show that $\omega_{n-1}\theta(x) \le \omega_{n-1}(N(x)-1)\sigma + s$ for any $s > 0$, where $N(x)$ is the smallest integer larger than $\theta(x)/\sigma$. This will be done by analyzing a blow-up procedure in which we rely on the fact that blow-ups around a point x in the support of V admitting an approximate tangent plane T_xV will converge to the varifold $\theta(x)v(T_xV)$.

Fix $x \in \operatorname{spt}\|V\|$ admitting an approximate tangent plane T_xV, which is the case for $\mathcal{H}^{n-1} \llcorner \operatorname{spt}\|V\|$ a.e. due to rectifiability. Without loss of generality we assume that $x = 0$ and $T_xV = \pi_n := \{x_n = 0\}$. For a sequence of scales $r_k \to 0$ we have that $(\eta_{r_k})_\# V \to \theta v(\pi_n)$ as varifolds with θ being the density at the origin and $\eta_r(x) := x/r$. Via a diagonal argument we can further assume (up to a subsequence) that $(\eta_{r_k})_\# V_k \to \theta v(\pi_n)$ and $\varepsilon_k/r_k \to 0$ given that $V_k \to V$ and $\varepsilon_k \to 0$. Then, defining $\hat{u}_k(x) := u_k(r_k x)$ and $\hat{\varepsilon}_k := \varepsilon_k/r_k$ we have that

$$\hat{\varepsilon}_k \Delta \hat{u}_k = \hat{\varepsilon}_k^{-1} W'(\hat{u}_k) \quad \text{and } \hat{\varepsilon}_k \overset{k\to\infty}{\longrightarrow} 0,$$

and the blow-up sequence satisfies the assumptions of (**B**)[5] on any fixed Lipschitz bounded domain containing the origin (for k large enough); for our purposes B_3 will suffice. We also define $\hat{w}_k = \Phi \circ \hat{u}_k$ and the discrepancy function $\hat{\xi}_k$ as before and $\hat{V}_k$ the associated varifold which coincides with $(\eta_{r_k})_\# V$ and has limit $\hat{V}$. In particular, we have that $\|\hat{V}_k\| = |\nabla \hat{w}_k| \, d^n x \to \theta \|v(\pi_n)\|$ with $\|v(\pi_n)\| = \mathcal{H}^{n-1} \llcorner \pi_n$. It is easy to check that $\hat{\xi}_k(x) = r_k \xi(r_k x)$ so by Proposition 4.7

$$\sup_{B_3} \left(\frac{\hat{\varepsilon}_k |\nabla \hat{u}_k|^2}{2} - \frac{W(\hat{u}_k)}{\hat{\varepsilon}_k} \right) \le \mathcal{O}(r_k) \overset{k\to\infty}{\longrightarrow} 0. \tag{4.11}$$

Moreover, given that the sequence of rescaled critical points satisfies the same assumptions, Proposition 4.14 gives

$$\lim_{k\to\infty} \int_{B_3} \left| \frac{\hat{\varepsilon}_k |\nabla \hat{u}_k|^2}{2} - \frac{W(\hat{u}_k)}{\hat{\varepsilon}_k} \right| = 0, \tag{4.12}$$

[4] Indeed, first take $\limsup_{r\to 0}$ at both sides of the inequality and then $\liminf_{R\to 0}$ to find that $\Theta_*^{n-1} \ge \Theta^{*,n-1}$.

[5] The uniform bound on $|\hat{u}_k|$ clearly holds for the same value of C_0 and the uniform bound on the energy is a direct consequence of the monotonicity formula (4.7).

and due to Theorem 4.17 both $\frac{\hat{\varepsilon}_k}{2}|\nabla \hat{u}_k|^2 \,\mathrm{d}^n x$ and $\hat{\varepsilon}_k^{-1} W(\hat{u}_k)\,\mathrm{d}^n x$ also converge to $\mathcal{H}^{n-1} \llcorner \pi_n$.

With all of this, we can write

$$\omega_{n-1}\theta = \lim_{k\to\infty} \int_{B_1} |\nabla \hat{w}_k|,$$

which is the integral we are going to bound to show that $\theta/\sigma = N$. For this we are going to have to split the integral, first into a low energy part $\{|u| \geq 1-b\}$ and a high energy part $\{|u| \leq 1-b\}$ and then split the high energy part into a good and a bad set, the latter of which we will see has vanishing measure in the limit.

To deal with the low energy part we use the following result which says that the potential energy term goes to 0 as $k \to \infty$ around regions of low energy.

Proposition 4.19 *Let $s > 0$ and $\{u_k\}_k$ a sequence satisfying* (**B**)*, then there exists $b > 0$ and $k_0 \in \mathbb{N}$ depending only on c_0, E_0, W and s such that*

$$\int_{B_1\cap\{|u_k|\geq 1-b\}} \frac{W(u_k)}{\varepsilon_k} \leq s$$

whenever $k \geq k_0$.

Proof See Proposition 5.1 and Lemmas 5.2 and 5.3 in [HT00]. □

In combination with (4.11), the previous proposition implies that for any $s > 0$ we can choose $b > 0$ such that

$$\int_{B_3\cap\{|\hat{u}_k|\geq 1-b\}} \left(\frac{\hat{\varepsilon}_k|\nabla \hat{u}_k|^2}{2} + \frac{W(\hat{u}_k)}{\hat{\varepsilon}_k} \right) \leq s, \tag{4.13}$$

which takes care of the low energy part.

Now, to study the high energy term we will need to control one more quantity in addition to the energy and discrepancy, namely the *tilt excess*. Define $\boldsymbol{\nu}^u = (\nu_1, \dots, \nu_n) := \nabla u/|\nabla u|$ whenever $|\nabla u| \neq 0$ and 0 otherwise. Let $\boldsymbol{\nu}^k = \boldsymbol{\nu}^{\hat{u}_k}$, then

$$\lim_{k\to\infty} \int_{B_3} (1-(\nu_n^k)^2) \frac{\hat{\varepsilon}_k|\nabla \hat{u}_k|^2}{2} = \lim_{k\to\infty} \int_{B_3} (1-(\nu_n^k)^2)|\nabla \hat{w}_k| = 0, \tag{4.14}$$

where the first equality comes from equipartition of energy for the rescaled sequence and the second is a consequence of $\hat{V}_k \to \theta\mathcal{H}^{n-1} \llcorner \pi_n$. In other words, we have that thanks to the varifold convergence towards the plane π_n the tilt excess vanishes in the limit. We also define the maps $T : \mathbb{R}^n \to \mathbb{R}^{n-1}$ the orthonormal projection onto $\pi_n \cong \mathbb{R}^{n-1}$ and $T^\perp : \mathbb{R}^n \to \mathbb{R}$ the orthonormal projection onto $\pi_n^\perp \cong \mathbb{R}$.

To ease notation, in the following propositions we drop all indices and hats and we also define

$$e_\varepsilon = \frac{\varepsilon|\nabla u|^2}{2} + \frac{W(u)}{\varepsilon} \quad \text{and} \quad \xi_\varepsilon = \frac{\varepsilon|\nabla u|^2}{2} - \frac{W(u)}{\varepsilon}. \tag{4.15}$$

The following two propositions will allow us to treat the good and bad sets mentioned previously and obtain our desired bound. The first one says that we can control the energy ratios around a set of points of high energy confined to $\pi_n^\perp$ by the global energy ratio as long as the tilt excess and gradient energy scale down faster than r^{n-1}.

Proposition 4.20 *For each R, E_0, s and N such that $0 < R < \infty$, $0 < E_0 < \infty$, $0 < s < 1$ and $N \in \mathbb{N}$, there exists $\eta > 0$ with the following property:*

Assume that (see Fig. 4.2):

(i) $Y \subset \mathbb{R}^n$ has no more than $N+1$ elements, $T(y) = 0$ for all $y \in Y$, $a > 0$, $|y - z| > 3a$ for all $y, z \in Y$ and $\operatorname{diam}(Y) \leq \eta R$.

(ii) On $\{x \in \mathbb{R}^n \mid d(x, Y) < R\}$, u satisfies the Allen–Cahn equation $(\mathrm{AC}_\varepsilon)$ with $|u| \leq 2$ and $\xi_\varepsilon \leq \eta$.

(iii) For each $y \in Y$ and $a \leq r \leq R$

$$\int_{B_r(y)} |\xi_\varepsilon| + (1 - \nu_n^2)\varepsilon|\nabla u|^2 \leq \eta r^{n-1} \quad \textit{and} \quad \int_{B_r(y)} \varepsilon|\nabla u|^2 \leq \eta E_0 r^{n-1}.$$

Then

$$\sum_{y \in Y} \frac{1}{a^{n-1}} \int_{B_a(y)} e_\varepsilon \leq s + \frac{1+s}{R^{n-1}} \int_{d(x,Y)<R} e_\varepsilon.$$

Proof See Lemma 5.4 and Proposition 5.5 in [HT00]. $\square$

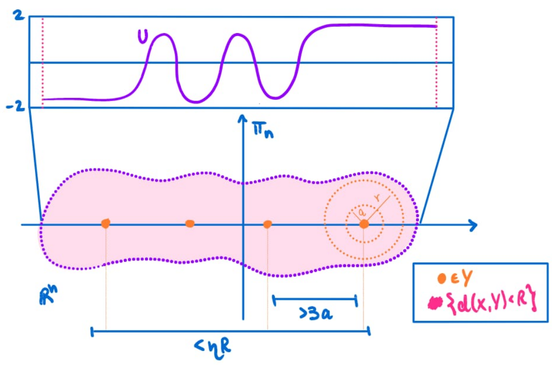

Figure 4.2: Sketch of the assumptions in Proposition 4.20

The second proposition says that the smallness of the discrepancy measure and tilt excess imply that the solution is close to the one dimensional solution interpolating between -1 and 1. In particular, it says that the value $u(0)$ is only achieved at $x = 0$ and the energy ratio is close to that of the 1D solution, on small enough balls.

Proposition 4.21 *Given $0 < s < 1$ and $0 < b < 1$, there exist $0 < \eta < 1$ and $1 < L < \infty$ with the following property:*

Assume $0 < \varepsilon < 1$ and u satisfies the Allen–Cahn equation $(\mathrm{AC}_\varepsilon)$ and $\xi_\varepsilon \le \eta$ on $B_{4\varepsilon L}$ with $|u(0)| \le 1 - b$ and

$$\int_{B_{4\varepsilon L}} |\xi_\varepsilon| + (1 - \nu_n^2)\varepsilon|\nabla u|^2 \le \eta(4\varepsilon L)^{n-1}.$$

Then, we have $T^{-1}(0) \cap \{x \in B_{3L\varepsilon} \mid u(x) = u(0)\} = \{0\}$ and

$$\left|\frac{1}{\omega_{n-1}(L\varepsilon)^{n-1}} \int_{B_{L\varepsilon}} e_\varepsilon - 2\sigma\right| \le s. \tag{4.16}$$

Proof See Proposition 5.6 in [HT00]. □

Now, with the choices of s and b given by Proposition 4.19 to have the bound in (4.13) and $R = 1$, we choose η and L via the previous two propositions (taking the smaller η of the two). Then, for large enough k we define the good sets as those having high energy with a controlled discrepancy and tilt excess on B_2

$$G_k := B_2 \cap \{|\hat{u}_k| \le 1 - b\} \cap$$
$$\left\{x \;\middle|\; \int_{B_r(x)} \left|\frac{\hat{\varepsilon}_k|\nabla\hat{u}_k|^2}{2} - \frac{W(\hat{u}_k)}{\hat{\varepsilon}_k}\right| + (1 - (\nu_n^k)^2)\hat{\varepsilon}_k|\nabla\hat{u}_k|^2 \le \eta r^{n-1} \text{ if } r \in [4\hat{\varepsilon}_k L, 1]\right\}.$$

For our purposes, we need to bound both the $\hat{V}_k$-measure of the bad set and the $\mathcal{L}^{n-1}$-measure of its pushforward under T and to do this we will use Besicovitch's Covering Theorem (see Theorem 1.27 in [EG15]). In the first case, we have by construction that for any element in the complement $x \in B_2 \cap \{|\hat{u}_k| \le 1 - b\} \setminus G_k$ there is $r_x \in [4\hat{\varepsilon}_k L, 1]$ such that

$$\eta^{-1} \int_{B_{r_x}(x)} \left|\frac{\hat{\varepsilon}_k|\nabla\hat{u}_k|^2}{2} - \frac{W(\hat{u}_k)}{\hat{\varepsilon}_k}\right| + (1 - (\nu_n^k)^2)\hat{\varepsilon}_k|\nabla\hat{u}_k|^2 > r_x^{n-1}.$$

With this we associate to each x in the bad set the ball $B_{r_x}(x)$ and given the uniform bound $r_x \le 1$ we use the Besicovitch Covering Theorem to extract a countable subcover $\{B_j\}_{j \in J}$ such that no point is covered by more than some fixed finite number N_n of balls. Now, on each of these balls we have that

$$\|\hat{V}_k\|(B_j) \le c r_j^{n-1} \le c\eta^{-1} \int_{B_j} \left|\frac{\hat{\varepsilon}_k|\nabla\hat{u}_k|^2}{2} - \frac{W(\hat{u}_k)}{\hat{\varepsilon}_k}\right| + (1 - (\nu_n^k)^2)\hat{\varepsilon}_k|\nabla\hat{u}_k|^2$$

where the first inequality follows from the monotonicity formula in Proposition 4.8 and c depends on W, n and s. Since all the centers of the balls are contained in B_2 and the balls are of radius ≤ 1 we have that

$$\begin{aligned}
&\|\hat{V}_k\|(B_2 \cap \{|\hat{u}_k| \leq 1-b\} \setminus G_k) \leq \sum_{j\in J} \|\hat{V}_k\|(B_j) \\
&\leq N_n c\eta^{-1} \int_{B_3} \left| \frac{\hat{\varepsilon}_k |\nabla \hat{u}_k|^2}{2} - \frac{W(\hat{u}_k)}{\hat{\varepsilon}_k} \right| + (1-(\nu_n^k)^2)\hat{\varepsilon}_k |\nabla \hat{u}_k|^2 \xrightarrow{k\to\infty} 0
\end{aligned} \tag{4.17}$$

with the limit in the last term being a consequence of (4.12) and (4.14). Analogously, we obtain

$$\mathcal{L}^{n-1}(T(B_2 \cap \{|\hat{u}_k| \leq 1-b\} \setminus G_k)) \xrightarrow{k\to\infty} 0. \tag{4.18}$$

At this point we already have all the ingredients required to obtain the integrality.

Proof (of Integrality) First note that

$$\lim_{k\to\infty} \int_{B_1} |\nabla \hat{w}_k| \leq \lim_{k\to\infty} \int_{B_1 \cap \{|\hat{u}_k \leq 1-b|\} \cap G_k} |\nabla \hat{u}_k| \sqrt{W(\hat{u}_k)/2} + s$$

which follows from the parts of B_1 which have low energy or are part of the bad set vanishing in the limit thanks to (4.13) and (4.17), respectively. The Cauchy-Schwarz inequality and the expression for the gradient of w in terms of u and W have also been used.

Now, for $x \in B_1^{\pi_n} := B_1(0) \cap \pi_n$ and $|t| \leq 1-b$ define $Y := T^{-1}(x) \cap G_k \cap \{\hat{u}_k = t\}$. If we choose $a = L\hat{\varepsilon}_k$, Proposition 4.21 implies that the elements of Y are separated by at least $3L\hat{\varepsilon}_k$. We claim that Proposition 4.20 implies that Y has no more than $N-1$ elements for any $x \in B_1^{\pi_n}$. Assuming this, we obtain our desired conclusion

$$\begin{aligned}
\omega_{n-1}\theta &\leq \lim_{k\to\infty} \int_{B_1 \cap \{|\hat{u}_k \leq 1-b|\} \cap G_k} |\nabla \hat{u}_k| \sqrt{W(\hat{u}_k)/2} + s \\
&\leq \lim_{k\to\infty} \int_{-1+b}^{1-b} \|T_\#(v(\{\hat{u}_k\} \cap G_k))\|(B_1^{\pi_n}) \sqrt{W(t)/2}\, \mathrm{d}t + s \\
&\leq \omega_{n-1}(N-1) \int_{-1+b}^{1-b} \sqrt{W(t)/2}\, \mathrm{d}t + s \\
&\leq \omega_{n-1}(N-1)\sigma + s
\end{aligned}$$

where the second inequality follows from the co-area formula and $\|T_\# \hat{V}_k\| \to \|\hat{V}\|$ combined with (4.18).

Thus, to finish the proof we only need to show that Y does indeed have at most $N-1$ elements. Indeed, for large enough k

$$\sup_{x\in B_1^{\pi_n}} \frac{1}{\omega_{n-1}} \int_{B_1(x)} \left(\frac{\hat{\varepsilon}_k |\nabla \hat{u}_k|^2}{2} + \frac{W(\hat{u}_k)}{\hat{\varepsilon}_k} \right) \leq 2\theta + s,$$

by definition of θ. This means that the RHS in the estimate of Proposition 4.20 is smaller than $s + (1+s)(2\theta + s)$, given that $R = 1$. Also, by the estimate (4.16) in Proposition 4.21, the LHS in the estimate of Proposition 4.20 is greater than $2\sigma N - Ns$ if we assume that Y contains at least N elements . All in all,

$$2\sigma N \leq (N+1)s + (1+s)(2\theta + s)$$

if Y has at least N elements, which is a contradiction with $\theta/\sigma < N$ when choosing s small enough. $\square$

We conclude this section by pointing out some interesting facts which have not been covered. First of all, it can be showed that the density $\theta(x) = \sigma N(x)$ satisfies

$$N(x) = \begin{cases} \text{odd } \mathcal{H}^{n-1} \text{ a.e. x} \in M^\infty, \\ \text{even } \mathcal{H}^{n-1} \text{ a.e. x} \in \operatorname{spt}\|V\| \setminus M^\infty, \end{cases}$$

where M^∞ is the reduced boundary of $\{u^\infty = 1\}$. Although it can be assumed that $M^\infty \subset \operatorname{spt}\|\partial\{u^\infty = 1\}\|$, at this point it is not known if $\mathcal{H}^{n-1}(\operatorname{spt}\|\partial\{u^\infty = 1\}\| \setminus M^\infty) = 0$. If this were the case then an even amount of folds is excluded a.e. which correspond to a spike in the limit. Indeed, integer multiplicities larger than 1 may arise from the two different scenarios depicted in Fig. 4.3, the second of which seems less likely.

Secondly, in the constrained case the limiting varifold is no longer stationary but is still rectifiable, integral and has generalized mean curvature

$$\boldsymbol{H}(x) = \begin{cases} \frac{\lambda_\infty}{\theta(x)}\boldsymbol{\nu}^\infty(x) & \mathcal{H}^{n-1} \text{ a.e. } x \in M^\infty, \\ 0 & \mathcal{H}^{n-1} \text{ a.e. } x \in \operatorname{spt}\|V\|M^\infty, \end{cases}$$

where λ_∞ is the limit of the Lagrange multipliers and $\boldsymbol{\nu}^\infty$ the inward normal of M^∞.

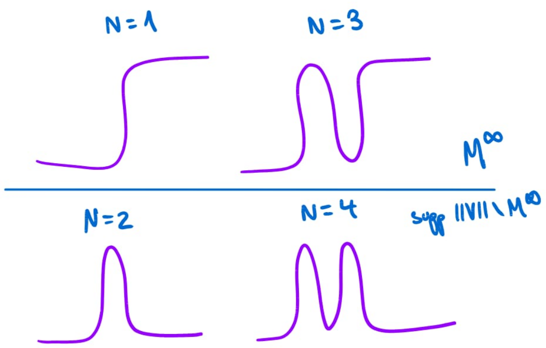

Figure 4.3: Integer multiplicities can occur from an even or odd amount of folding of the level sets onto themselves.

Chapter 5

Stable Critical Points & 2D Regularity

In this chapter we focus on the case of a stable critical points of E_ε, studied by Tonegawa in the paper [Ton05]. Precisely, we assume the following:

The sequence $\{u_k\}_{k\geq 1}$ *in* (**B**) *is stable, that is*

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\bigg|_{t=0} E_{\varepsilon_k}(u_k + t\phi; \Omega) = \int_\Omega \varepsilon_k |\nabla\phi|^2 + \frac{W''(u_k)}{\varepsilon_k}\phi^2 \geq 0, \ \forall k \geq 1. \tag{C}$$

With this additional assumption we then show that the limiting stationary varifold obtained in the previous chapter is also stable. For this, we first look at what it means for a varifold to be stable, in particular we introduce the notion of generalized second fundamental form of a varifold, first developed by Hutchinson in [Hut86]. We then prove that the limit of a stable sequence has a generalized second fundamental form satisfying the varifold analog of the stability inequality for hypersurfaces. Finally, we use the structure theory of one dimensional stationary varifolds [AA76] to prove the regularity of limit interfaces when $n = 2$, which tells us that $\operatorname{spt} V$ consists of disjoint straight line segments.

5.1 The Generalized Second Fundamental Form

As mentioned at the end of Section 2.2, obtaining an explicit expression for the second variation of a varifold is not as straightforward as it is for the first. This is because we expect the formula to involve the derivatives of an orthonormal frame of the tangent space $T_x V$; however, this does not necessarily make sense for general varifolds. In this section, we present the approach of Hutchinson [Hut86] for defining varifolds with a generalized second fundamental form, which will allow us to characterize stable varifolds.

To motivate the definition of the generalized second fundamental form we first consider $M \subset \mathbb{R}^n$ a smooth submanifold. For $x \in M$, let $[(S_x)_{ij}]_{1\leq i,j\leq n}$

be the orthogonal projection matrix onto T_xM. We recall that the second fundamental form of M at x is the symmetric bilinear map defined by

$$\begin{aligned} \boldsymbol{B} : TM \times TM &\to NM \\ (X, Y) &\mapsto (\partial_{\widetilde{X}} \widetilde{Y})^{\perp}. \end{aligned}$$

The domain of $\boldsymbol{B}$ can be extended to the whole of $\mathbb{R}^n \times \mathbb{R}^n$ by first projecting $X, Y \in T\mathbb{R}^n$ onto TM, that is $\boldsymbol{B}(X, Y) = \boldsymbol{B}(S_xX, S_xY)$. Let $e_1, \ldots, , e_n$ denote the standard orthonormal basis of $\mathbb{R}^n$, we define the components of $\boldsymbol{B}$ as

$$B_{ij}^k := \langle \boldsymbol{B}(e_i, e_j), e_k \rangle.$$

For example, the symmetry of $\boldsymbol{B}$ is expressed as $B_{ij}^k = B_{ji}^k$ in components. Using the Einstein summation convention for repeated indices, the components of the mean curvature are given by $H^k = B_{ii}^k$ and thus $\boldsymbol{H} = B_{ii}^k e_k$. Moreover, if we define

$$\delta_i \phi(x) := S_{ij}(x) \partial_j \phi(x)$$

for $\phi \in \mathcal{C}_c^1(M)$ (strictly speaking, ϕ has to be extended smoothly in a neighborhood of x), such that $e_i \delta_i$ corresponds to the tangential gradient operator, we have the following.

Proposition 5.1 *At each $x \in M$ the following hold:*

(i) $B_{ij}^k = S_{lj} \delta_i S_{kl}$

(ii) $\delta_i S_{jk} = B_{ij}^k + B_{ik}^j$

(iii) $H^i = \delta_j S_{ij}$

The previous identities follow from short computations using the definition of B_{ij}^k, S_{ij} and δ_i. The main takeaway is that the first 2 imply that we can recover B_{ij}^k from $\delta_i S_{jk}$ and vice versa, thus we will focus on the latter to define a weak notion of curvature. This also emphasizes the precise issue with defining the generalized second fundamental form as the $\delta_i S_{jk}$ are exactly the derivatives of the tangent plane T_xM when we identify it as its own projection.

Now, let $\Omega \subset \mathbb{R}^n$ open such that $\partial M \cap \Omega = \emptyset$ and $\phi \in \mathcal{C}^1(\Omega \times \mathbb{R}^{n^2})$ a test function such that ϕ is compactly supported in Ω w.r.t. the x variables. Denote the derivatives with respect to x_i and S_{ij} by $\partial_i \phi$ and $\partial_{ij}^* \phi$, respectively. If we let $X(x) = \phi(x, S_x) e_i$, then an application of the divergence theorem gives

$$0 = \int_M \operatorname{div}_M(X^\top) = \int_M \delta_r(S_{ir}\phi) = \int_M S_{ij}\partial_j \phi + (\delta_i S_{jk}) \partial_{jk}^* \phi + (\delta_j S_{ij}) \phi,$$

where $X^\top$ denotes the tangential component of X in T_xM. Motivated by the pervious identity we make the following definition.

Definition 5.2 Let $\Omega \subset \mathbb{R}^n$ open and $V \in \boldsymbol{IV}_k(\mathbb{R}^n)$. Then V has generalized curvature in Ω if there exist real-valued functions A_{ijk} for $1 \leq i,j,k \leq n$ defined $V \llcorner \Omega$-a.e. in Ω, such that

(i) $(V \llcorner \Omega, A_{ijk})$ is a measure-function pair over $G_k(\Omega)$ for $1 \leq i,j,k \leq n$,

(ii) for all $\phi \in \mathcal{C}^1(\Omega \times \mathbb{R}^{n^2})$ with compact support in the x variables there holds
$$\int_{G_k(\Omega)} P_{ij}\partial_j\phi + A_{ijk}\partial^*_{jk}\phi + A_{jij}\phi \, \mathrm{d}V(x,P) = 0. \tag{5.1}$$

We say V is a curvature varifold in Ω and write $V \in \boldsymbol{CV}_k(\mathbb{R}^n, \Omega)$.

We note that we have not yet defined the notion of measure-function pair which appears in the first item, this is a technical point which we postpone until the end of this section.

Proposition 5.3 *Let $V \in \boldsymbol{CV}_k(\mathbb{R}^n, \Omega)$, then any two version of the generalized curvature of V are equal $V \llcorner \Omega$ almost everywhere.*

Proof See Proposition 5.2.2. in [Hut86]. □

From the previous result we see that the curvature is unique whenever it exists, combining this with the first item in Proposition 5.1 we can define the generalized second fundamental form.

Definition 5.4 Let $V \in \boldsymbol{CV}_k(\mathbb{R}^n, \Omega)$, the generalized second fundamental form $\boldsymbol{B} : G_k(\Omega) \to \mathbb{R}^3$ of V is defined V-a.e. by
$$B^k_{ij}(x,P) = S_{lj}A_{ikl}(x,P)$$
for $1 \leq i,j,k \leq n$. We also write $|\boldsymbol{B}|^2 = \sum_{i,j,k=1}^n (B^k_{ij})^2$.

Remark 5.5 *It is not a consequence of the definition that the generalized second fundamental form is symmetric.*

The following covers the notion of measure-function pair used in the definition of generalized curvature and a compactness result which will be useful later on.

Definition 5.6 Suppose μ is a Radon measure on Ω and $f : \Omega \to \mathbb{R}^\alpha$ is locally integrable with respect to μ. Then the pair (μ, f) is called a *measure-function pair* over Ω.

Definition 5.7 Suppose $\{(\mu_k, f_k)\}_{k\geq 1}$ and (μ, f) are measure-function pairs over Ω such that $\mu_k \to \mu$ as Radon measures in Ω. We say (μ_k, f_k) *converges weakly* to (μ, f) if
$$\lim_{k\to\infty} \int_\Omega \langle f_k, \phi \rangle \, \mathrm{d}\mu_k = \int_\Omega \langle f, \phi \rangle \, \mathrm{d}\mu$$
for all $\phi \in \mathcal{C}_c(\Omega, \mathbb{R}^\alpha)$.

Theorem 5.8 *If $\{(\mu_k, f_k)\}_{k\geq 1}$ is a sequence of measure-function pairs over Ω such that $\liminf \mu_k(\Omega) < \infty$ and $\liminf \int_\Omega |f_k|^2 \, \mathrm{d}\mu_k < \infty$, then there exists a subsequence converging weakly to a measure-function pair (μ, f). Moreover,*

$$\int_\Omega |f|^2 \, \mathrm{d}\mu \leq \liminf_{k\to\infty} \int_\Omega |f_k|^2 \, \mathrm{d}\mu_k.$$

Proof See Theorem 4.4.2. in [Hut86] and take $F(y, q) = |q|^2$. □

5.2 Stability in the Limit

In this section we show that the limiting varifold obtained from a sequence of stable critical points has a generalized second fundamental form and is stable in the following sense.

Theorem 5.9 (Stability) *Assume hypothesis* (**A**), (**B**) *and* (**C**) *hold, then the limit varifold V given by Theorem 4.3 has generalized second fundamental form $\boldsymbol{B}$ and*

$$\int_{G_{n-1}(\Omega)} |\boldsymbol{B}|^2 \phi^2 \, \mathrm{d}V \leq \int_\Omega |\nabla \phi|^2 \, \mathrm{d}\|V\| \tag{5.2}$$

for all $\phi \in C_c^1(\Omega)$. Moreover, we have

$$\sum_{j=0} B_{jj}^k = 0 \text{ and } B_{ij}^k = B_{ji}^k$$

for $1 \leq i, j, k \leq n$ and V-a.e. $(x, P) \in G_{n-1}(\Omega)$.

Indeed, inequality (5.2) corresponds to the stability inequality of a minimal hypersurface (2.4) in Euclidean space. This correspondence is made even more explicit using the fact that from Theorem 4.3 we know V is rectifiable and therefore $\boldsymbol{B}$ can be regarded as a function on Ω by considering $\boldsymbol{B}(x) = \boldsymbol{B}(x, T_x M)$ such that

$$\int_\Omega |\boldsymbol{B}|^2 \phi^2 \, \mathrm{d}\|V\| \leq \int_\Omega |\nabla \phi|^2 \, \mathrm{d}\|V\|.$$

Remark 5.10 *We note that the RHS of the previous inequality incorporates the full gradient instead of the tangential gradient (unlike* (2.4)*). This is not an issue as we will later see but it is good to keep in mind.*

The proof of Theorem 5.9 is straightforward. First a stability inequality is established for the u_k's and the components of their second fundamental form is defined. After, it is shown that the components satisfy (5.1) except for a term which will vanish in the limit. Then the generalized second fundamental form of the limit interface is obtained using the compactness result for measure-function pairs in Theorem 5.8.

Before proceeding to the main argument, we slightly redefine the sequence of varifolds V_k to simplify our computations. Redefine

$$V_k(\phi) := \frac{1}{\sigma} \int_{\Omega \cap |\nabla uk| > 0} \phi(x, I - \boldsymbol{\nu_k} \otimes \boldsymbol{\nu_k}) \frac{\varepsilon_k}{2} |\nabla u_k|^2 \, \mathrm{d}^n x. \tag{5.3}$$

We claim that this sequence is equivalent to the previous one. This equivalence is a consequence of the equipartition of energy, which allows us to replace the measure $\frac{\varepsilon_k}{2}|\nabla u_k|^2 \, \mathrm{d}^n x$ with $|\nabla w_k| \, \mathrm{d}^n x$ while preserving the limit. Applying the coarea formula then yields

$$\begin{aligned} &\int_{\Omega \cap |\nabla uk| > 0} \phi(x, I - \nu_k(x) \otimes \nu_k(x)) |\nabla w_k| \, \mathrm{d}^n x \\ &\qquad = \int_{\mathbb{R}} \left(\int_{\Omega \cap |\nabla u_k| > 0 \cap w_k = t} \phi(x, T_x w_k = t) \, \mathrm{d}\mathcal{H}^{n-1} \right) \mathrm{d}t, \end{aligned}$$

where we used $\nabla u_k / |\nabla u_k| = \nabla w_k / |\nabla w_k|$ to identify $I - \nu_k(x) \otimes \nu_k(x) = T_x\{w_k = t\}$. By Riesz duality, this expression coincides with the original definition of V_k given in Section 4.1. Therefore, the redefined varifold sequence shares the same limit while providing a more manageable expression for computations and absorbing the σ factor so that the limit varifold is integral.

Using the redefined sequence we begin the proof of the stability in the limit by obtaining a prelimit analog of the stability inequality. In the following we drop the indices indicating the subsequence and instead just write ε.

Lemma 5.11 *For $\phi \in C_c^1(\Omega)$ and u_ε a stable critical point of the Allen–Cahn equation*

$$\int\limits_{\Omega \cap \{|\nabla u_\varepsilon| > 0\}} \left[\sum_{i,j=1}^n (\partial_i \partial_j u_\varepsilon)^2 - \frac{1}{|\nabla u_\varepsilon|^2} \sum_{j=1}^n \left(\sum_{i=1}^n \partial_i u_\varepsilon \partial_i \partial_j u_\varepsilon \right)^2 \right] \phi^2 \le \int\limits_{\Omega} |\nabla \phi|^2 |\nabla u_\varepsilon|^2 \tag{5.4}$$

Proof Testing the stability condition (**C**) against $\phi|\nabla u_\varepsilon|$ we find

$$\begin{aligned} \int\limits_{\Omega \cap \{|\nabla u_\varepsilon| > 0\}} \varepsilon \left[\phi^2 \sum_j \frac{\left(\sum_i \partial_i u_\varepsilon \partial_i \partial_j u_\varepsilon\right)^2}{|\nabla u_\varepsilon|^2} + 2\phi \sum_{i,j} \partial_i u_\varepsilon \partial_j \phi \partial_i \partial_j u_\varepsilon + |\nabla u_\varepsilon|^2 |\nabla \phi|^2 \right] \\ + \frac{W''(u_\varepsilon)}{\varepsilon} |\nabla u_\varepsilon|^2 \phi^2 \ge 0. \end{aligned}$$

Now, differentiating the Allen–Cahn equation (AC_ε) w.r.t x_j and multiplying by $\partial_j u_\varepsilon \, \phi^2$ we find

$$\varepsilon \Delta(\partial_j u_\varepsilon) u_\varepsilon \phi^2 = \frac{W''(u_\varepsilon)}{\varepsilon} (\partial_j u_\varepsilon)^2 \phi^2.$$

Summing over j, integrating over $\Omega \cap \{|\nabla u_\varepsilon| > 0\}$ and applying integration by parts to the LHS

$$\int \frac{W''(u_\varepsilon)}{\varepsilon}|\nabla u_\varepsilon|^2\phi^2 = -\varepsilon \int \sum_{i,j}\left[\left(\partial_i\partial_j u_\varepsilon\right)^2\phi^2 + 2\phi\partial_i u_\varepsilon \partial_j\phi\partial_i\partial_j u_\varepsilon\right], \tag{5.5}$$

which can be substituted into the first inequality to finish the proof. □

Now, for $(x, P) \in G_{n-1}(\Omega)$ such that x is a regular point of u_ε, i.e. $|\nabla u_\varepsilon| > 0$, we define the tensors $\boldsymbol{A}^\varepsilon$ and $\boldsymbol{B}^\varepsilon$ with components

$$A^\varepsilon_{ijk}(x,P) := -\delta_i(\nu^\varepsilon_j\nu^\varepsilon_k) = -P_{il}\partial_l(\nu^\varepsilon_j\nu^\varepsilon_k) \quad \text{and} \quad (B^k_{ij})^\varepsilon(x,P) := P_{lj}A^\varepsilon_{ijk}(x,P) \tag{5.6}$$

where ν^ε_j are the components of the normal vector to the level sets $\nu_\varepsilon = \nabla u_\varepsilon/|\nabla u_\varepsilon|$. Note that if $S = I - \nu_\varepsilon \otimes \nu_\varepsilon$ then $B(x, S)$ is the second fundamental form of the hypersurfaces coming from level sets of u_ε. We also define

$$H^\varepsilon_i(x,P) = H^\varepsilon_i(x) := \frac{1}{\varepsilon|\nabla u_\varepsilon|^2}\,\partial_i\left(\frac{\varepsilon}{2}|\nabla u_\varepsilon|^2 - \frac{W(u)}{\varepsilon}\right)$$

and note that it is constructed from a derivative of the discrepancy function, so we can already expect that it will vanish in the limit (see Remark 4.4). The following result shows that these functions are the candidates for the pre-limit of the generalized second fundamental of V.

Lemma 5.12 *For $\phi \in \mathcal{C}^1(\Omega \times \mathbb{R}^{n^2})$ compactly supported in the $x \in \Omega$ variables, we have*

$$\int_{G_{n-1}(\Omega)} P_{ij}\partial_j\phi + A^\varepsilon_{ijk}\partial^*_{jk}\phi + H^\varepsilon_i\phi \,\mathrm{d}V_\varepsilon(x,P) = 0$$

for $i = 1, \ldots, n$.

Proof Fix i and let $\psi \in \mathcal{C}^1_c(\Omega)$. Multiplying the Allen–Cahn equation (AC_ε) by $\psi\partial_i u_\varepsilon$ and integrating by parts twice (as in the proof of 4.6) we obtain

$$\int_\Omega \frac{\varepsilon}{2}|\nabla u_\varepsilon|^2\partial_i\psi - \varepsilon\partial_i u_\varepsilon\sum_j \partial_j u_\varepsilon\partial_j + \frac{W(u_\varepsilon)}{\varepsilon}\partial_i\psi = 0.$$

Splitting the first term and integrating by parts the second term we find

$$\int_\Omega (\partial_i\psi - \nu^\varepsilon_i\sum_j \nu^\varepsilon_j\partial_j\psi)\varepsilon|\nabla u_\varepsilon|^2 + \partial_i\left(\frac{\varepsilon}{2}|\nabla u_\varepsilon|^2 - \frac{W(u)}{\varepsilon}\right)\psi = 0.$$

Now, let $\phi \in \mathcal{C}^1(\Omega \times \mathbb{R}^{n^2})$ compactly supported in the x variables and for $s > 0$ define $\phi^s(x) = \phi\left(x, I - \frac{\nabla u_\varepsilon \otimes \nabla u_\varepsilon}{s^2 + |\nabla u_\varepsilon|^2}\right) \in \mathcal{C}^1_c(\Omega)$. Using the chain rule

$$\partial_i\phi^s = \partial_i\phi + \sum_{l,k}\partial^*_{lk}\phi\,\partial_i\left[\left(I - \frac{\nabla u_\varepsilon \otimes \nabla u_\varepsilon}{s^2 + |\nabla u_\varepsilon|^2}\right)_{lk}\right]$$

and writing $\partial_i \phi = \sum_j \delta_{ij} \partial_j \phi$, we obtain by setting $\psi = \phi^s$ and letting $s \to 0$ that

$$\int_\Omega (I - \nu_\varepsilon \otimes \nu_\varepsilon)_{ij} (\partial_j \phi - \partial_j (\nu_l^\varepsilon \nu_k^\varepsilon) \partial_{lk}^* \phi) \varepsilon |\nabla u_\varepsilon|^2 + \partial_i \left(\frac{\varepsilon}{2} |\nabla u_\varepsilon|^2 - \frac{W(u)}{\varepsilon} \right) \phi = 0$$

in the Einstein summation convention. Given that

$$\phi(x, P) \, \mathrm{d}V_\varepsilon(x, P) = \sigma^{-1} \phi(x, I - \nu_\varepsilon \otimes \nu_\varepsilon) \varepsilon |\nabla u_\varepsilon|^2 / 2 \, \mathrm{d}x$$

and the definitions of A_{ijk}^ε and H_i^ε we conclude. □

The following result is the last ingredient towards the proof of the main theorem which will allow us to extract convergent subsequences. Before proving it, it is very useful to observe that the integrand in the LHS of (5.4) is invariant under rotations, this is due to the fact that it is a scalar function constructed from contracting tensors with the (Euclidean) metric. Thus, if for a fixed $x \in \Omega$ we choose an orthonormal basis $\bar{e}_1, \ldots, \bar{e}_n$ of (the tangent space of) $\mathbb{R}^n$ such that $\bar{e}_n = \nu_\varepsilon(x)$ we have

$$\begin{aligned} &\sum_{i,j=1}^n \left(\partial_i \partial_j u_\varepsilon\right)^2 - \frac{1}{|\nabla u_\varepsilon|^2} \sum_{j=1}^n \left(\sum_{i=1}^n \partial_i u_\varepsilon \partial_i \partial_j u_\varepsilon \right)^2 \\ &\qquad = \sum_{i,j=1}^n \left(\partial_i \partial_j u_\varepsilon\right)^2 - \sum_{j=1}^n \left(\partial_n \partial_j u_\varepsilon\right)^2 = \sum_{i,j=1}^{n-1} (\partial_i \partial_j u_\varepsilon)^2 + \sum_{i=1}^{n-1} (\partial_n \partial_i u_\varepsilon)^2. \end{aligned}$$

Given that $\boldsymbol{A}^\varepsilon$ is a tensor its norm is also invariant under rotations. Applying the product rule and the fact that $S = I - \nu_\varepsilon \otimes \nu_\varepsilon$ is diagonal in this basis with $S_{jj} = 1$ for $j \leq n-1$ and $S_{nn} = 0$ we find

$$\sum_{i,j,k=1}^n |A_{ijk}^\varepsilon|^2 = \frac{2}{|\nabla u_\varepsilon|^2} \sum_{i,j=1}^{n-1} (\partial_i \partial_j u_\varepsilon)^2.$$

Similar computations show that $2|\boldsymbol{B}^\varepsilon|^2 = |\boldsymbol{A}^\varepsilon|^2$.

Lemma 5.13 *For $\phi \in C_c^1(\Omega)$, we have*

$$\int_\Omega \sum_{i,j,k=1}^n |A_{ijk}^\varepsilon|^2 \phi^2 \, \mathrm{d}V_\varepsilon \leq 2 \int_\Omega |\nabla \phi|^2 \, \mathrm{d}\|V_\varepsilon\|,$$

$$\int_\Omega \sum_{i,j,k=1}^n |(B_{ij}^k)^\varepsilon|^2 \phi^2 \, \mathrm{d}V_\varepsilon \leq \int_\Omega |\nabla \phi|^2 \, \mathrm{d}\|V_\varepsilon\|, \tag{5.7}$$

$$\int_\Omega \sum_{i=1}^n |H_i^\varepsilon|^2 \phi^2 \, \mathrm{d}\|V_\varepsilon\| \leq (n-1) \int_\Omega |\nabla \phi|^2 \, \mathrm{d}\|V_\varepsilon\|.$$

Proof The first two inequalities are consequences of the previous observations, Lemma 5.11 and the definition of $\mathrm{d}V_\varepsilon$. The third inequality follows from

$$
\begin{aligned}
\varepsilon|\nabla u_\varepsilon|^2(H_{i,\varepsilon})^2 &= (\varepsilon|\nabla u_\varepsilon|^2)^{-1}\left(\varepsilon\partial_j u_\varepsilon\partial_i\partial_j u_\varepsilon - \frac{W'(u_\varepsilon)}{\varepsilon}\partial_i u_\varepsilon\right)^2 \\
&= \varepsilon|\nabla u_\varepsilon|^{-2}(\partial_j u_\varepsilon\partial_i\partial_j u_\varepsilon - \Delta u_\varepsilon\partial_i u_\varepsilon)^2 \\
&= \begin{cases} \varepsilon\left(\sum_{j=1}^{n-1}\partial_j^2 u_\varepsilon\right)^2 & i = n, \\ \varepsilon(\partial_i\partial_n u_\varepsilon)^2 & i \neq n, \end{cases}
\end{aligned}
$$

where the first equality follows from u_ε solving the Allen–Cahn equation. To conclude we apply Lemma 5.11 again and the Cauchy-Schwarz inequality such that

$$
\left(\sum_{j=1}^{n-1}\partial_j^2 u_\varepsilon\right)^2 \leq (n-1)\sum_{j=1}^{n-1}\left(\partial_j^2 u_\varepsilon\right)^2. \qquad \square
$$

Proof (of Theorem 5.9) The inequalities of Lemma 5.13 and the uniform energy bound on the sequence V_ε allows us to invoke Theorem 5.8 to obtain functions A_{ijk}, B_{ij}^k and H_i defined V-a.e. and a subsequence ε_l such that $(A_{ijk}^{\varepsilon_l}, V_{\varepsilon_l})$, $((B_{ij}^k)_l^\varepsilon, V_{\varepsilon_l})$ and $(H_i^{\varepsilon_l}, V_{\varepsilon_l})$ converge weakly to (A_{ijk}, V), (B_{ij}^k, V) and (H_i, V), respectively. Additionally, for $i = 1, \dots, n$ and $\phi \in C_c^1(\Omega \times \mathbb{R}^{n^2})$

$$
\int_{G_{n-1}(\Omega)} P_{ij}\partial_j\phi + A_{ijk}\partial_{jk}^*\phi + H_i\phi \,\mathrm{d}V(x,P) = 0 \tag{5.8}
$$

thanks to Lemma 5.12. If $\phi \in C_c^1(\Omega)$, then, given that $\phi(x,P)\,\mathrm{d}V_\varepsilon(x,P) = \sigma^{-1}\phi(x, I - \nu_\varepsilon\otimes\nu_\varepsilon)\varepsilon|\nabla u_\varepsilon|^2/2\,\mathrm{d}x$,

$$
\begin{aligned}
\int_{G_{n-1}(\Omega)} H_i^{\varepsilon_l}(x,P)\phi(x)\,\mathrm{d}V_{\varepsilon_l}(x,P) &= \int_\Omega \partial_i\left(\frac{\varepsilon_l}{2}|\nabla u_{\varepsilon_l}|^2 - \frac{W(u_{\varepsilon_l})}{\varepsilon_l}\right)\phi \\
&= -\int_\Omega\left(\frac{\varepsilon_l}{2}|\nabla u_{\varepsilon_l}|^2 - \frac{W(u_{\varepsilon_l})}{\varepsilon_l}\right)\partial_i\phi \xrightarrow{l\to\infty} 0
\end{aligned}
$$

with the vanishing in the limit coming from the equipartition of energy (Prop. 4.14). Now, testing equation (5.8) (without the H_i term) against $\sum_m \phi(x)P_{mi}e_m$ ($\{e_m\}$ are the canonical basis vectors) we find

$$
\begin{aligned}
0 &= \int_{G_{n-1}(\Omega)} P_{ij}P_{mi}e_m\partial_j\phi + A_{ijk}\phi e_m\partial_{jk}^*P_{mi}\,\mathrm{d}V(x,P) \\
&= \int_{G_{n-1}(\Omega)} P_{mj}e_m\partial_j\phi + A_{imi}e_m\phi\,\mathrm{d}V(x,P) = \int_{G_{n-1}(\Omega)} A_{imi}e_m\phi\,\mathrm{d}V(x,P)
\end{aligned}
$$

using in the first equality that $P^2 = P$ and $\partial_{jk}^*P_{mi} = \delta_{jm}\delta_{ki}$ and in the second equality that V is stationary (2.7). As ϕ is arbitrary and $\{e_m\}$ a basis, we find $A_{imi} = 0$ for each m so the A_{ijk}'s satisfy the defining property of the curvature components in (5.1) with

$$\int_{G_{n-1}(\Omega)} P_{ij}\partial_j\phi + A_{ijk}\partial^*_{jk}\phi \, \mathrm{d}V(x,P) = 0.$$

so V is a curvature varifold. We note that by definition of weak convergence it also holds that $B^k_{ij} = S_{lj}A_{ikl}$ so B^k_{ij} is the generalized second fundamental form of V. The stability inequality (5.2) follows from the second point in Theorem 5.8 and (5.7) such that

$$\begin{aligned}\int_{G_{n-1}(\Omega)} \phi^2 \sum_{i,j,k} |B^k_{ij}|^2 \, \mathrm{d}V &\le \liminf_{l\to\infty} \int_{G_{n-1}(\Omega)} \phi^2 \sum_{i,j,k} |(B^i_{jk})^{\varepsilon_l}|^2 \, \mathrm{d}V^{\varepsilon_l} \\ &\le \liminf_{l\to\infty} \int_\Omega |\nabla\phi|^2 \, \mathrm{d}\|V^{\varepsilon_l}\| = \int_\Omega |\nabla\phi|^2 \, \mathrm{d}\|V\|.\end{aligned}$$

Finally, the symmetry of B^k_{ij} follows from the symmetry of $(B^k_{ij})^\varepsilon$ and $B^k_{jj} = 0$ follows from testing against $P_{pm}P_{mi}\phi(x)$ and using that $A_{jij} = 0$ and $P^2 = P$.□

Remark 5.14 *It is also proved in [Ton05] that*

$$\int_\Omega |\boldsymbol{B}|^2\phi^2 \, \mathrm{d}\|V\| \le c \int_{G_{n-1}(\Omega)} |\nabla\phi|^2 \operatorname{tr}((I-P)\cdot T) \, \mathrm{d}(x,P)$$

for any $T \in G(n-1,n)$, $\phi \in C^1_c(\Omega)$ and a constant $c > 0$. This inequality is the direct varifold analog of the classical inequality of Schoen for minimal hypersurfaces [Sch77]

$$\int_M |\boldsymbol{B}|^2\phi^2 \le c \int_M (1-(\nu\cdot\nu_0)^2)|\nabla\phi|^2$$

used in the regularity theory of stable minimal hypersurfaces.

5.3 Regularity in 2D

In this part we prove that the support of the limiting varifold in dimension 2 consists of non-intersecting line segments and, in particular, is smooth. Additionally, this means that the regularity of the minimal hypersurfaces obtained from limit interfaces of Allen–Cahn is better than that of general stationary varifolds in 2D, as these admit singularities in the form of junctions. The proof relies on the knowledge of the structure of stationary varifolds in one dimension which is why it does not apply to the general case.

The proof is quite simple and only relies on two preliminary results. The first of the two says that regions of high energy have a uniform lower bound on the gradient. This result can be seen as the equivalent for the kinetic energy to Proposition 4.19, however it is a pointwise bound.

Lemma 5.15 *Given $0 < b < 1$ and $\Omega' \Subset \Omega$, there exists constants $c > 0$ and $\bar{\varepsilon}$ depending only on W and b such that*

$$\frac{c}{\varepsilon} \leq |\nabla u_\varepsilon(x)|$$

for all $x \in \{|u_\varepsilon| < 1 - b\} \cap \Omega'$ and $\varepsilon < \bar{\varepsilon}$.

Sketch of Proof The result follows from a contradiction argument. Assuming there exists a point x_0 such that the bound does not hold, one can show that the discrepancy function satisfies $\xi_\varepsilon \gtrsim \varepsilon^{-1}$ in a ball $B_{\varepsilon r}(x_0)$. Then it is shown using the Poincaré inequality that the L^2-norm of the discrepancy function on $B_{\varepsilon r}(x_0)$ is controlled from above by the L^2-norm on an appropriate ball $B_R(x_0)$ such that

$$\left(\int_{B_R(x_0)} |\xi_\varepsilon|^2\right)^{1/2} \geq \left(\int_{B_{\varepsilon r}(x_0)} |\xi_\varepsilon|^2\right)^{1/2} \geq \sqrt{\pi}(\varepsilon r)\frac{2c^2}{\varepsilon} = 2c^2 r\sqrt{\pi}$$

which is a contradiction as the LHS converges to 0. The fact that $n = 2$ is essential here as in the second inequality the balls scale with ε^n and only $n = 2$ will make the RHS a positive constant (instead of going to 0 if $n \geq 3$). For the full details see Lemma 2 in [Ton05]. □

The following result is technical and will allow us to ensure that at least one of the level sets of u_ε has curvature in L^2.

Lemma 5.16 *For $0 < b < 1$ and small enough ε there exists a constant C depending only on Ω, Ω', W and E_0 such that*

$$\int_{-1+b}^{1-b} \left(\int_{\{u_\varepsilon = t\}\cap\Omega'} \kappa_\varepsilon^2 \, \mathrm{d}\mathcal{H}^1\right) \mathrm{d}t \leq C$$

where κ_ε is the curvature of the level curve of u_ε.

Proof We begin by pointing out that $|\boldsymbol{B}^\varepsilon|^2 = \kappa_\varepsilon^2$ for $n = 2$. Now, by inequality (5.7) and choosing ϕ to be a smooth approximation of the indicator of Ω' we find

$$\int_{\Omega'} |\boldsymbol{B}^\varepsilon|^2 \, \mathrm{d}\|V_\varepsilon\| \leq C$$

with C depending on Ω, Ω' and E_0. Since $\mathrm{d}\|V_\varepsilon\| = \frac{\varepsilon}{2\sigma}|\nabla u_\varepsilon|^2 \, \mathrm{d}x$, by Lemma 5.16

$$\int_{\Omega'\cap\{|u_\varepsilon|<1-b\}} |\boldsymbol{B}^\varepsilon|^2 |\nabla u_\varepsilon| \, \mathrm{d}^2 x \leq C$$

for a different constant C. We conclude with a direct application of the coarea formula. □

Using these two results we can prove the regularity in 2 dimensions. For this we will use that the structure of stationary one dimensional varifolds is well-known. The classification was proven by Allard and Almgren in [AA76] and says that the support of a stationary varifold of one dimension in a manifold is composed of geodesic segments which have at most a finite number of intersections at a given point in the form of junctions.

Theorem 5.17 ($n = 2$ regularity) *For any open ball $B \Subset \Omega$, $\operatorname{spt}\|V\| \cap B$ consists of a finite number of disjoint line segments $\bigcup_{i=1}^{N}(a_i, b_i)$, with a_j, $b_j \in \partial B$ and $[a_i, b_i] \cap [a_j, b_j] = \emptyset$ for $i \neq j$.*

Proof By Lemma 5.16 and Fatou's Lemma

$$\int_{-1+b}^{1-b} \liminf_{k\to\infty} \left(\int_{\{u_k=t\}\cap\Omega'} \kappa_{\varepsilon_k}^2 \, \mathrm{d}\mathcal{H}^1 \right) \, \mathrm{d}t < \infty$$

so there exists $t \in [-1+b, 1-b]$ such that

$$\liminf_{k\to\infty} \int_{\{u_k=t\}\cap\Omega'} \kappa_{\varepsilon_k}^2 \, \mathrm{d}\mathcal{H}^1 < \infty.$$

By Lemma 5.15 we know that the points of $\{u_k = t\} \cap \Omega'$ are regular and thus constitute a finite number of disjoint curves. We can then assume that locally $\{u_k = t\} \cap \Omega'$ is the union of a finite number of graphs of functions $g_1^{\varepsilon_k} < \cdots < g_m^{\varepsilon_k}$. Moreover, given the Frenet relation $\tau' = k\nu$ the previous bound tells us that the tangent vectors of these graphs are all in $W^{1,2}$ and by the Morrey embedding $g_j^\varepsilon \in \mathcal{C}^{1,1/2}$ with a uniform bound.

Now, we know by Proposition 4.18 that $\{u_k = t\} \cap \Omega'$ converge towards $\operatorname{spt}\|V\| \cap \Omega'$ in the Hausdorff distance therefore (up to a subsequence) we know $\operatorname{spt}\|V\| \cap \Omega'$ is a finite union of graphs of functions $g_1 \leq \cdots \leq g_m$ in $\mathcal{C}^{1,1/2}$. By the structure of stationary one dimensional varifolds the support is either a line segment or a junction, however the latter is impossible since that would imply some g_j is not differentiable at a point, see Fig. 5.1. $\square$

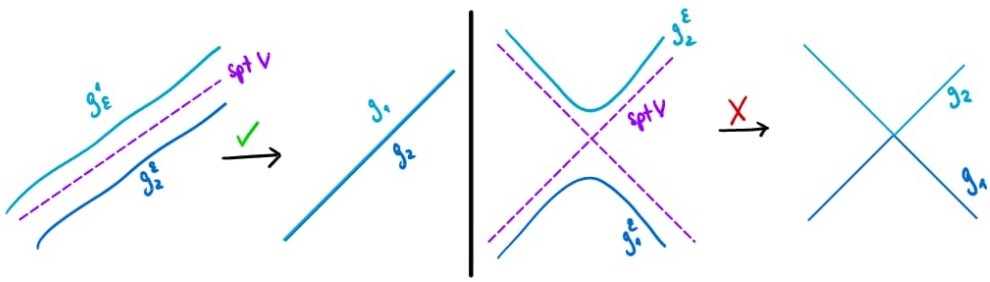

Figure 5.1: Depiction of the two scenarios which lead to the support of V being a straight line or a junction, the latter of which is not possible as it leads to a singularity of smooth functions.

Chapter 6

Regular Stable Limiting Interfaces on Manifolds

In this chapter we cover the regularity theory of the limiting interfaces obtained in Chapter 4 under the assumption of stability of Chapter 5 in [TW12].

The first section introduces the basics of the regularity theory for stable stationary integral varifolds in codimension 1. We discuss the theorem of Schoen and Simon [SS81], which relies on a singular set hypothesis, and the more recent theorem of Wickramasekera [Wic14], which utilizes the α-Structural Hypothesis. The latter framework is employed in the subsequent section to present the result of Tonegawa and Wickramasekera [TW12]; that stable stationary integral varifolds, obtained as limits of stable critical points of the Allen–Cahn equation, are optimally regular classical minimal surfaces. In the final section, following the outline provided by Guaraco [Gua18], we generalize the theory developed in Chapters 4 and 5 and Section 6.2 to the setting of an ambient closed Riemannian manifold.

6.1 Regularity Theory of Stable Minimal Hypersurfaces

In this section, we establish the fundamental regularity theory for stable, codimension 1 integral varifolds. We begin by defining the regular and singular sets of a varifold and state Allard's interior regularity theorem, which provides an initial partial regularity criterion under bounded generalized mean curvature. Next, we introduce the classical regularity theory of Schoen and Simon for stable minimal hypersurfaces. Finally, we present Wickramasekera's general regularity theorem for stable integer-multiplicity varifolds, which serves as the primary analytical tool in the upcoming section to prove the smooth regularity of stable limit interfaces.

Definition 6.1 Let V be an integral k-varifold on $\Omega \subset \mathbb{R}^n$. We say $x \in \operatorname{spt}\|V\|$ is a regular point of V if there exists $r > 0$ such that $\operatorname{spt}\|V\| \cap \overline{B}_r(x) \subset \Omega$ is a compact, connected, embedded smooth k-dimensional submanifold with boundary contained in $\partial B_r(x)$. The set of all regular points is denoted by $\operatorname{reg} V$ and $\operatorname{sing} V := (\operatorname{spt}\|V\| \setminus \operatorname{reg} V) \cap \Omega$ is the set of singular points.

By definition, $\operatorname{reg} V$ is a relatively open subset of $\operatorname{spt}\|V\|$, therefore the singular set $\operatorname{sing} V$ is closed in Ω. The following theorem by Allard gives sufficient conditions which guarantee that a point of an integral varifold with generalized mean curvature pertains to the regular set. The following is a version of this theorem involving the tilt excess which can be found in [De 18].

Theorem 6.2 (Allard's ε-Regularity Theorem) *Let $V = (M, \theta)$ be an integral k-varifold in an open set $\Omega \subset \mathbb{R}^n$ with generalized mean curvature H. There exist positive constants $\alpha, \varepsilon, \gamma \in (0,1)$ depending only on n and k such that if, for some ball $B_r(x_0) \subset \Omega$, the following two hypotheses are satisfied:*

(i) $\|V\|(B_r(x_0)) < (\omega_k + \varepsilon) r^k$ *and* $\|H\|_{L^\infty(B_r(x_0))} < \varepsilon r^{-1}$;

(ii) There exists a k-dimensional plane π such that the excess relative to π is small, precisely

$$r^{-k} \int_{B_r(x_0)} \|T_y M - \pi\|^2 \, \mathrm{d}\|V\|(y) < \varepsilon$$

where $\|T_y M - \pi\|^2$ is the Hilbert-Schmidt norm when identifying the planes with their orthogonal projections.

Then, $M \cap B_{\gamma r}(x_0)$ is a $C^{1,\alpha}$ submanifold of $B_{\gamma r}(x_0)$ without boundary, and $\theta \equiv 1$ on $B_{\gamma r}(x_0) \cap M$.

The first hypothesis requires that the mass ratio of the varifold in a local ball is close to that of a flat k-dimensional disk. The second hypothesis measures the L^2-deviation of the approximate tangent planes $T_y M$ from a fixed reference plane π (the tilt-excess). As a matter of fact, the smallness of the tilt-excess is actually redundant and is implied by the preceding assumptions. The advantage of this version is that its proof is simpler as it does not rely on the theory of general varifolds and it is also sufficient to prove the main conclusions of the theorem. Moreover, the assumption that H is locally bounded can also be relaxed to $H \in L^p_{\mathrm{loc}}$ for $p > k$, although this clearly does not matter in the stationary case.

Remark 6.3 *In the case of stationary varifolds, the Allard Regularity Theorem leads to much better regularity than the initial $C^{1,\alpha}$. Indeed, once $M \cap B_{\gamma r}(x_0)$ is known to be a $C^{1,\alpha}$ graph, the condition $H = 0$ implies that this graph solves the minimal surface equation, e.g.*

$$\operatorname{div}\left(\frac{\nabla f}{\sqrt{1 + |\nabla f|^2}}\right) = 0$$

if $k = n - 1$. Since this is a quasilinear elliptic PDE, standard elliptic regularity [GT01] bootstraps the regularity to $\mathcal{C}^\infty$.

The main conclusions of Theorem 6.2 are the following interior regularity results.

Corollary 6.4 *Let α be as in Theorem 6.2 and $V = (M, \theta)$ be an integral k-varifold with bounded generalized mean curvature in $\Omega \subset \mathbb{R}^n$. Then $\operatorname{reg} V$ is a $\mathcal{C}^{1,\alpha}$ submanifold and is dense in $\operatorname{spt}\|V\| \cap \Omega$. If, in addition, there exists $m \in \mathbb{N} \setminus \{0\}$ such that $\theta = m$ $\|V\|$-a.e., then $\|V\|(\operatorname{sing} V) = 0$.*

Remark 6.5 *The fact that $\operatorname{reg} V$ is dense in $\operatorname{spt} V$ alone does not imply that $\|V\|(\operatorname{sing} V) = 0$ in general. As a matter of fact, an example of Brakke [Bra78] shows that it can have positive measure under the same assumptions.*

We know that if $V = (M, \theta)$ is stationary and integral then it admits an approximate tangent plane $\mathcal{H}^k \llcorner M$-a.e. (Theorem 2.17), therefore by Theorem 6.2 the points which do not have this tangent plane or the multiplicity is not close to 1 are the potential singularities of V. Such potential singularities are studied with the use of tangent cones. We recall that, thanks to the Monotonicity Formula (Theorem 2.24), the density $\Theta^k(V, x)$ is well defined for every $x \in \Omega$ and the following can also be said (see [Sim84]).

Theorem 6.6 (Existence of Tangent Cones) *Let V be a stationary integral k-varifold, $x \in \Omega$ and $\hat{V}_{x,r} = (\eta_{x,r})_\# V$ the varifold sequence associated to the blow-up around x, where $\eta_{x,r}(y) = (y - x)/r$. Then there exists a subsequence of scales $\{r_j\}_{j \geq 1}$ and a stationary integral k-varifold C such that $\hat{V}_{x,r_j} \to C$ as varifolds. Moreover:*

- *(i) C is a cone, i.e. $(\eta_{0,r})_\# C = C$ for all $r > 0$;*
- *(ii) The density of V at x coincides with the (constant) density of C around the origin $\Theta^k(V, x) = \|C\|(B_r)/(\omega_k r^k)$ for all $r > 0$.*

C is known as a tangent cone *of V at x.*

Remark 6.7 *The existence of a sub-sequential limit (as measures) in Theorem 6.6 is a consequence of the Banach-Alaoglu Theorem. Because of this, there could be other converging subsequences to possibly different limits. The question of whether a tangent cone is unique and independent of the sequence of scales remains open in full generality.*

The main drawback of Allard's theory is that it requires the mass ratio to be sufficiently close to 1. Consequently, if a sequence of smooth surfaces develops folds as it converges to a limit with integer multiplicity $m \geq 2$, Allard's regularity hypotheses are no longer satisfied. Indeed, without further assumptions, little more can be said about the regularity of general stationary integral varifolds. To control the dimension of the singular set and handle these higher multiplicities, we must turn to stronger conditions, such as the area-minimizing property. Building on the work of De Giorgi, Fleming, Almgren, and Simons, Federer proved in 1970 that the singular set of an n-dimensional area-minimizing current has Hausdorff dimension at most $n-7$ (Theorem 3.25). Here, however, we focus on the milder assumption of stability since, as explained in Section 2.1, we cannot expect arbitrary manifolds to admit area-minimizing hypersurfaces.

Definition 6.8 Let $V = (M, \theta)$ be a stationary integral k-varifold on an open set $\Omega \subset \mathbb{R}^n$. We say V is *stable* in Ω if its regular set $\operatorname{reg} V$ satisfies the stability inequality,

$$\int_{\operatorname{reg} V} |\boldsymbol{B}|^2 \phi^2 \, \mathrm{d}\|V\| \leq \int_{\operatorname{reg} V} |\nabla^{\operatorname{reg} V} \phi|^2 \, \mathrm{d}\|V\|$$

for all $\phi \in \mathcal{C}^1_c(\operatorname{reg} V)$. Here $\boldsymbol{B}$ denotes the (classical) second fundamental form of $\operatorname{reg} V$ and $\nabla^{\operatorname{reg} V}$ is the tangential gradient operator on $\operatorname{reg} V$.

Integrating against the varifold weight $\mathrm{d}\|V\| = \theta \, \mathrm{d}\mathcal{H}^k$ accounts for the multiplicity of the sheets. However, this expression can be simplified with the Constancy Theorem (see Chapter 8 in [Sim84]) which says that a stationary varifold with smooth support has constant density.

Theorem 6.9 (Constancy Theorem) *Suppose V is a stationary k-varifold in Ω, and* $\operatorname{spt}\|V\| \cap \Omega \subset M$*, where M is a connected, k-dimensional $\mathcal{C}^2$ embedded submanifold of Ω. Then $V = \theta_0 v(M)$ for some constant $\theta_0 \geq 0$.*

If $\operatorname{reg} V$ is connected, then the previous theorem together with Corollary 6.4 mean that the stability inequality in Definition 6.8 simplifies to

$$\int_{\operatorname{reg} V} |\boldsymbol{B}|^2 \phi^2 \, \mathrm{d}\mathcal{H}^k \leq \int_{\operatorname{reg} V} |\nabla^{\operatorname{reg} V} \phi|^2 \, \mathrm{d}\mathcal{H}^k \tag{6.1}$$

for all $\phi \in \mathcal{C}^1_c(\operatorname{reg} V)$. Utilizing this stability inequality, Schoen and Simon [SS81] proved a groundbreaking partial regularity result for stable minimal hypersurfaces. This result makes the additional assumption that the singular set is small. Precisely, if M a closed set of Hausdorff codimension 1 then M satisfies the *Singular Set Hypothesis* in Ω if

$M \cap \Omega$ is a smooth embedded hypersurface outside of a closed set S with $\mathcal{H}^{n-3}(S) = 0$. (**SS**)

Theorem 6.10 (Schoen-Simon) *Let $V = (M, \theta)$ be a stable codimension 1 integral varifold in an open set $\Omega \subset \mathbb{R}^n$. Suppose M satisfies the* (**SS**) *hypothesis in Ω. Then* $\operatorname{reg} V$ *is an embedded analytic hypersurface, and the singular set satisfies*

$$\dim_{\mathcal{H}}(\operatorname{sing} V) \leq n - 8.$$

In particular, $\operatorname{sing} V = \emptyset$ *if $n \leq 7$, and* $\operatorname{sing} V$ *is a discrete set of points if $n = 8$.*

It is well known that Theorem 6.10 provides optimal regularity, as demonstrated by the Simons cone [Sim68] in $\mathbb{R}^8$

$$C = \{(x, y) \in \mathbb{R}^4 \times \mathbb{R}^4 \mid |x|^2 = |y|^2\}$$

which is a stable, minimal hypersurface with a single singularity at the origin. As a matter of fact, Theorem 6.10 relies on Simons' classification of stable stationary cones.

Theorem 6.11 (Simons [Sim68]) *Let C be a stationary, stable, minimal hypercone in $\mathbb{R}^n$. If $n \leq 7$, then C is a flat hyperplane.*

The singularities which arise in dimension $n \geq 8$ are usually referred to as *generally unavoidable singularities*.

Definition 6.12 We say that a stationary integral varifold V of codimension 1 has *optimal regularity* if $\dim_{\mathcal{H}}(\operatorname{sing} V) \leq n - 8$. Precisely, $\operatorname{sing} V$ is empty for $2 \leq n \leq 7$, it is a discrete set of points if $n = 8$ and if $n \geq 9$ then $\mathcal{H}^{n-8+\gamma}(\operatorname{sing} V) = 0$ for any $\gamma > 0$ and $\operatorname{reg} V$ is an embedded smooth stable minimal hypersurface

The (**SS**) hypothesis in Theorem 6.10 requires $\dim_{\mathcal{H}}(\operatorname{sing} V) \leq n - 3$ to begin with. This is because in the proof of Theorem 6.10, the (**SS**) hypothesis is used as a technical tool to employ the curvature estimates derived from the stability inequality (6.1). However, this assumption is very hard to check in practice and now known to be far from optimal[1]. In 2009, Wickramasekera [Wic14] showed that the (**SS**) hypothesis can be replaced with a much more general geometric assumption.

Definition 6.13 (α-Structural Hypothesis) An integral $(n-1)$-varifold V satisfies the *α-Structural Hypothesis* if for each $x \in \operatorname{sing} V$, there exists no $r > 0$ such that $\operatorname{spt} \|V\| \cap B_r(x)$ is equal to the union of a finite number of n-dimensional embedded $C^{1,\alpha}$ submanifolds with boundary of $B_r(x)$ all having common boundary in $B_r(x)$ equal to an $(n-2)$-dimensional embedded $C^{1,\alpha}$ submanifold of $B_r(x)$ containing x, and no two intersecting except along their common boundary.

If a varifold V corresponds to a smooth embedded submanifold it is clear that it satisfies the previous hypothesis however the converse is not immediately clear. For example, the Simons cone presents a singularity at the origin while still satisfying the α-Structural Hypothesis. Therefore, what makes the following result so surprising is that it is enough to rule out the singularities of this type to ensure a stable varifold enjoys the best possible regularity.

[1] In fact, it is enough to assume in the proof of Theorem 6.10 that the Hausdorff measure of the singularity set is finite $\mathcal{H}^{n-2}(\operatorname{sing} V) < \infty$, however this is still too restrictive of a condition.

Theorem 6.14 (Wickramasekera) *Let $n \geq 3$ and $V \in \boldsymbol{IV}_{n-1}(B_1(0))$ a stationary integral varifold such that $0 \in \operatorname{spt}\|V\|$ and $\|V\|(B_1(0)) < \infty$ which is stable (in the sense of Definition 6.8) and satisfies the α-Structural Hypothesis for some $\alpha \in (0,1)$. Then V has optimal regularity in $B_1(0)$.*

Since smoothness is a local property, Theorem 6.14 directly implies that the (**SS**) hypothesis in the Schoen-Simon theory can be replaced by the α-Structural Hypothesis. A very important consequence of this comes from the fact that the α-Structural Hypothesis is trivially implied by $\mathcal{H}^{n-2}(\operatorname{sing} V) = 0$. Thus the (**SS**) can be relaxed by letting the singularity set be of one dimension larger. We can then conclude that the weakest size hypothesis on the singular sets is precisely $\mathcal{H}^{n-2}(\operatorname{sing} V) = 0$. A major advantage of Wickramasekera's result is that it is purely geometric and does not involve the multiplicity directly or indirectly, this makes it what makes it applicable in a wider range of problems.

6.2 Regularity of Limit Interfaces

In this section we cover the contents of the paper [TW12] by Tonegawa and Wickramasekera in which optimal regularity is shown for the limit interface in Theorem 4.3 assuming stability for any $n \geq 3$. Precisely, the full result we are after is the following.

Theorem 6.15 (Limit Interface Regularity) *Let $n \geq 2$ and V_k be the sequence of varifolds in $\Omega \subset \mathbb{R}^n$ defined by (4.2) and assume hypothesis (**A**), (**B**) and (**C**). Then the limiting varifold V given by Theorem 4.3 is a stable minimal hypersurface with optimal regularity.*

As mentioned, we will be using the Theorem 6.14 of Wickramasekera to prove the regularity for $n \geq 3$ as the case $n = 2$ is already covered by Theorem 5.17. Before getting to the details, we mention that we will be using again the redefined varifold sequence in (5.3) to simplify computations. In addition, to simplify notation, for $u \in \mathcal{C}^2(\Omega)$ we will denote by $\boldsymbol{B}_u$ the non-negative function defined by

$$\boldsymbol{B}_u^2 := \frac{1}{|\nabla u|^2} \sum_{i,j=1}^{n} \left(\partial_i \partial_j u\right)^2 - \frac{1}{|\nabla u|^4} \sum_{j=1}^{n} \left(\sum_{i=1}^{n} \partial_i u \partial_i \partial_j u \right)^2 \tag{6.2}$$

if $|\nabla u| > 0$ and $\boldsymbol{B}_u = 0$ elsewhere. We note that if u_k is part of the sequence of stable critical points satisfying assumptions (**B**) and (**C**) then $\boldsymbol{B}_{u_k}|\nabla u|^2$ is the integrand in the stability inequality for the prelimit (5.4), in this case we set $\boldsymbol{B}_k := \boldsymbol{B}_{u_k}$. We make two remarks involving the $\boldsymbol{B}_k$ which will be useful later. On one hand, we have the following inequality

$$|\nabla \xi_k| \leq \varepsilon_k \sqrt{n-1} |\nabla u_k|^2 \boldsymbol{B}_k \tag{6.3}$$

where ξ_k denotes the discrepancy function corresponding to ε_k. This inequality is a consequence of the following general inequality for an $n \times n$ symmetric matrix M and a vector $m \in \mathbb{R}^n$

$$|M \cdot m - (\operatorname{tr} M)m| \leq \sqrt{n-1}(\operatorname{tr}(M^2) - m^t \cdot M^2 \cdot m)^{1/2}$$

applied to the Hessian $M = D^2 u_k$ and $m = \nabla u$ together with the facts that that u_k solves the (AC_ε) and $\Delta u_k = \operatorname{tr} D^2 u_k$. On the other hand, by Lemma 5.11 and the uniform energy bound in assumption (**B**) the L^1-norm of $\varepsilon_k \boldsymbol{B}_k^2 |\nabla u_k|^2$ is locally uniformly bounded, hence there exists a Radon measure β on Ω such that

$$\beta(\phi) = \lim_{k \to \infty} \int_\Omega \phi\, \varepsilon_k \boldsymbol{B}_k^2 |\nabla u_k|^2 \,\mathrm{d}^n x$$

for $\phi \in \mathcal{C}_c(\Omega)$, up to a subsequence.

Now, we recall that to apply the regularity Theorem of Wickramasekera it is enough to show that $\mathcal{H}^{n-2}(\operatorname{sing} V) = 0$ since this implies the α-Structural Hypothesis. For this reason, it suffices to verify the α-Structural Hypothesis in the complement of a set Z such that $\mathcal{H}^{n-2}(Z) = 0$ which is the first reduction we make. The second reduction comes from noting that the α-Structural Hypothesis can be equivalently stated at the tangent cone level by taking blow-ups. With these two observations it is clear how establishing the following result is sufficient to prove Theorem 6.15.

Proposition 6.16 *Let V as in Theorem 6.15. There exists a (possibly empty) Borel set $Z \subset \operatorname{spt}\|V\| \cap \Omega$ with $\mathcal{H}^{n-2}(Z) = 0$ such that for each $x \in (\operatorname{spt}\|V\| \setminus Z) \cap \Omega$ and each tangent cone C to V at x, C is not equal to a union of three or more half-hyperplanes of $\mathbb{R}^n$ meeting along an $(n-2)$-dimensional affine subspace.*

Assuming Proposition 6.16 the proof of Theorem 6.15 is straightforward.

Proof (of Theorem 6.15) Let $x \in \operatorname{spt}\|V\| \cap \Omega$, $r \in (0, d(x, \partial\Omega))$ and define the rescaled V so that it is in B_1 by $\hat{V} := (\eta_{x,r})_\# V$ where $\eta_{x,r}(y) = (y - x)/r$ as usual. We note that since V is stationary (Theorem 4.3) so is $\hat{V}$ in B_1. Moreover, since V is stable in the sense of Theorem 5.9 and the generalized second fundamental form coincides with the classical fundamental form on $\operatorname{reg} V$, for any $\phi \in \mathcal{C}_c^1(\operatorname{reg} V)$ we can choose an extension $\widetilde{\phi} \in \mathcal{C}_c^1(B_1 \setminus \operatorname{sing} \hat{V})$ of ϕ such that $\nabla \widetilde{\phi} = \nabla^{\operatorname{reg} V} \phi$. Testing $\widetilde{\phi} \circ \eta_{x,r}$ in inequality (5.2) shows that $\hat{V}$ is stable in the sense of Definition 6.8.

Finally, suppose that there exists $\alpha \in (0,1)$, $y \in \operatorname{spt}\|V\| \cap \Omega$ and $r \in (0, d(y, \partial\Omega))$ such that $\operatorname{spt}\|V\| \cap B_r(y)$ is a union of three or more embedded $\mathcal{C}^{1,\alpha}$ hypersurfaces-with-boundary meeting along a common $(n-2)$-dimensional $\mathcal{C}^{1,\alpha}$ submanifold L of $B_r(y)$ containing y. Using the Hopf boundary point Lemma for divergence-form elliptic equations it can be seen that the hypersurfaces meet transversely at every point of L (see Remark

6.17). This implies that the tangent cone to V at any point in L is supported by a union of three or more half-hyperplanes meeting transversely along a common $(n-2)$-dimensional affine subspace. Now, if $L \setminus Z \neq \emptyset$ we have a contradiction with Proposition 6.16, however $L \setminus Z$ has to been non-empty given that $\mathcal{H}^{n-2}(L) > 0$ and $\mathcal{H}^{n-2}(Z) = 0$. Hence, V satisfies the α-Structural Hypothesis and so does $\hat{V}$, so that Theorem 6.14 can be applied to $\hat{V}$ to deduce optimal regularity of V. $\square$

Remark 6.17 *Roughly speaking, the Hopf boundary point Lemma says that if u is a non-negative $\mathcal{C}^{1,\alpha}$ solution to an elliptic PDE whose coefficients in divergence-form are Hölder continuous up to the boundary of a $\mathcal{C}^{1,\gamma}$ connected domain Ω, then if there is $x \in \partial\Omega$ such that $u(x) = 0$ either*

$$u \equiv 0 \text{ on } \Omega \quad \text{or} \quad \langle \nu_{\partial\Omega}, \nabla u(x) \rangle > 0.$$

In other words, if u coincides up to first order at x then it has to take that value everywhere which is why this result can be regarded as a variation of the maximum principle. For the precise statement and proof see Lemma 10.1 in [HS79]. The claim in the proof can be established pairwise for each of the surfaces by considering them locally as graphs of functions $u \leq w$ solving the corresponding minimal surface PDE. Suitably applying the lemma to $w - u$ shows that either $w \equiv u$ or the surfaces meet transversely, with the former case being excluded by assumption.

In order to give a complete proof of the main theorem, the rest of this section is devoted to proving Proposition 6.16. For this, we define Z as the points in $\|V\| \cap \Omega$ where β has positive upper $(n-3)$-density $\Theta^{*,n-3} > 0$, i.e.

$$Z := \left\{ x \in \operatorname{spt}\|V\| \cap \Omega \;\middle|\; \limsup_{r \to 0} \frac{\beta(B_r(x))}{r^{n-3}} > 0 \right\}.$$

We claim that the set Z is precisely the set Z in Proposition 6.16. Indeed, using standard arguments in geometric measure theory[2] one can check that $\mathcal{H}^{n-3+\gamma}(Z) = 0$ for any $\gamma > 0$ which implies $\mathcal{H}^{n-2}(Z) = 0$ so Z satisfies the first requirement of the Proposition. The idea behind this choice of Z is that we are choosing to "throw away" the set where the limit curvature as measured by β is large so that we are left to check the α-Structural Hypothesis on a subset which is, in some appropriate sense, flatter and hence better behaved.

Now, for $x \in (\operatorname{spt}\|V\| \setminus Z) \cap \Omega$ we will prove that each tangent cone C to V at x, C is not equal to a union of $N \geq 3$ half-hyperplanes of $\mathbb{R}^n$ meeting along an $(n-2)$-dimensional subspace or *spine*[3]. For the sake of contradiction and

[2]This is a consequence of the general fact that the set of points with positive upper k-density of a Radon measure on $\mathbb{R}^n$ always has $\mathcal{H}^s$-measure 0 if $s > k$. The proof is completely analogous to the one of Theorem 2.10 in [EG15].

[3]Geometrically, the spine of a cone C is the maximal linear subspace $S \subset \mathbb{R}^n$ along which C is invariant under translations. The analysis of the dimension of the spine of tangent cones is a very useful tool in the regularity theory of stationary varifolds.

without loss of generality, assume that the tangent cone C to V at $x = 0$ is as previously mentioned with the half-hyperplanes meeting at the subspace $\{0\} \times \mathbb{R}^{n-2}$ such that

$$\operatorname{spt}\|C\| = \bigcup_{j=1}^{N} P_j$$

where

$$P_j := \{t\boldsymbol{p}_j \mid t \geq 0\} \times \mathbb{R}^{n-2}$$

and $\boldsymbol{p}_j \in \mathbb{R}^2$ are distinct for each $1 \leq j \leq N$ with $|\boldsymbol{p}_j| = 1/2$, see Fig. 6.1. From here on out, we will use the notation $(y, z) \in \mathbb{R}^n$ such that $y = (y_1, y_2) \in \mathbb{R}^2$ and $z = (z_1, \ldots, z_{n-2}) \in \mathbb{R}^{n-2}$.

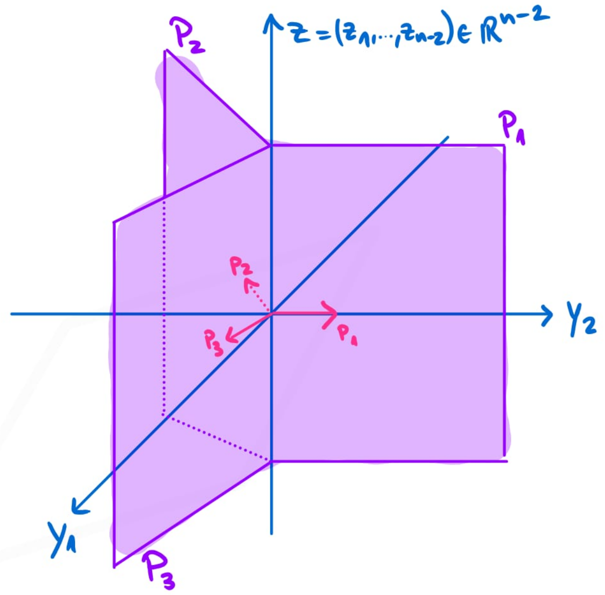

Figure 6.1: Depiction of the setting for contradiction for $N = 3$.

As we just hinted at, the contradiction we seek will come from the fact that the "curvature" measured by β vanishes in the limit in $\operatorname{spt}\|V\| \setminus Z$. This is the reason why Proposition 6.16 is stated at the tangent cone level since it allows us to take full advantage of the vanishing. Precisely, we consider, as in the proof of integrality (Section 4.4), a sequence of scales r_k and a (relabeled) subsequence ε_k such that $\hat{\varepsilon}_k := \varepsilon_k / r_k \to 0$. We then have that $\hat{V}_k \to C$ with $\hat{V}_k := (\eta_{r_k})_\# V_k$ and for $\phi \in C_c(G_{n-1}(r_k^{-1}\Omega))$

$$\hat{V}_k = \frac{1}{\sigma} \int_{\{|\nabla \hat{u}_k| > 0\}} \phi(x, I - \hat{\boldsymbol{v}}_k \otimes \hat{\boldsymbol{v}}_k) \frac{\hat{\varepsilon}_k}{2} |\nabla \hat{u}_k| \, \mathrm{d}^n x$$

with $\hat{u}_k(x) := u_k(r_k x)$ and $\hat{\boldsymbol{v}}_k := \frac{\nabla \hat{u}_k}{|\nabla \hat{u}_k|}$ as before. We already know that the rescaled sequence will satisfy assumption (**B**) on any bounded domain containing 0 (for large enough k), for our purposes B_2 will suffice. It is straightforward to check that the stability assumption (**C**) is also satisfied. Moreover, thanks to the fact that $\Theta^{n-3}(\beta, x) = 0$, we also have that (up to a further subsequence)

$$\lim_{k\to\infty} \int_{B_2} \hat{\varepsilon}_k \hat{\boldsymbol{B}}_k^2 |\nabla \hat{u}_k| \, \mathrm{d}^n x = 0, \tag{6.4}$$

with $\hat{\boldsymbol{B}}_k = \boldsymbol{B}_{\hat{u}_k}$.

For the proof of Proposition 6.16 we will be using the rescaled sequence to obtain a contradiction. For this we will be needing three technical Lemmas which will allow us to generalize the argument used in the $n = 2$ case, the first two of which will play the role of Lemma 5.15. Indeed, despite not having a uniform pointwise lower bound on $|\nabla \hat{u}_k|$ for $n > 2$ (as mentioned in the proof of 5.15) the following Lemma tells us that the set where the gradient is bounded (near points of high energy) covers the entire measure of the disk.

Lemma 6.18 *Let $c > 0$ be such that $4c^2 = \min_{|t|\le 3/4} W(t)$ and*

$$D_k := \operatorname{int}\left\{ z \in B_1^{n-2} \;\middle|\; |\nabla \hat{u}_k(y,z)| \ge \frac{c}{\hat{\varepsilon}_k} \text{ for all } y \in \overline{B}_1^2 \text{ with } |\hat{u}_k(y,z)| \le \frac{1}{2} \right\},$$

then we have that

$$\lim_{k\to\infty} \mathcal{L}^{n-2}(B_1^{n-2} \setminus D_k) = 0.$$

Proof For each $k \in \mathbb{N}$ let $\{B_{2\hat{\varepsilon}_k}^{n-2}(\tilde{z}_{k,j})\}_{j=1}^{J_k}$ be a maximal pairwise disjoint collection of balls such that $\tilde{z}_{k,j} \in B_1^{n-2} \setminus D_k$ for $j = 1, \dots J_k$, which exist by continuity of $\hat{u}_k$ and boundedness of B_1^{n-2}. By maximality

$$B_1^{n-2} \setminus D_k \subset \bigcup_{j=1}^{J_k} B_{4\hat{\varepsilon}_k}^{n-2}(\tilde{z}_{k,j})$$

and by definition of D_k there exists for each $j = 1, \dots, J_k$ points $(y_{k,j}, z_{k,j}) \in \overline{B_1^2} \times B_{\hat{\varepsilon}_k}^{n-2}(\tilde{z}_{k,j})$ such that $|\hat{u}_k(y_{k,j}, z_{k,j})| \le 1/2$ and $|\nabla \hat{u}_k(y_{k,j}, z_{k,j})| \le c/\hat{\varepsilon}_k$. Additionally, the balls $B_{\hat{\varepsilon}_k}^{n-2}(z_{k,j})$ are pairwise disjoint for $j = 1, \dots, J_k$ and

$$B_1^{n-2} \setminus D_k \subset \bigcup_{j=1}^{J_k} B_{5\hat{\varepsilon}_k}^{n-2}(z_{k,j}).$$

By standard elliptic estimates[4] we have that

$$\sup_{B_{15/8}} |D^2 \hat{u}_k| \le \frac{C}{\hat{\varepsilon}_k^2} \sup_{B_2} (|\hat{u}_k| + |W'(\hat{u}_k)|) \le \frac{C'}{\hat{\varepsilon}_k^2}$$

[4]Letting $v(x) := \hat{u}_k(x_0 + \varepsilon_k x)$ such that $\Delta v = W'(v)$ we can apply for instance Theorem 4.6 or the more general Theorem 6.2 in [GT01] to obtain the estimate.

where $C = C(n, W)$ and the second constant $C' = C'(n, W, C_0)$ comes from the uniform bound on $|u|_k$ in assumption (**B**). Due to the uniform control on the Hessian, a Taylor expansion argument guarantees the existence of $r_0 \in (0, 1]$ such that

$$|\hat{u}_k(x)| \leq \frac{3}{4} \quad \text{and} \quad |\nabla \hat{u}_k(x)| < \frac{2c}{\hat{\varepsilon}_k}$$

for each $x \in B_{2r_0\hat{\varepsilon}_k}(y_{k,j}, z_{k,j})$, which gives

$$-\hat{\xi}_k := \frac{W(\hat{u}_k)}{\hat{\varepsilon}_k} - \frac{\hat{\varepsilon}_k |\nabla \hat{u}_k|^2}{2} \geq \frac{1}{\hat{\varepsilon}_k} \left(\min_{|t| \leq 3/4} W(t) - 2c^2 \right) \geq \frac{2c^2}{\hat{\varepsilon}_k}$$

on $B_{2r_0\hat{\varepsilon}_k}(y_{k,j}, z_{k,j})$. Since $B^2_{r_0\hat{\varepsilon}_k}(y_{k,j}) \times \{z\} \subset B_{2r_0\hat{\varepsilon}_k}(y_{k,j}, z_{k,j})$ for each $z \in B^{n-2}_{r_0\hat{\varepsilon}_k}(z_{k,j})$, we then have that

$$2c^2\sqrt{\pi} r_0 \leq \left(\int_{B^2_{r_0\hat{\varepsilon}_k}(y_{k,j})} (\hat{\xi}_k(y,z))^2 \, \mathrm{d}y \right)^{1/2}$$

for each $z \in B^{n-2}_{r_0\hat{\varepsilon}_k}(z_{k,j})$, moreover

$$\begin{aligned}
\left(\int_{B^2_{r_0\hat{\varepsilon}_k}(y_{k,j})} (\hat{\xi}_k(y,z))^2 \, \mathrm{d}y \right)^{1/2} &\leq \left(\int_{B^2_{1+r_0\hat{\varepsilon}_k}} (\hat{\xi}_k(y,z))^2 \, \mathrm{d}y \right)^{1/2} \\
&\leq C'' \int_{B^2_{1+r_0\hat{\varepsilon}_k}} |\hat{\xi}_k(y,z)| + |\nabla_y \hat{\xi}_k(y,z)| \, \mathrm{d}y \\
&\leq C'' \int_{B^2_{1+r_0\hat{\varepsilon}_k}} |\hat{\xi}_k(y,z)| \, \mathrm{d}y \\
&\quad + C\sqrt{n-1}\hat{\varepsilon}_k \int_{B^2_{1+r_0\hat{\varepsilon}_k}} \hat{\boldsymbol{B}}_k(y,z) |\nabla \hat{u}_k(y,z)|^2 \, \mathrm{d}y
\end{aligned}$$

where the second inequality follows from the Sobolev inequality (with constant C'') and the third inequality is a consequence of (6.3). We have found that

$$2c^2\sqrt{\pi} r_0 \leq C'' \int_{B^2_{1+r_0\hat{\varepsilon}_k}} |\hat{\xi}_k(y,z)| \, \mathrm{d}y + C\sqrt{n-1}\hat{\varepsilon}_k \int_{B^2_{1+r_0\hat{\varepsilon}_k}} \hat{\boldsymbol{B}}_k(y,z) |\nabla \hat{u}_k(y,z)|^2 \, \mathrm{d}y$$

for any $z \in B^{n-2}_{r_0\hat{\varepsilon}_k}(z_{k,j})$. Now, integrating the inequality over $B^{n-2}_{r_0\hat{\varepsilon}_k}(z_{k,j})$, summing over j and using the fact that the balls are disjoint we have that

$$\begin{aligned}
C'''\hat{\varepsilon}_k^{n-2} J_k &\leq C'' \int_{B^2_{1+r_0\hat{\varepsilon}_k} \times B^{n-2}_{1+r_0\hat{\varepsilon}_k}} |\hat{\xi}_k| \, \mathrm{d}x + C''\sqrt{n-1} \int_{B^2_{1+r_0\hat{\varepsilon}_k} \times B^{n-2}_{1+r_0\hat{\varepsilon}_k}} \hat{\varepsilon}_k \hat{\boldsymbol{B}}_k |\nabla \hat{u}_k|^2 \, \mathrm{d}x \\
&\leq C'' \int_{B_{15/8}} |\hat{\xi}_k| \, \mathrm{d}x + C''\sqrt{E_0}\sqrt{n-1} \left(\int_{B_{15/8}} \hat{\varepsilon}_k \hat{\boldsymbol{B}}_k^2 |\nabla \hat{u}_k|^2 \, \mathrm{d}x \right)^{1/2}
\end{aligned}$$

where C''' is independent of k and the second inequality follows from Cauchy-Schwarz and the uniform energy bound in (**B**). Finally, we conclude by noting that the RHS goes to 0 thanks to (6.4) and equipartition of energy (Proposition 4.14) and the fact that $\mathcal{L}^{n-2}(B_1^{n-2} \setminus D_k) \leq J_k \omega_{n-2}(5\hat{\varepsilon}_k)^{n-2}$. □

We note that by simple logic, the set D_k contains the set of all points of low energy, i.e. $|\hat{u}_k(y,z)| > 1/2$, but as the next Lemma shows this set is small. We will omit the proof of the Lemma as it is long and technical, employing the same arguments as in the proof of integrality in Section 4.4.

Lemma 6.19 *Let $\delta > 0$ small enough such that $\{B^2_{2\delta}(\boldsymbol{p}_j)\}_{j=1}^N$ are disjoint and*

$$Q_k = \left\{ z \in B_1^{n-2} \,\middle|\, \forall t \in \left[-\frac{1}{2}, \frac{1}{2}\right], \forall 1 \leq j \leq N, \exists y \in B^2_\delta(\boldsymbol{p}_j) \text{ s.t. } \hat{u}_k(y,z) = t \right\},$$

then

$$\lim_{k\to\infty} \mathcal{L}^{n-2}(B_1^{n-2} \setminus Q_k) = 0.$$

Proof See Lemma 4.2 in [TW12]. □

The final lemma we need is a general geometric/analytic result which applies to our case whose setup we now describe (see Fig. 6.2). Let $u \in \mathcal{C}^2(\overline{B}_1^2 \times B_1^{n-2})$ and suppose that $\nabla u \neq 0$ on $(\overline{B}_1^2 \times D) \cap \{|u| \leq 1/2\}$, where D is a non-empty open subset of B_1^{n-2}. Thus for all t with $|t| \leq 1/2$, $M = u^{-1}(t) \cap (B_1^2 \times D)$ is, if non-empty, an $(n-1)$-dimensional embedded $\mathcal{C}^2$ submanifold of $B_1^2 \times D$. Fix $t \in [-1/2, 1/2]$ such that $M \neq \emptyset$ and let L be the set of points $z \in B_1^{n-2}$ satisfying the following two requirements:

(i) $T^{-1}(z) \cap M \neq \emptyset$;

(ii) z is a regular value of the map $T|_M : M \to \mathbb{R}^{n-2}$;

where $T : \mathbb{R}^n \to \mathbb{R}^{n-2}$ is the orthogonal projection. By Sard's theorem, $\mathcal{H}^{n-2}$-almost all points $z \in B_1^{n-2}$ are regular values of $T|_M$ and for each $z \in L$, $\ell_z := T^{-1}(z) \cap M$ is a $\mathcal{C}^2$ 1-manifold. For $z \in L$, let $\kappa_z(p)$ denote the curvature of ℓ_z at $p \in \ell_z$.

The role of this result in the proof of Proposition 6.16 will analogous to that of Lemma 5.16 for the $n = 2$ case, as it will allows us to control the curvature of a curve in the limit. Roughly speaking, the lemma says that the addition of the total curvature of each "slice" ℓ_z is bounded by the curvature of u as measured by $\boldsymbol{B}_u$ and the size of the set covered by all the slices.

Lemma 6.20 *With the previous assumptions, for any Borel set $G \subset L$ we have*

$$\int_G \left(\int_{\ell_z} |\kappa_z| \, \mathrm{d}s \right) \mathrm{d}z \leq \left(\int_{M\cap T^{-1}(G)} \boldsymbol{B}_u^2 \, \mathrm{d}\mathcal{H}^{n-1} \right)^{1/2} \left(\mathcal{H}^{n-1}(M \cap T^{-1}(G)) \right)^{1/2},$$

where $\boldsymbol{B}_u$ is as in (6.2).

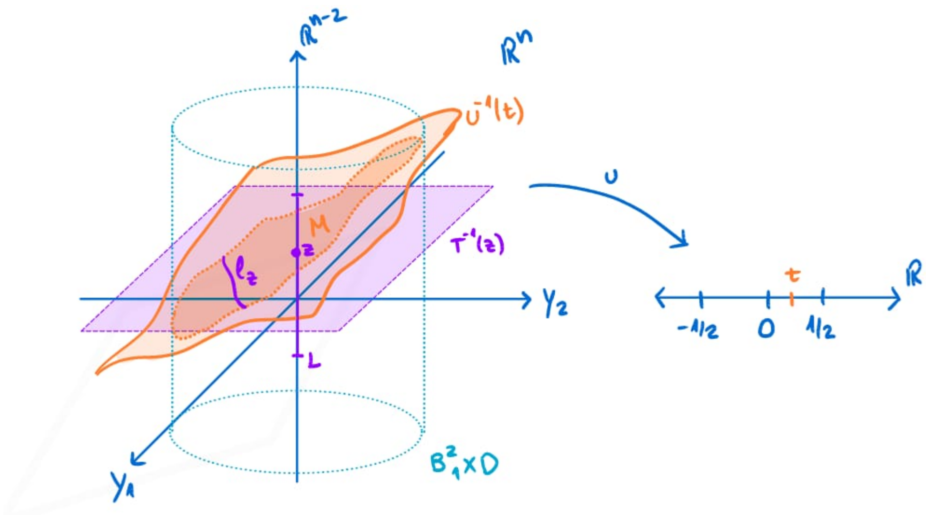

Figure 6.2: Depiction of the setting in Lemma 6.20.

Proof Set $\widetilde{M} := T^{-1}(L) \cap M$, then by our assumptions we have that $(u_{y_1}, u_{y_2}) \neq (0, 0)$ on $\widetilde{M}$. Let $x \in \widetilde{M}$, then by the implicit function theorem there exists $r > 0$ such that $\widetilde{M} \cap B_r(x)$ is the graph a $\mathcal{C}^2$ function v over an open subset U and without loss of generality $U \subset \{y_2 = 0\}$. Therefore $u(y_1, v(y_1, z), z) \equiv t$ for all $(y_1, z) \in U$ and in particular for each $z \in L \cap T(U)$ we have

$$\ell_z \cap B_r x = \{(y_1, v(y_1, z), z) \mid y_1 \in U \cap T^{-1}(z)\}$$

and hence[5] $\kappa_z = |v_{y_1 y_1}|(1 + v_{y_1}^2)^{-3/2}$. Since $u(y_1, v(y_1, z), z) \equiv t$ we find using implicit differentiation that

$$\kappa_z = \frac{|u_{y_2 y_2} u_{y_1}^2 - 2u_{y_1 y_2} u_{y_1} u_{y_2} + u_{y_1 y_1} u_{y_2}^2|}{(u_{y_1}^2 + u_{y_2}^2)^{3/2}}.$$

Since the length element $\mathrm{d}s$ is given by

$$\mathrm{d}s = \sqrt{1 + v_{y_1}^2}\, \mathrm{d}y_1 = \frac{\sqrt{u_{y_1}^2 + u_{y_2}^2}}{|u_{y_2}|}\, \mathrm{d}y_1,$$

we have that

$$|\kappa_z|\, \mathrm{d}s = \frac{|u_{y_2 y_2} u_{y_1}^2 - 2u_{y_1 y_2} u_{y_1} u_{y_2} + u_{y_1 y_1} u_{y_2}^2|}{|u_{y_2}|(u_{y_1}^2 + u_{y_2}^2)}\, \mathrm{d}y_1.$$

[5] Since z is constant we are essentially dealing with the graph of a function $v : \mathbb{R} \to \mathbb{R}$ whose curvature is given by $\kappa(t) = \frac{|v''(t)|}{(1+[v'(t)]^2)^{3/2}}$

Now, for a unit vector $\boldsymbol{m} \in \mathbb{R}^n$ with $\boldsymbol{m} \perp \nabla u$ and $M = D^2 u$, we have[6]

$$(\boldsymbol{m}^t M \boldsymbol{m})^2 \leq |M\boldsymbol{m}|^2 = \boldsymbol{m}^t M^2 \boldsymbol{m} \leq \operatorname{tr}(M^2) - \frac{1}{|\nabla u|^2} (\nabla u)^t M^2 (\nabla u) = |\nabla u|^2 |\boldsymbol{B}_u|^2.$$

Since $(u_{y_2}, -u_{y_1}, 0, \ldots, 0) \perp \nabla u$, we deduce that

$$|u_{y_2 y_2} u_{y_1}^2 - 2u_{y_1 y_2} u_{y_1} u_{y_2} + u_{y_1 y_1} u_{y_2}^2|^2 \leq (u_{y_1}^2 + u_{y_2}^2)^2 \boldsymbol{B}_u^2 |\nabla u|^2,$$

which implies

$$\frac{|u_{y_2 y_2} u_{y_1}^2 - 2u_{y_1 y_2} u_{y_1} u_{y_2} + u_{y_1 y_1} u_{y_2}^2|}{|u_{y_2}|(u_{y_1}^2 + u_{y_2}^2)} \leq \frac{\boldsymbol{B}_u |\nabla u|}{|u_{y_2}|}.$$

We also note that

$$\sqrt{1 + v_{y_1}^2 + v_{z_1}^2 + \cdots + v_{z_{n-2}}^2} = \frac{|\nabla u|}{|u_{y_2}|},$$

therefore

$$\frac{|u_{y_2 y_2} u_{y_1}^2 - 2u_{y_1 y_2} u_{y_1} u_{y_2} + u_{y_1 y_1} u_{y_2}^2|}{|u_{y_2}|(u_{y_1}^2 + u_{y_2}^2)} \leq \boldsymbol{B}_u \sqrt{1 + v_{y_1}^2 + v_{z_1}^2 + \cdots + v_{z_{n-2}}^2}.$$

We note that the term in the RHS involving the square root is the infinitesimal element of area of the graph of v; so covering $\widetilde{M}$ with a finite number of charts like U, integrating the previous inequality over $(y_1, z) \in U \cap T^{-1}(G)$ and summing over the charts with a suitable partition of unity we obtain

$$\int_G \left(\int_{\ell_z} |\kappa_z| \, \mathrm{d}s \right) \mathrm{d}z \leq \int_{M \cap T^{-1}(G)} \boldsymbol{B}_u \, \mathrm{d}\mathcal{H}^{n-1},$$

which gives the desired inequality after applying Cauchy-Schwarz. $\square$

At this point we are ready to achieve a contradiction to prove Proposition 6.16. The argument is divided into 2 steps, similarly to the argument in the proof of the $n = 2$ case. First, we extract a sequence of embedded curves coming from the sets D_k and Q_k whose total curvature we control in the limit. Then we arrive at a geometric contradiction using the properties of this sequence of curves and the fact that they live near the union of three or more half-hyperplanes of $\mathbb{R}^n$ meeting along an $(n-2)$-dimensional affine subspace

[6]The first inequality comes from Cauchy-Schwarz and $|\boldsymbol{m}| = 1$ and the second from the fact that the eigenvalues of M^2 are non-negative and $\operatorname{tr}(M^2) = \sum_{j=1}^n \boldsymbol{e}_j^t M^2 \boldsymbol{e}_j$ where $\{\boldsymbol{e}_j\}$ is an orthonormal basis obtained by completing $\boldsymbol{m}$ and $\nabla u / |\nabla u|$.

Proof (of Proposition 6.16) We have that $B_1^{n-2} \setminus (D_k \cap Q_k) = (B_1^{n-2} \setminus D_k) \cup (B_1^{n-2} \setminus Q_k)$ so by Lemmas 6.18 and 6.19

$$\lim_{k\to\infty} \mathcal{L}^{n-2}(B_1^{n-2} \setminus (D_k \cap Q_k)) = 0. \tag{6.5}$$

Letting $T : B_1^2 \times B_1^{n-2} \to B_1^{n-2}$ be the orthogonal projection, by the definition of D_k we have

$$\int_{T^{-1}(D_k\cap Q_k)\cap\{|\hat{u}_k|\le 1/2\}} \hat{\boldsymbol{B}}_k^2 |\nabla \hat{u}_k| \,\mathrm{d}^n x \le \frac{\hat{\varepsilon}_k}{c} \int_{B_1^2\times B_1^{n-2}} \hat{\boldsymbol{B}}_k^2 |\nabla \hat{u}_k|^2 \,\mathrm{d}^n x \overset{k\to\infty}{\longrightarrow} 0$$

where the limit comes from (6.4). By the co-area formula we then have

$$\lim_{k\to\infty} \int_{-1/2}^{1/2} \left(\int_{T^{-1}(D_k\cap Q_k)\cap\{\hat{u}_k=t\}} \hat{\boldsymbol{B}}_k^2 \,\mathrm{d}\mathcal{H}^{n-1} \right) \mathrm{d}t = 0.$$

We can then choose $t \in (-1/2, 1/2)$ such that $\{\hat{u}_k = t\} \cap B_1^2 \times D_k$ is a $\mathcal{C}^3$ hypersurface for each $k \in \mathbb{N}$,

$$\lim_{k\to\infty} \int_{T^{-1}(D_k\cap Q_k)\cap\{\hat{u}_k=t\}} \hat{\boldsymbol{B}}_k^2 \,\mathrm{d}\mathcal{H}^{n-1} = 0$$

and

$$\liminf_{k\to\infty} \mathcal{H}^{n-1}(T^{-1}(D_k \cap Q_k) \cap \{\hat{u}_k = t\}) < \infty.$$

Where the last condition is possible thanks to the following estimate coming from the coarea formula, the definition of D_k and Fatou's lemma

$$\begin{aligned} \int_{-1/2}^{1/2} \liminf_{k\to\infty} \mathcal{H}^{n-1}(T^{-1}(D_k \cap Q_k) \cap \{\hat{u}_k = t\}) \,\mathrm{d}t \\ \le \limsup_{k\to\infty} \int_{T^{-1}(D_k\cap Q_k)\cap\{|\hat{u}_k|\le 1/2\}} |\nabla \hat{u}_k| \,\mathrm{d}^n x \\ \le \limsup_{k\to\infty} \frac{\hat{\varepsilon}_k}{c} \int_{B_2} |\nabla \hat{u}_k|^2 \,\mathrm{d}^n x < E_0. \end{aligned}$$

Now, applying Lemma 6.20 with $u = \hat{u}_k$ and $G = D_k \cap Q_K \cap R_k$, where R_k is the set of regular values of $T|_{\{\hat{u}_k=t\}}$, we find

$$\int_{D_k\cap Q_K\cap R_k} \left(\int_{\ell_z} |\kappa_z| \,\mathrm{d}s \right) \mathrm{d}z \overset{k\to\infty}{\longrightarrow} 0 \tag{6.6}$$

thanks to (6.4). Because of (6.5) there must be a sequence $z_k \in D_k \cap Q_K \cap R_k$ such that

$$\int_{\ell^k_{z_k}} |\kappa^k_{z_k}| \,\mathrm{d}s \overset{k\to\infty}{\longrightarrow} 0 \tag{6.7}$$

as otherwise (6.6) would not hold, where $\ell^k_{z_k} = T^{-1}(z_k) \cap \{\hat{u}_k = t\} \cap (B^2_1 \times D_k)$ and $\kappa^k_{z_k}$ is the curvature of $\ell^k_{z_k}$.

Now, note that $\ell^k_{z_k}$ is the union of disjoint, connected, embedded planar curves with no boundary point inside $B^2_1 \times B^{n-2}_1$. Moreover, given that $z_k \in Q_k$ necessarily $\ell^k_{z_k}$ has a point in each of the disjoint balls $B^2_\delta(\boldsymbol{p}_j) \times \{z_k\}$ for each $j = 1, \ldots, N$. Given that $N \geq 3$, which means that at least two half hyperplanes are not aligned, and $\ell^k_{z_k}$ has no interior boundary point it must be that there exists a connected component γ_k which falls within one of the three possibilities in Fig. 6.3. We claim that either of these 3 possibilities is in contradiction with the limit in (6.7). Indeed, if ν_k is a continuous choice of unit normal to γ_k, we have that $\nu_k'(s) = \kappa^k_{z_k}(s)$ and thus the integral of the curvature over γ_k is bounded below by a fixed positive number in all 3 cases[7]. From this contradiction we see that no point outside of Z can violate the α-Structural Hypothesis which proves the Proposition. $\square$

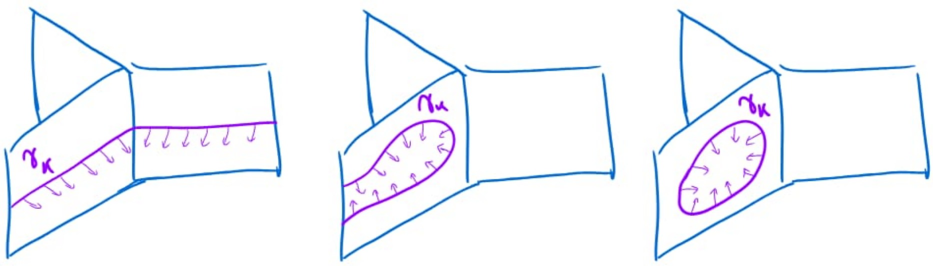

Figure 6.3: Depiction of the 3 possibilities at least one connected component of $\ell^k_{z_k}$ has to satisfy in the proof of Proposition 6.16, for the case $N = 3$.

6.3 Generalization to Closed Riemannian Manifolds

Our goal in this section is to outline how to generalize Theorems 4.3, 5.9, and 6.15 to a closed Riemannian manifold, which we will denote by (M, g) throughout. The following material is based on Appendix B of [Gua18].

We begin by noting that most of the arguments used in Chapters 4 and 5 and Section 6.2 rely exclusively on blow-up analysis and elliptic estimates, both of which are local in nature. Consequently, they can be adapted to our setting with only minor modifications by working in normal coordinates.[8] In

[7]In the case that γ_k is closed the lower bound is 2π, if the curve turns around it is π and if the curves traverses two planes which are not aligned then the lower bound is the angle formed by the normals of the planes.

[8]We recall that normal coordinates centered at $x \in M$ are local coordinates obtained via the exponential map $\exp_x$ such that the metric satisfies $g_{ij}(x) = \delta_{ij}$ and $\partial_k g_{ij}(x) = 0$.

these coordinates, the Laplace-Beltrami operator coincides at the origin with the standard Laplacian on $\mathbb{R}^n$. In most cases the resulting lower-order error terms can be controlled by restricting to a sufficiently small neighborhood of the point under consideration. Thus, we will be focusing on the differences which are not as simple to deal with.

The following statement summarizes the generalization, where the Euclidean assumptions (**B**) and (**C**) are suitably adapted to the manifold setting[9].

Theorem 6.21 *Let (M, g) be a closed Riemannian n.manifold ($n \geq 2$), and let V_k be the sequence of varifolds in M associated with u_k via w_k as defined in* (6.8). *Suppose that W and the sequence u_k satisfy hypotheses* (**A**) *and* (**B**), *respectively. Then, up to a subsequence, the following hold:*

(i) The associated varifolds V_k converge in the varifold sense to a stationary integral varifold V;

(ii) $\|V\| = \frac{1}{2\sigma} \lim_{k\to\infty} E_k(u_k)$;

(iii) If the critical points u_k are stable on an open set $\Omega \subset M$ (i.e., they satisfy (**C**)*), then $V \llcorner \Omega$ is stable and has optimal regularity.*

Allen–Cahn Energy

The first step towards generalizing the results in the previous chapters to Riemannian manifolds is to define the Allen–Cahn energy on (M, g).

Definition 6.22 Let (M, g) be a closed Riemannian manifold, then for each $\varepsilon > 0$ we define the *Allen–Cahn energy* on M as the functional $E_\varepsilon : H^1(M) \to [0, \infty]$ defined by

$$E_\varepsilon(u) := \int_M \varepsilon \frac{|\nabla u|^2}{2} + \frac{W(u)}{\varepsilon} \, \mathrm{d}\mu_g, \tag{ACE$_\varepsilon$}$$

where W is a double-well potential, $|\nabla u|^2 = g(\nabla u, \nabla u)$ and μ_g denotes the volume measure induced from the Riemannian structure[10].

Remark 6.23 *We recall that on a Riemannian manifold, the gradient of a smooth function $f \in \mathcal{C}^\infty(M)$ is defined as the unique vector field $\nabla f \in \mathfrak{X}(M)$ satisfying $g(\nabla f, X) = \mathrm{d}f(X)$ for any $X \in \mathfrak{X}(M)$. In local coordinates $\{x^i\}$, the gradient admits the expression*

$$\nabla f = g^{ij} \partial_j f \partial_i$$

[9]Precisely, assumption (**B**) is modified so that the sequence consists of critical points of the Allen–Cahn energy on the manifold (ACE$_\varepsilon$), replacing the bounded domain Ω with the closed manifold M. Furthermore, the stability condition (**C**) is replaced by its manifold equivalent (6.11).

[10]Which coincides with the n-dimensional Hausdorff measure defined from the geodesic distance which turns M into a metric space.

and consequently $|\nabla f|^2 = g^{ij}\partial_i f \partial_j f$. The Sobolev space $H^1(M)$ is defined as the completion of $\mathcal{C}^\infty(M)$ with respect to the norm

$$\|u\|^2_{H^1(M)} := \int_M |u|^2 + |\nabla u|^2 \, \mathrm{d}\mu_g.$$

For compact manifolds this definition is equivalent to the one obtained via local charts. As a consequence $H^1(M)$ inherits the fundamental properties of $H^1(\mathbb{R}^n)$, such as the Sobolev Embeddings and the Rellich-Kondrachov Compactness Theorem.

As before, we may take variations of the energy with respect to $\phi \in \mathcal{C}^1(M)$ and find that

$$\delta E_\varepsilon(u)(\phi) = \int_M \varepsilon g(\nabla u, \nabla \phi) + \frac{W'(u)}{\varepsilon}\phi \, \mathrm{d}\mu_g.$$

We then obtain that critical points of (ACE$_\varepsilon$) will weakly solve the Euler-Lagrange equation

$$\varepsilon \Delta_g u = \frac{W'(u)}{\varepsilon} \qquad (\mathrm{AC}_\varepsilon)$$

where Δ_g denotes the Laplace-Beltrami operator[11] on M.

As in Chapter 3, standard elliptic regularity theory implies that any weak solution to (AC$_\varepsilon$) is classical. Because the Laplace-Beltrami operator is strictly elliptic when read in any local coordinate chart, the local elliptic estimates and bootstrapping arguments in Proposition 3.5 carry over directly to the case of (M, g). Consequently, any weak solution u will be as regular as the potential W allows.

Varifold Sequence

If u is a critical point of the Allen–Cahn energy on M we want to associate a varifold to it like in Section 4.1 and for this we need to first look at varifolds on manifolds. The k-Grassmannian bundle on M is naturally defined as

$$G_k(M) = \{(x, S) \mid x \in M, S \in G(k, T_x M)\}$$

where $G(k, T_x M)$ denotes the collection of all unoriented k-dimensional subspaces of $T_x M$. There are several equivalent ways to define the topology on $G_k(M)$, for instance the one induced from local trivializations of the tangent bundle TM.

Definition 6.24 A *k-varifold* on a manifold M is a Radon measure on $G_k(M)$.

[11]Recall that the Laplace-Beltrami operator is defined by $\Delta_g f = \operatorname{div} \nabla f$. In local coordinates, the operator is given by $\Delta_g f = \frac{1}{\sqrt{|g|}}\partial_i \left(\sqrt{|g|} g^{ij}\partial_j f\right)$, where $|g| = \det(g_{ij})$.

With this, if u_k is a critical point of (ACE$_\varepsilon$) with parameter ε_k then we associate to it the $(n-1)$-varifold

$$V_k(A) := \frac{1}{\sigma}\int_{\mathbb{R}} v(\{w_k = t\})(A)\,\mathrm{d}t \tag{6.8}$$

as in Section 4.1, where $w_k = \Phi \circ u_k$ and $v(\{w_k = t\})$ is the varifold in M induced by the level set $\{w_k = t\}$ as in Definition 2.18 (with Ω replaced by M and $\mathcal{H}^k$ the corresponding measure on M).

Monotonicity Type Formula

The monotonicity type formula in Section 4.2 is the main tool used throughout [HT00] to prove that the limit varifold is rectifiable, stationary and integral. However, when dealing with a curved ambient space, we cannot obtain an expression of the exact same form as (4.5). This is expected, as this is also the case when dealing with stationary varifolds in manifolds, which no longer satisfy the monotonicity formula (2.8), but rather an almost-monotonicity formula[12]. Consequently, we expect to obtain an almost-monotonicity type formula for (ACE$_\varepsilon$).

For the rest of this chapter, $B_r(x)$ will denote the normal/geodesic ball centered at $x \in M$ with radius $r > 0$, constructed via the Riemannian distance function. Additionally, e_ε and ξ_ε will denote the pointwise Allen–Cahn energy density and discrepancy function adapted to manifolds, analogous to (4.15). The vector field $\boldsymbol{\nu}^u = \nabla u/|\nabla u|$ denotes the unit normal vector to the level sets of u. Any appearance of the divergence operator refers to the divergence on M induced by the metric g. Furthermore, we clarify that ∇_X denotes the Levi-Civita connection with respect to a vector field X, whereas ∇ without a subscript denotes the Riemannian gradient.

Before deriving the local monotonicity formula, we begin by establishing a useful preliminary lemma.

Lemma 6.25 *Suppose u solves* (AC$_\varepsilon$) *and let $X \in \mathfrak{X}(M)$, then*

$$\int_{\{|\nabla u|>0\}} (\operatorname{div} X - g(\nabla_{\boldsymbol{\nu}^u} X, \boldsymbol{\nu}^u))\varepsilon|\nabla u|^2 = \int_M \left(\varepsilon\frac{|\nabla u|^2}{2} - \frac{W(u)}{\varepsilon}\right)\operatorname{div} X. \tag{6.9}$$

Proof Multiply both sides of (AC$_\varepsilon$) by $g(\nabla u, X)$ and integrate each side by parts. We find for the RHS that

$$\int_M W'(u) g(\nabla u, X) = -\int_M W(u)\operatorname{div} X.$$

[12] Instead, one shows that there exists a constant $C > 0$, depending on the curvature bounds of M, such that the modified ratio $r \mapsto e^{Cr}\frac{\|V\|(B_r(x))}{r^{n-1}}$ is non-decreasing for sufficiently small radii.

For the LHS we first have that

$$\int_M \Delta_g u\, g(\nabla u, X) = -\int_M g(\nabla u, \nabla g(\nabla u, X))$$
$$= -\int_M g(\nabla_{\nabla u}\nabla u, X) + g(\nabla_{\nabla u} X, \nabla u)$$

where we have used the metric compatibility and definition of the gradient in the second equality. Using the identity

$$g(\nabla_{\nabla u}\nabla u, X) = \frac{1}{2} g(\nabla|\nabla u|^2, X)$$

and integrating by parts the term involving the previous identity we find

$$-\int_M g(\nabla_{\nabla u}\nabla u, X) + g(\nabla_{\nabla u} X, \nabla u) = \int_M \frac{1}{2}|\nabla u|^2 \operatorname{div} X - g(\nabla_{\nabla u} X, \nabla u).$$

Finally (6.9) is obtained combining both results and using that $\boldsymbol{\nu}^u = \nabla u / |\nabla u|$ if $|\nabla u| > 0$. □

Proposition 6.26 *Let $x \in M$ and suppose the sectional curvature of M is bounded by K on $B_R(x)$ for some $R > 0$. Then there exists $\bar{\varepsilon} > 0$ such that for any u solving* (AC$_\varepsilon$) *and* $0 < \rho \leq R$

$$\frac{\mathrm{d}}{\mathrm{d}\rho}\left[\frac{e^{C\rho}}{\rho^{n-1}} \int_{B_\rho(x)} e_\varepsilon(u)\right] \geq -\frac{e^{C\rho}}{\rho^n}\int_{B_\rho(x)} \xi_\varepsilon(u) \tag{6.10}$$

for any $\varepsilon < \bar{\varepsilon}$, where $C \geq 3\sqrt{K}$ is a constant.

Proof Let $\zeta : \mathbb{R} \to \mathbb{R}$ be a smooth function such that $\zeta(t) = 1$ if $t \leq 0$, $\zeta(t) = 0$ if $t \geq 1$ and $0 < \zeta(t) < 1$ for $x \in (0,1)$. Let $d : M \to \mathbb{R}$ be the geodesic distance to x and define the vector field $X = r\psi(r)\nabla r$, where $\psi(r) = \zeta\left(\frac{r-\rho}{\delta}\right)$ for $\rho > 0$. We note that $\psi(r)$ converges to $\chi_{B_\rho(x)}$ as $\delta \to 0$.

Now, we substitute X into (6.9) and for this we compute

$$\operatorname{div} X = r\psi(r)\Delta_g r + \psi(r) + r\psi'(r),$$

where we have used that $|\nabla r| = 1$. Also

$$g(\nabla_{\boldsymbol{\nu}^u} X, \boldsymbol{\nu}^u) = \big(\psi(r) + r\psi'(r)\big) g(\nabla r, \boldsymbol{\nu}^u)^2 + r\psi(r)\operatorname{Hess} r(\boldsymbol{\nu}^u, \boldsymbol{\nu}^u);$$

where $\operatorname{Hess} f(\cdot,\cdot)$ denotes the Riemannian Hessian[13], therefore

$$\operatorname{div} X - g(\nabla_{\boldsymbol{\nu}^u} X, \boldsymbol{\nu}^u) = r\psi(r)\big(\Delta_g r - \operatorname{Hess} r(\boldsymbol{\nu}^u, \boldsymbol{\nu}^u)\big)$$
$$+ \big(\psi(r) + r\psi'(r)\big)\big(1 - g(\nabla r, \boldsymbol{\nu}^u)^2\big).$$

[13]For $f \in C^\infty(M)$, the Riemannian Hessian is the symmetric $(0,2)$-tensor defined by $\operatorname{Hess} f(Y,Z) = g(\nabla_Y \nabla f, Z)$ for any vector fields $Y, Z \in \mathfrak{X}(M)$.

Now, we have the following bound which is a consequence of the Hessian comparison theorem[14]

$$\left| \operatorname{Hess} r(X,X) - \frac{1}{r}\big(|X|^2 - g(X,\nabla r)^2\big) \right| \le \sqrt{K},$$

for $|X| = 1$. The previous bound and the fact that $\Delta_g r$ can be computed as the trace of the Hessian in an orthonormal base completing $X = \nabla r$ gives

$$\Delta_g r \ge \frac{n-1}{r} - (n-1)\sqrt{K}.$$

We also find that

$$-\operatorname{Hess} r(\boldsymbol{\nu}^u, \boldsymbol{\nu}^u) \ge -\frac{1}{r}\big(1 - g(\nabla r, \boldsymbol{\nu}^u)^2\big) - \sqrt{K}$$

by choosing $X = \boldsymbol{\nu}^u$. We can then bound the integrand in the LHS of (6.9)

$$\begin{aligned}(\operatorname{div} X - g(\nabla_{\boldsymbol{\nu}^u} X, \boldsymbol{\nu}^u))\, \varepsilon|\nabla u|^2 \ge \psi(r)(n-1)\varepsilon|\nabla u|^2 - \sqrt{K} r n \varepsilon |\nabla u|^2 \psi(r) \\ + r\psi'(r)\varepsilon \left(|\nabla u|^2 - g(\nabla u, \nabla r)^2\right),\end{aligned}$$

where we have also used that $\boldsymbol{\nu}^u = \nabla u / |\nabla u|$.

Applying the previous bound to (6.9) tested against X and letting $\delta \to 0$ we find

$$\begin{aligned}-(n-1)\int_{B_\rho} e_\varepsilon(u) + \rho \int_{\partial B_\rho} e_\varepsilon(u) \ge -\int_{B_\rho} -\xi_\varepsilon(u) \\ + \varepsilon\rho \int_{\partial B_\rho} g(\nabla u, \nabla r)^2 - \sqrt{K}\int_{B_\rho} r\big(e_\varepsilon(u) + \varepsilon|\nabla u|^2\big),\end{aligned}$$

where we have also used $\Delta_g r \le \frac{n-1}{r} + (n-1)\sqrt{K}$ to bound the RHS of (6.9). The previous inequality implies that

$$-\left(n - 1 - 3\rho\sqrt{K}\right)\int_{B_\rho} e_\varepsilon(u) + \rho\int_{\partial B_\rho} e_\varepsilon(u) \ge -\int_{B_\rho} -\xi_\varepsilon(u)$$

since $r \le \rho$ and $\varepsilon|\nabla u|^2 \le 2e_\varepsilon(u)$. Finally, we obtain (6.10) by multiplying both sides of the previous inequality by $e^{C\rho}\rho^{-n}$ and using that $\frac{d}{d\rho}\int_{B_\rho} f = \int_{\partial B_\rho} f$.$\square$

By integrating inequality (6.10), decomposing the discrepancy measure as $\xi_\varepsilon = \xi_\varepsilon^+ - \xi_\varepsilon^-$, and applying the same argument used in Section 4.2, which relies only on local elliptic estimates for ξ_ε^+, we obtain the following almost-monotonicity type formula on Riemannian manifolds.

[14] See Lemma 7.1 in [CM11].

Corollary 6.27 (Almost-Monotonicity Type Formula) *In the setting of Proposition 6.26, there exists a constant $c > 0$ such that*

$$\frac{e^{CR}}{R^{n-1}}\int_{B_R(x)} e_\varepsilon(u) - \frac{e^{Cr}}{r^{n-1}}\int_{B_r(x)} e_\varepsilon(u) \geq \int_r^R \left[\frac{e^{C\rho}}{\rho^n}\int_{B_\rho(x)}\left(\frac{W(u)}{\varepsilon} - \varepsilon\frac{|\nabla u|^2}{2}\right)^+\right] \mathrm{d}\rho - cR$$

for R and ε small enough.

We note that we can always assume that the sectional curvature is bounded around any point in M and also e^{CR} can be made arbitrarily close to 1 by choosing a smaller R. Thus, the previous formula can be applied in the same way as in the Euclidean setting in combination with the appropriate local estimates to obtain the analogs of Propositions 4.12 and 4.14.

Rectifiability & Stationarity

The first variation $\delta V(X)$ of a varifold on a Riemannian manifold (M, g) with respect to a vector field $X \in \mathfrak{X}(M)$ is defined analogously to the Euclidean setting (see Section 1 in [SS81]). In our case, we have that

$$\delta V_k(X) = \frac{1}{\sigma}\int_{\{|\nabla u_k|>0\}} (\operatorname{div} X - g(\nabla_{\boldsymbol{\nu}_k} X, \boldsymbol{\nu}_k))|\nabla w_k| \, \mathrm{d}\mu_g,$$

where $\boldsymbol{\nu}_k = \boldsymbol{\nu}^{u_k}$. Hence, the same local estimates of Theorem 4.17 guarantee that $\delta V_k \to 0$. The rectifiability then follows from Allard's Theorem, which remains valid on general Riemannian manifolds [All72].

Integrality

The proof of integrality in Section 4.4 relies only on local estimates and a blow-up procedure, and thus can be generalized to (M, g). To perform this blow-up locally around a point $x \in M$ on a curved space, one utilizes the exponential map[15]. For a radius $\rho > 0$ smaller than the injectivity radius of x, we consider the restriction of the varifold to the normal coordinate ball $B_\rho(x)$ and push it forward to the tangent space $T_x M$ via the inverse exponential map

$$\widetilde{V} = (\exp_x^{-1})_\#(V \llcorner B_\rho(x)).$$

The blow-up is performed by applying the dilation map $\eta_r(y) = y/r$ to $\widetilde{V}$. Precisely, the rescaled varifolds are given by

$$\hat{V}_r := (\eta_r)_\#\widetilde{V} = (\eta_r \circ \exp_x^{-1})_\#(V \llcorner B_\rho(x)).$$

[15]Recall that the exponential map $\exp_x : T_x M \to M$ assigns to a tangent vector $v \in T_x M$ the point $\gamma_v(1) \in M$, where γ_v is the unique geodesic satisfying $\gamma_v(0) = x$ and $\gamma_v'(0) = v$. The injectivity radius at x, denoted $\mathrm{Inj}_M(x)$, is the supremum of all radii $r > 0$ such that $\exp_x$ restricted to the ball $B_r(0) \subset T_x M$ is a diffeomorphism onto its image.

As $r \to 0$, the pullback of the Riemannian metric converges to the standard flat Euclidean metric on T_xM, which allows the arguments in Section 4.4 to be adapted to show integrality in the limit.

Generalized Second Fundamental Form

We now assume that our sequence is composed of stable critical points, i.e.,

$$\delta^2 E_{\varepsilon_k}(u_k)(\phi) = \int_M \varepsilon_k |\nabla\phi|^2 + \frac{W''(u_k)}{\varepsilon_k}\phi^2 \, \mathrm{d}\mu_g \geq 0 \tag{6.11}$$

for all $\phi \in \mathcal{C}^1(M)$. As in Section 5.2, we want to show that the limit varifold admits a generalized second fundamental form in the sense of Hutchinson [Hut86]. For this, we first use the Nash embedding theorem to isometrically embed M into $\mathbb{R}^p$ for some $p \in \mathbb{N}$. We can do this because Hutchinson's framework for curvature varifolds is developed for the general case of a varifold in a smooth submanifold of Euclidean space. We will see that the embedded varifold admits a generalized second fundamental form in the ambient space $\mathbb{R}^p$ and hence can be restricted to M to obtain the same conclusion without the embedding.

Remark 6.28 *The Nash embedding is invoked at this stage, and not earlier, because extending the intrinsic Allen–Cahn equation to the ambient space $\mathbb{R}^p$ would introduce extrinsic curvature terms into the PDE, complicating the almost-monotonicity formula and the first variation. Once V is established as an integral, stationary varifold in M, we can isometrically embed $M \hookrightarrow \mathbb{R}^p$ to apply Hutchinson's framework. Note, however, that while V is stationary in M ($\boldsymbol{H} = 0$), it is generally not stationary in the ambient space $\mathbb{R}^p$. Its ambient mean curvature $\overline{\boldsymbol{H}}$ splits as*

$$\overline{\boldsymbol{H}} = \boldsymbol{H} + \mathrm{tr}_V(\overline{\boldsymbol{B}}) = \mathrm{tr}_V(\overline{\boldsymbol{B}}),$$

where $\overline{\boldsymbol{B}}$ is the second fundamental form of the embedding $M \hookrightarrow \mathbb{R}^p$.

In the following we drop the indices indicating the subsequence and instead just write ε. We will abuse notation and refer to the embedded manifold M (and related quantities such as the gradient or divergence) and varifold sequence with the same name. Additionally, for each $x \in M$ we let $S(x)$ denote the projection onto T_xM such that $S - \boldsymbol{\nu}_\varepsilon \otimes \boldsymbol{\nu}_\varepsilon$ is the projection onto the subspace orthogonal to $\boldsymbol{\nu}_\varepsilon$ in T_xM. We also denote by (x_i, P_{jk}) the coordinates in the ambient Grassmanian bundle $G_{n-1}(\mathbb{R}^p)$ and their respective partial derivatives by ∂_i and ∂^*_{jk}, using the standard basis $e_1, \dots, e_p$ of $\mathbb{R}^p$. Once again, we work with the redefined varifold sequence

$$V_k(\phi) = \frac{1}{\sigma}\int_{\Omega\cap|\nabla uk|>0} \phi(x, I - \boldsymbol{\nu_k} \otimes \boldsymbol{\nu_k})\frac{\varepsilon_k}{2}|\nabla u_k|^2 \, \mathrm{d}\mu_g$$

to simplify computations.

Our prelimit candidates for the generalized second fundamental form components are the same as in (5.6) with an additional term accounting for the curvature of M. Indeed, we define the tensors $\boldsymbol{A}^\varepsilon$, $\boldsymbol{B}^\varepsilon : G_{n-1}(\mathbb{R}^p) \to \mathbb{R}$ by

$$A^\varepsilon_{ijk}(x,P) := P_{li}\partial_l(S_{jk} - \nu^\varepsilon_j \nu^\varepsilon_k)$$

and

$$(B^k_{ij})^\varepsilon(x,P) := P_{lj}(A^\varepsilon_{ijk}(x,P) - P_{im}\partial_m S_{kl}),$$

for $1 \le i,j,k \le p$, whenever $|\nabla u_\varepsilon| > 0$. We will now see that $\boldsymbol{A}^\varepsilon$ satisfies the equivalent of Lemma (5.12) in $M \hookrightarrow \mathbb{R}^p$.

Lemma 6.29 *Let $\phi \in \mathcal{C}^1(\mathbb{R}^p \times \mathbb{R}^{p^2})$ be a function with compact support in the spatial variable $x \in \mathbb{R}^p$. For each $i = 1, \dots, p$, let $X_i \in \mathfrak{X}(M)$ be the vector field*

$$X_i = \phi(x, S - \boldsymbol{\nu}_\varepsilon \otimes \boldsymbol{\nu}_\varepsilon)(S - \boldsymbol{\nu}_\varepsilon \otimes \boldsymbol{\nu}_\varepsilon)(e_i).$$

Then,

$$\int\limits_{G_{n-1}(\mathbb{R}^p)} P_{ij}\partial_j\phi + A^\varepsilon_{ijk}\partial^*_{jk}\phi + A^\varepsilon_{jij}\phi \,\mathrm{d}V_\varepsilon(x,P) = \int\limits_M \left(\varepsilon\frac{|\nabla u|^2}{2} - \frac{W(u)}{\varepsilon}\right) \operatorname{div} X_i \,\mathrm{d}\mu_g \tag{6.12}$$

for any $i = 1, \dots, p$.

Sketch of Proof The proof proceeds exactly as that of Lemma 5.12, by testing the identity (6.9) against a smooth approximation X^s_i of X_i and passing to the limit as $s \to 0$.

It is worth pointing out that the present identity differs from the one in Lemma 5.12 because the ambient space M is now curved. More precisely, in the Euclidean setting, a subset of the components of $\boldsymbol{A}^\varepsilon$ is isolated and defined as H^ε_i. Due to the lack of ambient curvature, this term corresponds in the limit to the mean curvature of the varifold V, which vanishes by stationarity. However, when considering the isometric embedding $M \hookrightarrow \mathbb{R}^p$, the level sets of u_ε will generally not have vanishing mean curvature in the limit when viewed as submanifolds of the ambient $\mathbb{R}^p$, despite asymptotically having zero mean curvature as submanifolds of M. Thus, the present identity reflects the splitting of the global mean curvature into an intrinsic component within M (appearing on the RHS as the discrepancy term) and an extrinsic, non-vanishing contribution A^ε_{jij} arising from the embedding into $\mathbb{R}^p$. $\square$

Noting that the RHS of (6.12) vanishes in the limit, we wish to proceed as in Section 5.2 by extracting a convergent measure-function pair subsequence. For this, we need a control on the L^2-norm of $\boldsymbol{A}^\varepsilon$, which can be obtained from the following inequality

$$\int_M \left(\mathrm{Ric}_M(\nabla u_\varepsilon, \nabla u_\varepsilon) + |\operatorname{Hess} u_\varepsilon|^2 - \big|\nabla|\nabla u_\varepsilon|\big|^2\right)\phi^2 \le \int_M |\nabla u_\varepsilon|^2|\nabla\phi|^2, \tag{6.13}$$

for any $\phi \in \mathcal{C}^1(M)$. This inequality follows from u_ε solving (AC_ε), the stability condition (6.11) and applying the Bochner identity[16].

Proposition 6.30 *The limiting varifold V admits a generalized second fundamental form.*

Sketch of Proof Similar computations to those in Section 5.2 yield that

$$|\operatorname{Hess} u_\varepsilon|^2 - |\nabla|\nabla u_\varepsilon||^2 = \sum_{i,j=1}^{n-1} (\partial_i \partial_j u_\varepsilon)^2 + \sum_{i=1}^{n-1} (\partial_n \partial_i u_\varepsilon)^2.$$

and

$$\sum_{i,j,k}^{p} |A_{ijk}^\varepsilon|^2 = 2 \sum_{i,j}^{n-1} \frac{(\partial_i \partial_j u_\varepsilon)^2}{|\nabla u_\varepsilon|^2} + \sum_{i}^{n-1} \sum_{j,k \geq n+1} (\partial_i S_{jk})^2$$

in an appropriate coordinate system. Given that the last term in the previous sum is only dependent on the embedding we have

$$|\boldsymbol{A}^\varepsilon|^2 \leq \frac{|\operatorname{Hess} u_\varepsilon|^2 - |\nabla|\nabla u_\varepsilon||^2}{|\nabla u_\varepsilon|^2} + C_M$$

with $C_M > 0$ a constant. We then find that

$$\int_M |\boldsymbol{A}^\varepsilon|^2 \phi^2 \, \mathrm{d}V^\varepsilon \leq \frac{1}{\sigma} \int_M \left(|\operatorname{Hess} u_\varepsilon|^2 - |\nabla|\nabla u_\varepsilon||^2 \right) \varepsilon \phi^2 \, \mathrm{d}\mu_g + \int_M C_M \phi^2 \, \mathrm{d}V^\varepsilon.$$

for $\phi \in \mathcal{C}^1(M)$. Using (6.13) together with the fact that, since M is compact, there exists $K > 0$ such that $\mathrm{Ric}_M(\nabla u_\varepsilon, \nabla u_\varepsilon) \geq -K|\nabla u_\varepsilon|^2$, gives

$$\int_M |\boldsymbol{A}^\varepsilon|^2 \phi^2 \, \mathrm{d}V^\varepsilon \leq 2 \int_M |\nabla \phi|^2 \, \mathrm{d}V^\varepsilon + (2K + C_M) \int_M \phi^2 \, \mathrm{d}V^\varepsilon.$$

A similar bound holds for $|\boldsymbol{B}^\varepsilon|^2$, therefore the uniform energy bound on the sequence V_ε allows us to invoke Theorem 5.8 to obtain functions A_{ijk} and B_{ij}^k defined V-a.e. and a subsequence ε_l such that $(A_{ijk}^{\varepsilon_l}, V_{\varepsilon_l})$ and $((B_{ij}^k)_l^\varepsilon, V_{\varepsilon_l})$ converge weakly to (A_{ijk}, V) and (B_{ij}^k, V), respectively. In view of (6.12) we have

$$\int_{G_{n-1}(\mathbb{R}^p)} P_{ij} \partial_j \phi + A_{ijk} \partial_{jk}^* \phi + A_{jij} \phi \, \mathrm{d}V_\varepsilon(x, P) = 0$$

so $V \in \mathbf{CV}_{n-1}(\mathbb{R}^p)$ with generalized second fundamental form $\boldsymbol{B}$. $\square$

[16]For the proof see Theorem 7 in [FSV13].

Stability

To apply Wickramasekera's regularity theory, we require the limit varifold V to be stable in M. In the Euclidean setting, this corresponds to satisfying the standard stability inequality (5.2). For a varifold in a Riemannian manifold, stability is formulated by evaluating the stability of its pushforward to the tangent space T_xM via the exponential map. Specifically, one pushes forward the standard stability inequality for hypersurfaces (2.4) to T_xM, accounting for the error terms introduced by the curvature of M. This formulation is equivalent to the usual intrinsic requirement of stability within M; see [SS81] or [Wic14] for further details.

Definition 6.31 A stationary integral $(n-1)$-varifold V is *stable* in (M, g) if for each $x \in \operatorname{spt}\|V\|$ and each normal ball $B_\rho(x) \subset M$ the integral $(n-1)$-varifold

$$\hat{V} = (\exp_x^{-1})_\#(V \llcorner B_\rho(x))$$

on $B_\rho(0) \subseteq \mathbb{R}^n$ satisfies

$$\begin{aligned}\int_{\operatorname{reg}\hat{V}\cap B_\rho(0)} |\boldsymbol{B}|^2\phi^2 \, \mathrm{d}\mathcal{H}^{n-1} \leq \int_{\operatorname{reg}\hat{V}\cap B_\rho^n(0)} |\nabla\phi|^2 \, \mathrm{d}\mathcal{H}^{n-1} \\ + c_1\mu \int_{\operatorname{reg}\hat{V}\cap B_\rho^n(0)} c_2\mu\phi^2 + \phi|\nabla\phi| + \phi^2|\boldsymbol{B}| + |y||\nabla\phi|^2 + |y|\phi^2|\boldsymbol{B}|^2 \\ + c_2\mu|y|^2\phi^2|\boldsymbol{B}|^2 \, \mathrm{d}\mathcal{H}^{n-1}(y)\end{aligned} \tag{6.14}$$

for all $\phi \in C_c^1(\operatorname{reg}\hat{V})$, where c_1, c_2 and μ are constants depending only on M.

Now, proceeding as before, it can be shown that

$$|\boldsymbol{B}^\varepsilon|^2 \leq \frac{\sum_{i,j=1}^{n-1}(\partial_i\partial_j u_\varepsilon)^2}{|\nabla u_\varepsilon|^2} \leq \frac{|\operatorname{Hess} u_\varepsilon|^2 - |\nabla|\nabla u_\varepsilon||^2}{|\nabla u_\varepsilon|^2}$$

which gives

$$\int_M |\boldsymbol{B}^\varepsilon|^2\phi^2 \, \mathrm{d}V^\varepsilon \leq \int_M |\nabla\phi|^2 \, \mathrm{d}V^\varepsilon + K\int_M \phi^2 \, \mathrm{d}V^\varepsilon.$$

Passing to the limit $\varepsilon \to 0^+$ in the previous inequality, extending test functions on V to M and taking the pushforward with $\exp^{-1}$ we find (6.14).

Regularity

The proof of optimal regularity for stable limit interfaces when $n \geq 3$ in Section 6.2 involves analyzing the tangent cones of V. Because these arguments are purely local and rely exclusively on elliptic estimates and a blow-up procedure, they carry over directly to our Riemannian setting. In the case $n = 2$, optimal regularity is instead obtained via the Allard-Almgren structure theorem [AA76]; here too, the argument depends solely on local elliptic estimates and thus generalizes immediately to (M, g).

Chapter 7

Existence of Minimal Hypersurfaces in Riemannian Manifolds

In this final chapter, we present Guaraco's proof [Gua18] establishing the existence of minimal hypersurfaces in Riemannian n-manifolds, for $n \geq 3$. This framework builds on the foundations developed throughout Chapters 4, 5, and 6. As always, (M, g) will denote a smooth, closed Riemannian n-manifold.

7.1 Min-Max Theory & Minimal Hypersurfaces

Before getting to the argument of Guaraco, we begin with a brief overview of the classical Almgren–Pitts min-max theory to provide some context. Originally initiated by Almgren in the 1960s [Alm65; Alm62] and completed by Pitts in 1981 [Pit81], the Almgren–Pitts min-max theory was developed with the goal of proving the existence of minimal hypersurfaces in arbitrary closed Riemannian manifolds. The theory relies on applying a min-max procedure to the area functional over continuous sweepouts of the manifold M. Precisely, one considers a suitable family Λ of continuous sweepouts $\{\Sigma_t\}_{t\in[0,1]}$ of M by closed hypersurfaces. The Almgren–Pitts min-max width of M is defined as

$$\omega(M) = \inf_{\{\Sigma_t\}\in\Lambda} \max_{t\in[0,1]} \mathcal{H}^{n-1}(\Sigma_t).$$

Taking a minimizing sequence of sweepouts, one can extract a sequence of slices whose areas converge to $\omega(M)$. While these slices generally develop singularities in the limit, Almgren showed that the limit is a stationary integral $(n-1)$-varifold[1] V whose total mass satisfies $\|V\|(M) = \omega(M)$.

[1]Almgren originally established the existence of such stationary integral k-varifolds for any dimension $1 \leq k \leq n-1$.

Pitts later developed the regularity theory for the codimension-one case. By introducing the concept of almost minimizing varifolds, Pitts proved that for ambient dimensions $3 \leq n \leq 6$, the limit varifold is actually a smooth, closed, embedded minimal hypersurface. Schoen and Simon [SS81] subsequently refined this regularity theory to hold in arbitrary dimensions, establishing the following complete existence result.

Theorem 7.1 (Almgren, Pitts, Schoen–Simon) *Every closed Riemannian manifold of dimension $n \geq 3$ contains a minimal hypersurface with optimal regularity.*

As mentioned in Chapter 2, the min-max procedure is necessary because we cannot generally guarantee the existence of stable minimal hypersurfaces in arbitrary closed manifolds. However, the geometric min-max scheme in the classical Almgren–Pitts theory introduces significant technical difficulties turning it into a very complex theory.

Recently, relying on the theory presented in Chapters 4, 5 and 6, Guaraco [Gua18] provided an elegant alternative proof of this result. Instead of working directly with geometric sweepouts, the proof of Guaraco starts out by applying min-max theory to the much better-behaved Allen–Cahn energy functional and then taking the limit as $\varepsilon \to 0^+$.

The key to Guaraco's argument lies in extending the regularity theory of Tonegawa and Wickramasekera (Theorem 6.15), which requires global stability, to critical points with a uniformly bounded Morse index.

Definition 7.2 Let G be a Hilbert space and let $E : G \to \mathbb{R}$ be a twice Fréchet differentiable functional. If $u \in H$ is a critical point of E, the *Morse index* of u, denoted by $m(u)$, is defined as the supremum of the dimensions of all subspaces $X \subset G$ on which the second variation $E''(u)(\cdot,\cdot)$ is strictly negative definite.

In other words, while any small perturbation of a stable critical point will generally increase its energy, a critical point with Morse index m admits exactly m independent directions along which the energy strictly decreases. In particular, a globally stable critical point has a Morse index of zero.

In the upcoming sections, we present Guaraco's proof, which proceeds as follows:

(1) **Min-Max on the Allen–Cahn Energy:** By applying a Mountain Pass-type theorem to a modified Allen–Cahn energy one guarantees the existence of a sequence of critical points $\{u_\varepsilon\}_{\varepsilon>0}$ to (AC$_\varepsilon$) with bounded Morse index. In this process one has to ensure that the functional is smoothly defined and satisfies the Palais–Smale condition on $H^1(M)$.

(2) **Uniform Energy Bounds:** For the limiting varifold to be meaningful, uniform upper and lower bounds on the energy have to be satisfied by

the sequence of critical points

$$0 < c \leq E_\varepsilon(u_\varepsilon) \leq C < \infty.$$

The upper bound is required to apply the Allen–Cahn varifold theory, while the strictly positive lower bound ensures that the sequence does not converge towards a constant function (which corresponds to an empty limit).

(3) **Regularity:** Since we are dealing with non-stable critical points the stable regularity theory of Theorem 6.21 cannot be applied directly. However, a bounded Morse index implies that the sequence u_ε is locally stable. By applying the stability and regularity results on local domains, one deduces that the limit varifold is a regular minimal hypersurface away from a finite set of singular points. A subsequent removable singularity argument extends this regularity across the punctures to achieve regularity in low dimensions.

7.2 Min-Max for the Allen–Cahn Energy

As outlined, the first step in Guaraco's proof is to apply a min-max scheme to the Allen–Cahn energy to find a sequence of critical points with a uniformly bounded Morse index. In this section, we focus on constructing such a sequence. Specifically, we prove the following result.

Proposition 7.3 *For every $\varepsilon > 0$, there exists a smooth, unstable critical point u_ε of the Allen–Cahn energy E_ε such that $|u_\varepsilon| \leq 1$ and its Morse index satisfies $m(u_\varepsilon) \leq 1$.*

Finding unstable critical points of functionals is a recurrent task in functional analysis, as it has applications to many different problems such as the one at hand. While global minima can often be found through direct methods, non-linear PDEs frequently possess saddle-point solutions that represent unstable equilibrium states; the Mountain Pass Lemma gives a method for finding such critical points.

For a functional $F \in \mathcal{C}^1(G)$ on a Hilbert space G, let Γ denote the set of all continuous paths $\gamma : [0,1] \to G$ connecting two fixed points $u_0 = \gamma(0)$ and $u_1 = \gamma(1)$. The min-max value is given by

$$c = \inf_{\gamma \in \Gamma} \max_{t \in [0,1]} F(\gamma(t)).$$

The fundamental example of a functional for which c is potentially a critical value is one with a valley at 0 and a mountain range surrounding it, see Fig. 7.1. If additionally F satisfies what is known as the *Palais–Smale condition* then c will correspond to a critical value. Palais–Smale requires that any sequence $\{u_k\}_k$ such that $F(u_k) \to c$ and $\|DF(u_k)\|_{G^*} \to 0$ has a convergent subsequence which ensures that such sequences do not "escape to infinity". For an introduction to critical point theory and Non-Linear Methods see [Str00].

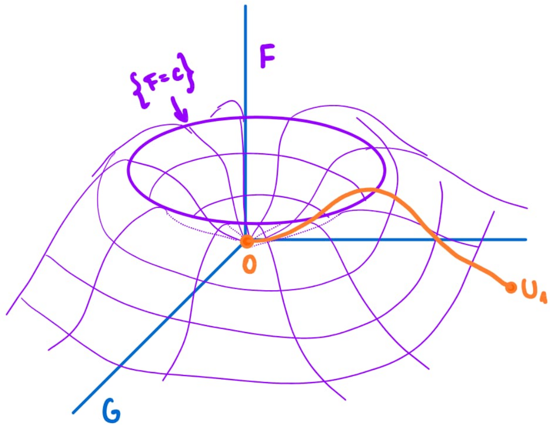

Figure 7.1: Example of a functional with a structure suited for applying the Mountain Pass Lemma.

Fortunately, there is a number of results of the Mountain-Pass type dealing with the problem of finding critical points with bounded Morse index. Coming back to the case of the Allen–Cahn energy, we begin by making the following important observations:

(a) The qualitative behavior of the family of functionals E_ε is the same for any $\varepsilon > 0$, which can be justified by absorbing the parameter ε into the potential W. Because of this, the following results can be proven without loss of generality for the case of $E := E_1$.

(b) The structure of the functional E resembles that of the example in Fig. 7.1 in the sense that there is an energy barrier. Indeed, since $u \equiv \pm 1$ are the only global minimizers and lie in the 1-dimensional subspace Y of constant functions in $H^1(M)$ it is straightforward to check that there is a global energy lower for any function in the orthogonal complement. Given that the orthogonal complement $X := Y^\perp$ is precisely the functions with 0 average we have the following result.

Lemma 7.4 *There exists a constant $E_X > 0$ such that $E(u) \geq E_X$ for any $u \in H^1(M)$ with $\int_M u = 0$.*

Proof It suffices to show that

$$E_X = \inf\left\{ E(u) \;\middle|\; \int_M u = 0 \right\}$$

is attained for some u, since $E_X = 0$ would imply $E(u) = 0$ which can only be the case if $u \equiv \pm 1$ and contradicts $\int_M u = 0$.

Indeed, let $\{u_k\}_k$ be a sequence in $H^1(M)$ such that $\int_M u_k = 0$ and $E(u_k) \searrow E_X$. By Poincaré's inequality[2] there exists $C > 0$ such that

$$C \|u_k\|^2_{L^2(M)} \leq \|\nabla u_k\|^2_{L^2(M)} \leq E(u_k)$$

which means $\{u_k\}_k$ is bounded in $H^1(M)$. Since M is compact, Rellich–Kondrachov compactness implies that there is a subsequence $\{u_{k_j}\}_j$ and $u \in H^1(M)$ such that $u_{k_j} \rightharpoonup u$ weakly in $H^1(M)$ and $u_{k_j} \to u$ strongly in $L^2(M)$. Finally, since E is lower semicontinuous[3] we find that $E(u) \leq \liminf E(u_{k_j}) = E_X$ and the L^2 convergence guarantees that $\int_M u = 0$, hence $E(u) \geq E_X$. □

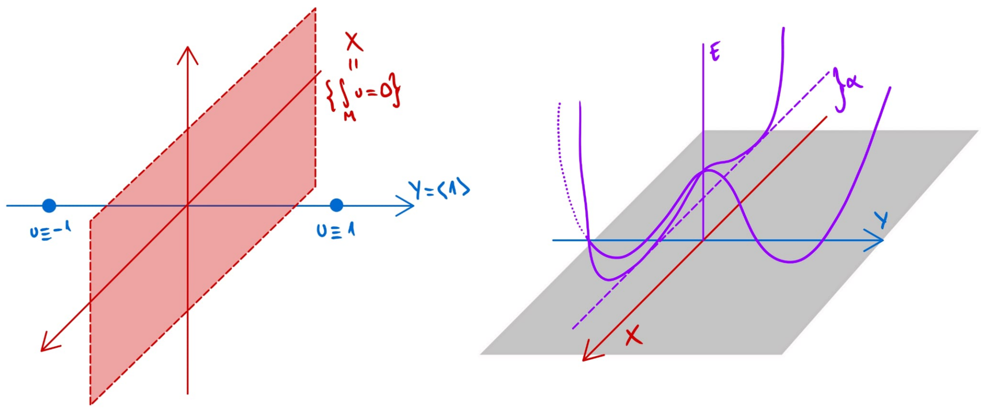

Figure 7.2: Depiction of the energy barrier along the functions of 0 average in $H^1(M)$.

The following theorem is the precise Mountain-Pass type result which applies to our situation; the much more general statement and proof can be found in Chapter 10 of [Gho93].

Theorem 7.5 *Let F be a C^2 functional on a Hilbert space $G = X \oplus Y$ with* $\dim Y = 1$ *and let $B_Y := B_1 \cap Y$, $S_Y := S_1 \cap Y = \{\pm v\}$ and*

$$\Gamma := \{\gamma : [0,1] \to G \mid \gamma \text{ continuous with } \gamma(0) = -v \text{ and } \gamma(1) = v\}$$

the set of continuous paths from $-v$ to v. Suppose

$$E_X = \inf_{u \in X} F(u) > \max_{u \in S_Y} F(u)$$

[2]**Poincaré inequality:** For $u \in H^1(M)$ such that $\int_M u = 0$ there exists $C = C(M) > 0$ such that $\|u\|_{L^2(M)} \leq C \|\nabla u\|_{L^2(M)}$.

[3]The lower semicontinuity is a consequence of the Dirichlet energy being l.s.c. and W being smooth in combination with Rellich-Kondrachov.

and set

$$c := \inf_{\gamma \in \Gamma} \max_{t \in [0,1]} F(\gamma(t)) \geq E_X; \quad K_c := \{u \in G \mid F(u) = c,\ F'(u) = 0\}.$$

Let $\{\gamma_k\}_k$ *in* Γ *be a* minimizing sequence, *i.e. such that*

$$\max_{t \in [0,1]} F(\gamma_k(t)) \stackrel{k \to \infty}{\longrightarrow} c,$$

and suppose F satisfies the Palais–Smale condition along $\{\gamma_k\}_k$*:*

> *For every* $\{u_k\}_k \subset G$ *that is a min-max subsequence for* $\{\gamma_k\}_k$ *(i.e.,* $d_G(u_k, \gamma_k([0,1])) \to 0$ *and* $E(u_k) \to c$ *as* $k \to \infty$*), if* $E'(u_k) \to 0$ *then* $\{u_k\}_k$ *has a strongly converging subsequence;* **(PS)**

and F'' *is Fredholm*[4] *on* K_c*.*

Then there exists a min-max subsequence for $\{\gamma_k\}_k$ *converging to a critical point* $u \in K_c$ *with Morse index* $m(u) \leq 1$*.*

To be able to apply Theorem 7.5 we will first need to check that E is $\mathcal{C}^2$ on $H^1(M)$, however, as discussed in Remark 3.4, this not actually the case in arbitrary dimension n for the standard choice of W. To fix this we introduce a modified potential $\widetilde{W}$ which is as smooth as W, coincides with W on $[-1,1]$, $\widetilde{W}(x) > 0$ for $|x| > 1$ and is constant on $\mathbb{R} \setminus [-2,2]$, see Fig. 7.3. We accordingly define $\widetilde{E}$ to be the energy associated to this new potential.

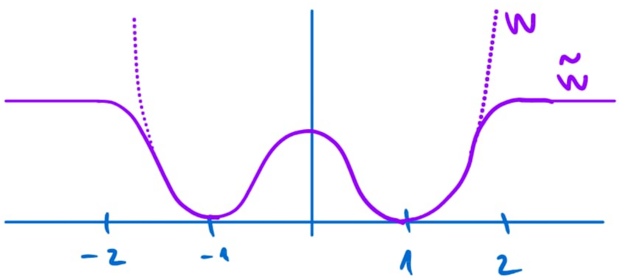

Figure 7.3: Comparison of the original potential W and the modified potential $\widetilde{W}$.

The key fact is that any critical point u of $\widetilde{E}$ such that $|u| \leq 1$ will also be a critical point of E and will have the same Morse index. Heuristically, if u lies between -1 and $+1$ it "does not see" the rest of the potential. However now $\widetilde{E}$ will be smooth while preserving the structure of E in Lemma 7.4.

[4] F'' is Fredholm if the linear map $u \in H \mapsto F''(u) \in H^* \cong H$ is Fredholm, i.e. has finite dimensional kernel and co-kernel and a closed range. This requirement is merely technical and not too restrictive.

Lemma 7.6 *The modified energy*

$$\widetilde{E}(u) := \int_M \frac{|\nabla u|^2}{2} + \widetilde{W}(u)$$

satisfies $\widetilde{E} \in \mathcal{C}^2(H^1(M))$ in the Fréchet sense and $\widetilde{E}''$ is Fredholm on $H^1(M)$.

Proof The fact that $\widetilde{E}$ is smooth comes from a general result concerning functionals of the form

$$F(u) = \int_\Omega \frac{1}{2}|\nabla u|^2 - P(x,u) \, \mathrm{d}x$$

where $\Omega \subset \mathbb{R}^n$ is a bounded domain with smooth boundary. Concretely, if

$$P(x,\xi) = \int_0^\xi p(x,t) \, \mathrm{d}t$$

with $p : \overline{\Omega} \times \mathbb{R} \to \mathbb{R}$ continuous and such that $|p(x,\xi)| \leq C(1 + |\xi|^{s-1})$ for some constants $C > 0$ and $0 \leq s < (n+2)/(n-2)$ then F is $\mathcal{C}^2$ (see Appendix B in [Rab86]). In our case given that $\widetilde{W}$ is bounded we can apply this result to $\widetilde{E}$ by considering a finite atlas covering M, which exists by compactness, and a subordinate partition of unity.

With regards to $\widetilde{E}''$ being Fredholm, the differential operator corresponding to the second variation at u is $v \mapsto -\Delta_g v + \widetilde{W}''(u)v$. It is a standard result that the Laplace–Beltrami operator is Fredholm and adding a compact operator, in our case the multiplication by $\widetilde{W}''$, does not change this fact. □

The second requirement in Theorem 7.5 that needs to be checked for $\widetilde{E}$ is that it satisfies the (**PS**) condition. As the following result shows, this is true whenever we are dealing with bounded sequences.

Proposition 7.7 *If $\{u_k\}_k$ is a sequence in $H^1(M)$ such that $\|u_k\|_{H^1(M)}$ and $\widetilde{E}(u_k)$ are bounded sequences and $\widetilde{E}'(u_k) \to 0$ then it contains a convergent subsequence. In particular, $\widetilde{E}$ satisfies the* (**PS**) *condition along norm and energy bounded sequences.*

Proof By Rellich-Kondrachov, there exists $u \in H^1(M)$ and a subsequence $\{u_{k_j}\}_j$ converging strongly in $L^2(M)$ and weakly in $H^1(M)$ so it suffices to show that $\nabla u_{k_j} \to \nabla u$ in $L^2(M)$.

For this we first note that u is a critical point of $\widetilde{E}$, indeed let $\phi \in \mathcal{C}_c^\infty(M)$ then thanks to the L^2 convergence

$$\widetilde{E}'(u)(\phi) = \lim_{j\to\infty} \widetilde{E}'(u_{k_j})(\phi) = 0$$

with the limit being 0 by hypothesis. We then have that

$$\int_M |\nabla u_{k_j} - \nabla u|^2 + \left[\widetilde{W}'(u_{k_j}) - \widetilde{W}'(u)\right](u_{k_j} - u)$$
$$= \widetilde{E}'(u_{k_j})(u_{k_j} - u) - \widetilde{E}'(u)(u_{k_j} - u) \overset{j\to\infty}{\longrightarrow} 0$$

and in particular $\nabla u_{k_j} \to \nabla u$ in L^2. □

With all this, to prove Proposition 7.3 we just need to find a bounded minimizing sequence and check that everything works-out.

Proof (of Prop. 7.3) Let $\{\gamma_k\}_k$ be a minimizing sequence of E towards $c \geq E_X$ and define for each $k \in \mathbb{N}$ the truncated path

$$\overline{\gamma}_k(t) = \begin{cases} \gamma_k(t) & \text{if } |\gamma_k(t)| \leq 1 \\ \operatorname{sign}(\gamma_k(t)) & \text{if } |\gamma_k(t)| > 1 \end{cases}$$

By construction $|\overline{\gamma}_k| \leq 1$ and $\widetilde{E}(\overline{\gamma}_k(t)) \leq \widetilde{E}(\gamma_k(t))$ so $\{\overline{\gamma}_k\}_k$ is also minimizing. For the same reason we have $\|\nabla\overline{\gamma}_k(t)\|_{L^2(M)}$ is uniformly bounded and since $\overline{\gamma}_k \in L^\infty(M)$ and M is compact we also have $\|\overline{\gamma}_k(t)\|_{L^2(M)} \leq (\mu_g(M))^{1/2} < \infty$. Hence, any min-max sequence of $\{\overline{\gamma}_k\}_k$ will also be bounded in $H^1(M)$ and $\widetilde{E}$ and by Proposition 7.7 the (**PS**) condition is satisfied. By Theorem 7.5 applied to $\widetilde{E}$ there exists a critical point $u \in K_c$ with Morse index $m(u) \leq 1$, therefore u is also as such for E. The smoothness of u follows from elliptic regularity and in particular it is a classical solution of the Euler-Lagrange equation (AC$_\varepsilon$). □

7.3 Bounding the Limit Energy

In order to apply Theorem 6.21 and have a non-empty limit we need to control the energy of the sequence $\{u_\varepsilon\}$ given by Proposition 7.3. Throughout this section we define $c_\varepsilon := E_\varepsilon(u_\varepsilon)$ for simplicity of notation.

7.3.1 The Lower Bound

Bounding the energy from below is straightforward and only relies on an isoperimetric type inequality on $H^1(M)$, originally due to De Giorgi (c.f. Lemma 3.15 in [FR22]). In its turn, this inequality is obtained using the properties of the isoperimetric profile function.

Definition 7.8 Let (M, g) be a smooth Riemannian manifold. The *isoperimetric profile function* of M is the function defined by

$$I_M(v) := \inf\{\mathcal{H}^{n-1}(\partial\Omega) \mid \Omega \subseteq M \text{ has finite perimeter and } \mu_g(\Omega) = v\},$$

for $v \in [0, \mu_g(M)]$.

For each v, $I_M(v)$ corresponds to the measure of the smallest possible hypersurface in M with the constraint that it bounds a set of measure v. For example, the isoperimetric profile of euclidean space is completely understood thanks to the isoperimetric inequality and $I_{\mathbb{R}^n}(v) = n\omega_n^{1/n} v^{\frac{n-1}{n}}$.

It is easy to see from the definition that $I_M(0) = 0$ in any manifold M. Furthermore, if M is compact then $\mu_g(M) < \infty$ and $I_M : [0, \mu_g(M)] \to [0, \infty)$. In this case, since $\mu_g(\Omega^c) = \mu_g(M) - \mu_g(\Omega)$ and $\partial\Omega = \partial\Omega^c$ then I_M is symmetric about $v = \mu_g(M)/2$ and also vanishes at the upper end. A bit more work yields the following useful properties.

Lemma 7.9 *Let M be a closed Riemannian manifold and $v \in (0, \mu_g(M))$, then:*

(i) $\exists \Omega \subset M$ of finite perimeter and volume v such that $I_M(v) = \mathcal{H}^{n-1}(\partial\Omega)$;

(ii) I_M is continuous;

(iii) if M is connected then $I_M(v) > 0$.

Proof See Chapter 3 in [Rit23]. □

Remark 7.10 *The previous properties hold on non-compact manifolds assuming they admit a strictly convex Lipschitz continuous exhaustion function. Such a function exists in manifolds with strictly positive sectional curvature for example; for more details see Chapter 4 in [Rit23].*

Since we are dealing with M compact and connected we can use the continuity and positivity on the interior of the interval $(0, \mu_g(M))$ to prove the following.

Lemma 7.11 (Isoperimetric inequality for $H^1(M)$) *Let $u \in H^1(M)$ such that $\mu_g(\{u < a\}) > \delta$ and $\mu_g(\{b < u\}) > \delta$ for some $\delta > 0$ and $a < b$ reals. Then there exists $C = C(\delta, M) > 0$ such that*

$$C(b-a) \leq \sqrt{\mu_g(\{a \leq u \leq b\})}\ \|\nabla u\|_{L^2(M)}$$

Proof Let $v \in (a, b)$ and define $\Omega_v := \{u \leq v\}$, then $\delta < \mu_g(\Omega_v) < \mu_g(M) - \delta$ and since I_M is continuous and strictly positive outside of the ends there exists a constant such that $0 < C \leq I_M(v)$. Since $u \in H^1(M)$, Ω_v has finite perimeter for almost every v (see Th. 5.9 in [EG15]) hence $\mathcal{H}^{n-1}(\partial\Omega_v) \geq C$ for a.e. v and we find

$$C(b-a) \leq \int_a^b \mathcal{H}^{n-1}(\partial\Omega_v)\, \mathrm{d}v.$$

Using the coarea formula and Hölder inequality we obtained the desired inequality

$$\int_a^b \mathcal{H}^{n-1}(\partial\Omega_v)\, \mathrm{d}v = \int_{a\leq u \leq b} |\nabla u| \leq \sqrt{\mu_g(\{a \leq u \leq b\})}\ \|\nabla u\|_{L^2(M)}. \quad \square$$

Finally, using this inequality we can show that the limiting energy must be strictly positive. The strategy relies on getting a contradiction by finding a point in $H^1(M)$ between -1 and $+1$ whose energy is close to that of a minimizing sequence but does not decrease to 0.

Proposition 7.12 (Energy lower bound) $\liminf_{\varepsilon\to 0^+} c_\varepsilon > 0$

Proof For the sake of contradiction, suppose that there exists a subsequence $\varepsilon_k \to 0$ such that $c_{\varepsilon_k} \to 0$. Fix $k > 0$ and let $\gamma : [0,1] \to H^1(M)$ be a continuous path from -1 to 1 such that $|\gamma(t)| \leq 1$ and $E_{\varepsilon_k}(\gamma(t)) \leq c_{\varepsilon_k} + \varepsilon_k$ for all $t \in [0,1]$ which we know exists as in the proof of Proposition 7.3. Since γ is continuous there exist $t_0 \in (0,1)$ such that $u := \gamma(t_0)$ has zero average $\int_M u = 0$; we now show that u has high energy.

Fix $0 < a < 1$, then by the general shape of W there exists a constant $C = C(a, W) > 0$ such that $W(u) \geq C$ on $\{-a \leq u \leq a\}$, hence

$$C\mu_g(\{-a \leq u \leq a\}) \leq \int_{\{-a\leq u\leq a\}} W(u) \leq \varepsilon_k E_{\varepsilon_k}(u) \leq \varepsilon_k(c_{\varepsilon_k} + \varepsilon_k).$$

Since $|u| \leq 1$ we find the inequality

$$0 = \int_M u \,\mathrm{d}\mu_g \leq -a\mu_g(\{u < -a\}) + \mu_g(\{u > a\}) + C^{-1}\varepsilon_k(c_{\varepsilon_k} + \varepsilon_k)$$

and also

$$\mu_g(M) \leq \mu_g(\{u < -a\}) + \mu_g(\{u > a\}) + \frac{\varepsilon_k}{C}(c_{\varepsilon_k} + \varepsilon_k).$$

Combining both inequalities we find

$$\begin{aligned}\mu_g(M) &\leq a^{-1}(\mu_g(\{u > a\} + \frac{\varepsilon_k}{C}(c_{\varepsilon_k})) + \mu_g(\{u > a\}) + \frac{\varepsilon_k}{C}(c_{\varepsilon_k} + \varepsilon_k)\\ &\leq 2a^{-1}(\mu_g(\{u > a\}) + \frac{\varepsilon_k}{C}(c_{\varepsilon_k} + \varepsilon_k))\end{aligned}$$

thus

$$\mu_g(\{u > a\}) \geq \frac{a}{2}\mu_g(M) - \frac{\varepsilon_k}{C}(c_{\varepsilon_k} + \varepsilon_k)$$

so that for ε_k small enough $\mu_g(\{u > a\}) \geq \frac{a}{3}\mu_g(M)$. Making the first inequality in the opposite direction we find

$$0 \geq -\mu_g(\{u < -a\}) + a\mu_g(\{u > a\}) - \frac{\varepsilon_k}{C}(c_{\varepsilon_k} + \varepsilon_k)$$

which yields in an analogous way that $\mu_g(\{u < -a\}) \geq \frac{a}{3}\mu_g(M)$. With this, we can apply Lemma 7.11 to find there exists $C' > 0$ such that

$$\begin{aligned}0 < 2aC' &\leq \sqrt{\mu_g(\{-a \leq u \leq a\})}\ \|\nabla u\|_{L^2(M)}\\ &\leq \sqrt{\frac{\varepsilon_k(c_{\varepsilon_k} + \varepsilon_k)}{C}\,\|\nabla u\|^2_{L^2(M)}} \leq \sqrt{\frac{c_{\varepsilon_k} + \varepsilon_k}{C} 2E_{\varepsilon_k}(u)} = \sqrt{\frac{2}{C}}(c_{\varepsilon_k} + \varepsilon_k)\end{aligned}$$

contradicting the fact that $c_{\varepsilon_k} \to 0$ as $k \to \infty$. □

7.3.2 The Upper Bound

We now move onto obtaining an upper bound for the limiting energy which is done using some techniques coming from the Almgren-Pitts theory. The strategy is to explicitly construct, for each $\varepsilon > 0$, a continuous path $h_\varepsilon(t)$ in $H^1(M)$ connecting $u \equiv -1$ to $u \equiv 1$ such that the maximal energy along this path is bounded independently of ε. For a fixed ε, this path is constructed by continuously sweeping out the manifold M with a family of closed hypersurfaces Σ_t. Each hypersurface divides M into two disjoint regions; we assign the value $+1$ to one region and -1 to the other, and then smoothly interpolate the transition across a small tubular neighborhood of Σ_t using the optimal 1D optimal energy profile. Formalizing this sweepout procedure requires tools from Morse theory.

A function $f \in \mathcal{C}^\infty(M)$ is a *Morse function* if it has no degenerate critical points, i.e., whenever $\nabla f_p = 0$ then $\det(\operatorname{Hess} f_p) \neq 0$. Morse's lemma states that if p is a non-degenerate critical point of f then there exist local coordinates $(x_1, \ldots, x_n)$ and an integer $0 \leq i \leq n$ such that $f(x) = f(p) - x_1^2 - \cdots - x_i^2 + x_{i+1}^2 + \cdots + x_n^2$; i is known as the *Morse index* of f at p.

An *ambient isotopy* of M is a smooth map $\psi : M \times [0,1] \to M$ such that for each $s \in [0,1]$ the map $\psi_s := \psi(\cdot, s) : M \to M$ is a diffeomorphism and $\psi_0 = \mathrm{Id}$. If $\mathrm{Diff}(M)$ denotes the set of all diffeomorphisms of M then $\mathrm{Is}(M) := \mathcal{C}^\infty([0,1], \mathrm{Diff}(M))$ is the set of all isotopies. Using the notions of Morse functions and isotopies we can define the set Λ of sweep-outs of M which we will consider

$$\Lambda := \{\{\Sigma_t\}_{t\in[0,1]} \mid \Sigma_t = \psi_t(f^{-1}(t))\}, \tag{7.1}$$

with $\psi \in \mathrm{Is}(M)$ and f a surjective Morse function with values in $[0,1]$. We note that the sweep-outs in Λ are a bit more general than those obtained from just the level sets of a Morse function as we allow for smooth deformations of each level set. Clearly other families of sweep-outs can be defined, this particular choice is justified by the fact that the properties of Λ will suit our needs. The *width* of the family Λ is defined by the min-max area

$$\omega(\Lambda) := \inf_{\{\Sigma_t\}\in\Lambda} \max_{t\in[0,1]} \mathcal{H}^{n-1}(\Sigma_t).$$

Intuitively, $\omega(\Lambda)$ measures the minimum area of the "widest" hypersurface encountered across all possible topological sweepouts of the manifold.

With this, the goal of this section is to show that

$$\limsup c_\varepsilon \leq 2\sigma\omega(\Lambda), \tag{7.2}$$

and we note that $\omega(\Lambda) < \infty$ because $\mathcal{H}^{n-1}(\Sigma_t)$ is a continuous function of t.

Constructing the Path

There are two main tools involved in constructing the path described at the beginning of the section, the first of which is the heteroclinic solution to the Allen–Cahn equation in $\mathbb{R}$.

Recall from Section 3.2 that for $\varepsilon = 1$ the 1D Allen–Cahn equation admits a unique monotone increasing solution $\varphi : \mathbb{R} \to (-1, 1)$ interpolating between the two wells of W such that $\varphi(0) =: \gamma \in (-1, 1)$ which satisfies:

(i) $sW(\varphi(s)) \to 0$ as $s \to \pm\infty$;

(ii) $E(\varphi) = \int_{\mathbb{R}} \frac{(\varphi')^2}{2} + W(\varphi) = 2\sigma$.

The corresponding solution for any $\varepsilon > 0$ can be obtained by considering $\varphi_\varepsilon(x) = \varphi(x/\varepsilon)$.

The idea now is to place the profile of the 1-dimensional solution in M such that the transition region lies on a hypersurface coming from a sweep-out. To do this we employ the *signed distance* which is the second main tool we will need. First, we recall that the distance to a subset $A \subset M$ is defined as

$$d_A(x) := \inf_{y \in A} d(x, y)$$

and if A is compact the infimum is always achieved. The triangle inequality implies that the distance function is Lipschitz thus, by Rademacher's theorem[5], it is differentiable a.e. in M. It is easily checked that $d_A \in W^{1,\infty}(M)$, $\nabla d_A(x) = 0$ for a.e. $x \in A$ and outside of A it satisfies a.e. the Eikonal equation $|\nabla d_A| = 1$ (see Appendix A in [Gua18]). For a given t, we can define the signed distance function on Σ_t by

$$\begin{cases} d_{\Sigma_t}(x) & x \in \psi_t(f^{-1}([0, t])) \\ -d_{\Sigma_t}(x) & x \in \psi_t(f^{-1}((t, 1])), \end{cases}$$

i.e., such that it measures the distance as positive on one side of Σ_t and negative on the other, see Fig. 7.4. Without risk of confusion we will refer to the signed distance by d_{Σ_t} as well. Doing as such for each $t \in [0, 1]$, we obtain a family of distances $t \mapsto d_{\Sigma_t}$ varying continuously in $H^1(M)$.

Remark 7.13 *The continuity of d_{Σ_t} in t is due to the fact that $t \mapsto \Sigma_t$ is continuous with respect to the Hausdorff distance on the compact subsets of M which in turn is due to $\{\Sigma_t\}_t$ being constructed out of Morse functions and isotopies.*

With both tools at hand, it is quite clear how we can proceed. Fix $\varepsilon > 0$ and $\delta > 0$ which will simultaneously act as the width of the tubular neighborhood

[5]**Rademacher's theorem:** Any locally Lipschitz function on $\mathbb{R}^n$ is differentiable $\mathcal{L}^n$-almost everywhere.

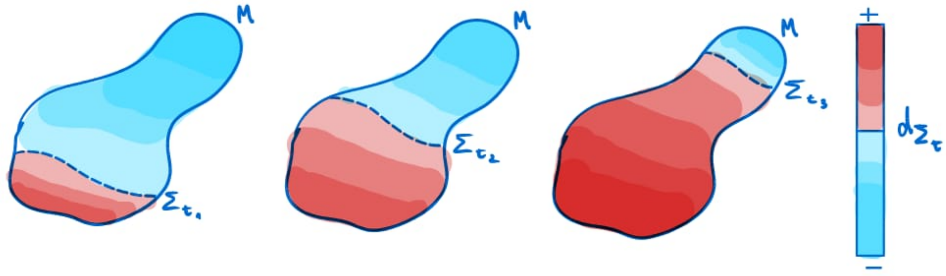

Figure 7.4: Sketch of the signed distance at different sweep-out slices.

in which the transition will occur and control the point-wise bound for the function. Then for a given sweep-out $\{\Sigma_t\}_t$ we define for each $t \in [0,1]$

$$v_{\varepsilon,\delta}(\Sigma_t, x) := \begin{cases} \varphi(d_{\Sigma_t}(x)/\varepsilon) & \text{if } |d_{\Sigma_t}(x)| \leq \delta, \\ \varphi(\operatorname{sign}(d_{\Sigma_t}(x))\delta/\varepsilon) & \text{if } |d_{\Sigma_t}(x)| > \delta; \end{cases}$$

as in Fig. 7.5. For each fixed value of t, ε and δ the function $v_{\varepsilon,\delta}(\Sigma_t, \cdot)$ belongs to $H^1(M)$, indeed this is because it is constructed from a piecewise gluing of Lipschitz pieces agreeing on $|d_{\Sigma_t}(x)| = \delta$ and in turn the pieces are Lipschitz because they are a composition of Lipschitz functions. Moreover, as t varies the map $g(t) := v_{\varepsilon,\delta}(\Sigma_t, \cdot)$ is continuous in $H^1(M)$ given that d_{Σ_t} is.

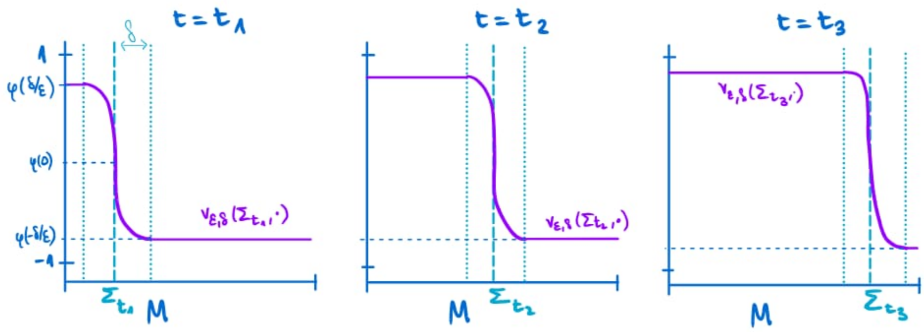

Figure 7.5: Depiction of the function $v_{\varepsilon,\delta}(\Sigma_t, \cdot)$ at different times for fixed values of ε and δ.

We note however that g does not join the constant functions ± 1 as wanted, to fix this we have to perform slight modifications at the ends. We define

$$f^+(t) := (1-t) + tv_{\varepsilon,\delta}(\Sigma_0, \cdot) \quad \text{and} \quad f^-(t) := -(1-t) + tv_{\varepsilon,\delta}(\Sigma_1, \cdot)$$

which join the endpoints of g to the constant functions 1 and -1, respectively. Then we can define $h(t)$ to be the concatenation of f^+, g and f^- in this order to obtain a continuous path with the desired properties.

Controlling the Energy

Now that we have constructed a candidate path (one for each sweep-out as a matter of fact), we can work towards obtaining the upper bound (7.2). There are two steps involved in the procedure; first we show that the modification h of g that we made to construct an appropriate path does not increase the energy. This is harder to write down than to visualize if one imagines how the paths $f^+(t)$ and $f^-(t)$ flatten out the function as t goes from 0 to 1 towards absolute minima of W, see Fig. 7.6. Formally we have the following.

Lemma 7.14 *Let g and h be the paths as previously defined corresponding to a fixed choice of $\{\Sigma_t\}_t \in \Lambda$, then*

$$\max_{t\in[0,1]} E_\varepsilon(h(t)) = \max_{t\in[0,1]} E_\varepsilon(g(t)).$$

Proof We show that the energy of $f^+(t)$ is controlled by $E_\varepsilon(g(0))$ and similarly $f^-(t)$ is controlled by $E_\varepsilon(g(1))$. This implies the claim since h is the concatenation of f^+, g and f^- .

The control on the gradient term comes from the fact that f interpolates with a constant function hence $|\nabla f^+(t)| \leq |\nabla v_{\varepsilon,\delta}(\Sigma_0, \cdot)| = |\nabla g(0)|$.

For the potential term, first notice that Σ_0 consists of a finite number of points. Indeed this is because 0 is the global extrema of the Morse function defining the sweep-out, hence a critical point due to $\partial M = \emptyset$, and Morse functions can only have a finite number of critical points on a compact manifold (due to Morse's lemma). This has the consequence that, by our choice of signs, d_{Σ_0} is non-negative everywhere on M and thus $\gamma \leq v_{\varepsilon,\delta}(\Sigma_0, x) < 1$ for all $x \in M$. This corresponds to the interval $[0, 1]$ on the domain of W where it is a decreasing function, hence $W(f^+(t)) \leq W(v_{\varepsilon,\delta}(\Sigma_0, x))$.

Combining both terms we find $E_\varepsilon(f^+(t)) \leq E_\varepsilon(v_{\varepsilon,\delta}(\Sigma_0, x))$ and with a completely analogous argument $E_\varepsilon(f^-(t)) \leq E_\varepsilon(v_{\varepsilon,\delta}(\Sigma_1, x))$. □

The step that remains is to control the energy of g, for this we use the following technical lemma which will allow us to estimate the size of hypersurfaces "parallel" to Σ_t.

Lemma 7.15 *Let $\{\Sigma_t\}_t \in \Lambda$ with Λ as in* (7.1)*, then there exist constants C, $\eta > 0$ such that*

$$\mathcal{H}^{n-1}(\{d_{\Sigma_t} = s\}) \leq (1 + C|s|^n)\mathcal{H}^{n-1}(\Sigma_t)$$

for every $|s| \leq \eta$.

Remark 7.16 *Proving Lemma 7.15 involves distinguishing between Σ_t being defined by a regular value of f or a critical point. In the first case it is a consequence of a much more general result since Σ_t corresponds to a regular submanifold. The second case is a bit more technical and it involves analyzing the behavior of the*

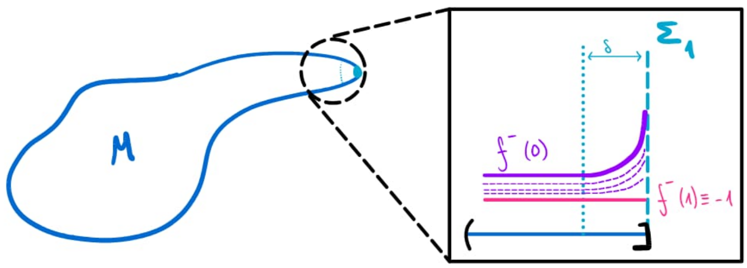

Figure 7.6: Representation of the homotopy f^- going from the phase transition solution to the constant $u \equiv -1$ function near the slice Σ_1

parallel hypersurfaces near nondegenerate critical points (such as those of Morse functions). For the details see Appendix A in [Gua18].

Proposition 7.17 (Energy upper bound) $\limsup c_\varepsilon \leq 2\sigma\omega(\Lambda)$.

Proof By definition, the energy of $v_{\varepsilon,\delta}(\Sigma_t, \cdot)$ is

$$E_\varepsilon(v_{\varepsilon,\delta}(\Sigma_t,\cdot)) = \int_M \varepsilon\frac{|\nabla v_{\varepsilon,\delta}(\Sigma_t,x)|^2}{2} + \frac{W(v_{\varepsilon,\delta}(\Sigma_t,x))}{\varepsilon}\, \mathrm{d}\mu_g(x).$$

and the integral can be split into the sets $\{|d_{\Sigma_t}| > \delta\}$ and $\{|d_{\Sigma_t}| \leq \delta\}$ which we treat separately.

Since $v_{\varepsilon,\delta}(\Sigma_t,\cdot)$ is constant on $\{|d_{\Sigma_t}| > \delta\}$ we have $\nabla v_{\varepsilon,\delta}(\Sigma_t,\cdot) = 0$ such that

$$\begin{aligned}\int_{\{|d_{\Sigma_t}|>\delta\}} \varepsilon\frac{|\nabla v_{\varepsilon,\delta}(\Sigma_t,x)|^2}{2} + \frac{W(v_{\varepsilon,\delta}(\Sigma_t,x))}{\varepsilon}\, \mathrm{d}\mu_g(x) \\ \leq \frac{\mu_g(M)}{\varepsilon}\left[W(\varphi(-\delta/\varepsilon)) + W(\varphi(\delta/\varepsilon))\right] =: R(\varepsilon)\end{aligned}$$

with $R(\varepsilon) \to 0$ as $\varepsilon \to 0^+$ due to property (i) of the heteroclinic solution φ.

Since $|\nabla d_{\Sigma_t}| = 1$ and φ is non-decreasing we have that

$$|\nabla v_{\varepsilon,\delta}(\Sigma_t,x)| = \frac{1}{\varepsilon}\varphi'(d_{\Sigma_t}(x)/\varepsilon)$$

on $\{|d_{\Sigma_t}| \leq \delta\}$. Using the coarea formula

$$\begin{aligned}\int_{\{|d_{\Sigma_t}|\leq\delta\}} \varepsilon\frac{|\nabla v_{\varepsilon,\delta}(\Sigma_t,x)|^2}{2} + \frac{W(v_{\varepsilon,\delta}(\Sigma_t,x))}{\varepsilon}\, \mathrm{d}\mu_g(x) \\ = \int_{-\delta}^{\delta} \frac{1}{\varepsilon}\left[\frac{\varphi'(s/\varepsilon)^2}{2} + W(\varphi(s/\varepsilon))\right] \mathcal{H}^{n-1}(\{d_{\Sigma_t} = s\})\, \mathrm{d}s\end{aligned}$$

and by Lemma 7.15 there exists C, $\eta > 0$ such that if $\delta \leq \eta$

$$\begin{aligned}\int_{-\delta}^{\delta} \frac{1}{\varepsilon} \left[\frac{\varphi'(s/\varepsilon)^2}{2} + W(\varphi(s/\varepsilon)) \right] \mathcal{H}^{n-1}(\{d_{\Sigma_t} = s\})\, \mathrm{d}s \\ \leq (1 + C\delta^n)\mathcal{H}^{n-1}(\Sigma_t) \int_{-\delta}^{\delta} \frac{1}{\varepsilon} \left[\frac{\varphi'(s/\varepsilon)^2}{2} + W(\varphi(s/\varepsilon)) \right] \mathrm{d}s \\ \leq 2\sigma(1 + C\delta^n)\mathcal{H}^{n-1}(\Sigma_t)\end{aligned}$$

where the last inequality is consequence of property (ii) and a change of variable.

Combining the bounds on both sets we find that

$$E_\varepsilon(g(t)) \leq 2\sigma(1 + C\delta^n)\mathcal{H}^{n-1}(\Sigma_t) + R(\varepsilon),$$

and using Lemma 7.14

$$c_\varepsilon \leq \max_{t \in [0,1]} E_\varepsilon(h(t)) + R(\varepsilon) \leq 2\sigma(1 + C\delta^n) \max_{t \in [0,1]} \mathcal{H}^{n-1}(\Sigma_t) + R(\varepsilon)$$

which holds for any $\delta > 0$ small enough. Taking the limit as $\varepsilon \to 0^+$ and observing that the LHS is independent of the sweep-out we conclude. □

7.4 Regularity of Unstable Limits with Finite Morse Index

In this section we conclude the proof of Guaraco by showing that the limiting varifold obtained from a sequence of critical points with uniformly bounded Morse index has optimal regularity. Concretely, we prove the following result which directly implies Theorem 7.1.

Theorem 7.18 *Let (M, g) be an n-dimensional, $n \geq 3$, closed Riemannian manifold and let $\{u_\varepsilon\}_\varepsilon$ be the family of critical points of* (ACE$_\varepsilon$) *in Proposition 7.3 and V the limiting varifold obtained as in Theorem 6.21. Then V has optimal regularity.*

The proof of Theorem 7.18 relies on the following simple observation.

Observation: If u_ε is a critical point with Morse index $m > 0$ and $\{\Omega_j\}_{j=1}^{m+1}$ is a family of $m+1$ pairwise disjoint subsets of M, then u_ε has to be stable in at least one the Ω_j's. This is just because otherwise one could find $\phi_j \in C_c^1(\Omega_j)$ which span an $m+1$ dimensional subspace of $H^1(M)$ where $E''_\varepsilon(u_\varepsilon)$ is negative definite. Since the number of open sets is finite, we can find a subsequence $\{u_{\varepsilon_k}\}_k$ such that every term is stable in a fixed Ω_{j_0}. By Theorem 6.21 this implies optimal regularity of the limiting varifold V in Ω_{j_0}.

Using the previous observation we can show that regularity in small annuli centered at an arbitrary point of M for $n \geq 3$. For $0 < \tau < \tau'$ we denote by

$$An(x, \tau, \tau') := B_{\tau'}(x) \setminus \overline{B_\tau(x)}$$

and for $r > 0$

$$\mathcal{AN}(x, r) := \{An(x, \tau, \tau') \mid 0 < \tau < \tau' < r\}.$$

The upcoming arguments are adapted from De Lellis and Tasnady [LT13], who utilized them to largely simplify the classical Almgren–Pitts proof of Theorem 7.1. In fact, a close inspection of the proofs reveals that the specific origin of the limiting varifold V (namely, that it comes from a sequence of critical points of the Allen–Cahn energy) does not play an essential role. The subsequent regularity theory holds for any stationary integral varifold with a finite Morse index, provided it satisfies the property that local stability implies local optimal regularity.

Lemma 7.19 *There is a positive function $r : M \to [0, \infty)$ such that in every annulus $An \in \mathcal{AN}(x, r(x))$ the varifold V in Theorem 7.18 is stable with optimal regularity.*

Proof Assume for the sake of contradiction that there is $x \in M$ and $0 < \rho < \mathrm{Inj}_M(x)$ such that for any $0 < r < \rho$ there exists $AN \in \mathcal{AN}(x, r)$ such that V does not have optimal regularity in AN. Then by choosing r small enough $m + 1$ times we can find $m + 1$ concentric disjoint annuli centered at x such that V is not optimally regular in any of them, which is a contradiction. □

We observe that the preceding lemma establishes that V is stable and optimally regular within the punctured ball $B_{r(x)}(x) \setminus \{x\}$ for every $x \in M$. Consequently, the set of singularities of V which the lemma does not account for is at most a discrete set of isolated points (having Hausdorff dimension zero). For ambient dimension $n \geq 8$, the definition of optimal regularity allows a singular set of dimension up to $n - 8$, therefore these potential singularities can be "hidden" within the generally unavoidable ones. However, for ambient dimensions $3 \leq n \leq 7$, optimal regularity requires that the hypersurface must be everywhere smooth. Thus, to complete the proof of Theorem 7.18, it remains to show that these potential singularities are actually not singularities at all. For this we will need the following useful result coming from the Schoen—Simon curvature estimates (c.f. Theorem 2 in Section 6 of [SS81]).

Theorem 7.20 *Let Ω be an orientable open subset of a manifold and $\{g^k\}_k$ and $\{\Gamma^k\}_k$, respectively, sequences of smooth metrics on Ω and of hypersurfaces satisfying* (**SS**) *in Ω. Assume that the metrics $\{g^k\}_k$ converge smoothly to a metric g, each Γ^k is stable and minimal relative to the metric g^k , and $\sup_k \mathcal{H}^{n-1}(\Gamma^k) < \infty$. Then* *there are a subsequence of $\{\Gamma^k\}_k$ (not relabeled), a stable stationary varifold V in Ω (relative to the metric g), and a closed set S of Hausdorff dimension at most $n-8$ such that:*

(i) V is a smooth embedded hypersurface in $\Omega \setminus S$;

(ii) $\Gamma^k \to V$ as varifolds in Ω;

(iii) Γ^k converges smoothly to V on every open $\Omega' \Subset \Omega \setminus S$.

Remark 7.21 *The smooth convergence of the subsequence $\{\Gamma^k\}$ in the last point is understood in the following sense: take an open set $\Omega'' \subset \Omega'$ where the varifold V is an integer multiple N of a smooth oriented surface Σ. Then, for k sufficiently large, $\Gamma^k \cap \Omega''$ is the union of N disjoint surfaces Γ_i^k, $i = 1, \dots, N$, that are normal graphs over Σ of functions $f_i^k \in C^\infty(\Sigma)$. The convergence is smooth, in the sense that, for every $l \in \mathbb{N}$ and $\varepsilon > 0$, $\|f_i^k\|_{C^l} < \varepsilon$, if k is sufficiently large.*

Finally, we show that the singularities are removable in low dimensions. The idea is to consider a blow-up procedure to show that the minimal surface can be expressed locally as the union of minimal graphs of Lipschitz functions. Using Simons' classification of stable hypercones, we conclude that all graphs coincide up to first order at the singularity and are disjoint elsewhere; hence, by an application of the strong maximum principle, there can only be a single graph. Heuristically, one first discards singularities arising from crossings, and subsequently eliminates those arising from foldings.

Proposition 7.22 *If $3 \le n \le 7$, for every $x \in \operatorname{spt} V$, any tangent cone to V at x is an integer multiple of a hyperplane. Furthermore, x is in the regular set of V.*

Sketch of Proof Let $\rho > 0$ be small enough such that V is supported on a stable embedded minimal hypersurface Σ in $B_\rho(x) \setminus \{x\}$. Denote by $T_\rho : B_1(0) \subset T_xM \to B_\rho(x)$ the rescaled exponential map defined by $T_\rho(z) := \exp_x(\rho z)$. If $\{\rho_k\}$ is a sequence of scales such that $\rho_k \to 0$ as $k \to \infty$, then using Theorem 7.20, we find that the rescaled surfaces $\Sigma_k := T_{\rho_k}^{-1}(\Sigma)$ converge to a stable minimal hypercone C in $B_1(0) \setminus \{0\} \subset T_xM$. Since $n \le 7$, by Simons' Theorem, C must be an integer multiple of a hyperplane.

Choose dyadic scales $\rho_k = 2^{-k}$. For any $c > 0$, provided k is large enough, there exists a hyperplane π_k such that $\Sigma_k \cap (B_1 \setminus \overline{B_{1/2}})$ is the union of $m(k)$ disjoint graphs $\Gamma_1^k, \dots, \Gamma_m^k$ over π_k of Lipschitz functions $f_1, \dots, f_m$ with $\mathrm{Lip}(f_i) \le c$, and each graph has multiplicity $j_i(k)$ such that $\sum_i j_i(k) = N$.

Because $\Sigma_k \to C$ and C is flat, the tilt between consecutive planes π_k and π_{k+1} vanishes as $k \to \infty$. Consequently, an inductive patching procedure across the concentric annuli implies that $\Sigma \setminus \{x\}$ is the union of m disjoint smooth embedded minimal hypersurfaces $\Gamma_1, \dots, \Gamma_m$, each homeomorphic to a punctured disk, with multiplicities summing to N. Since each Γ_i is a minimal surface with density 1 and possesses flat tangent cones, it follows from Allard's regularity theorem that each Γ_i extends smoothly across the puncture to become a regular disk (see Fig. 7.7). Finally, because all the Γ_i coincide at x up to first order but are strictly disjoint in the punctured neighborhood, the strong maximum principle for minimal surfaces (see Remark 6.17) dictates that they cannot touch unless they are the exact same surface everywhere. Therefore, $m = 1$, meaning there is only a single sheet, and x is a regular point of Σ. □

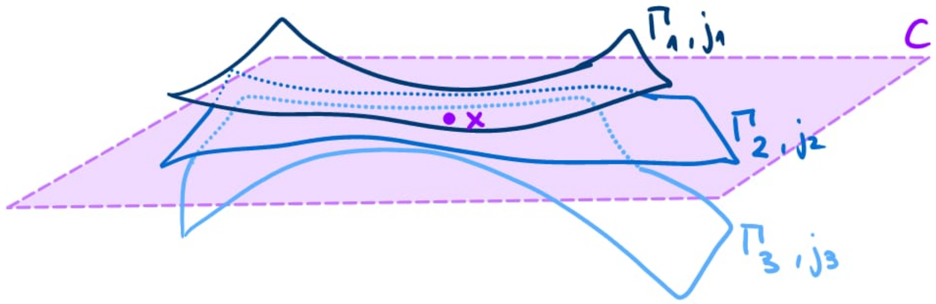

Figure 7.7: Sketch of the situation before applying the maximum principle in the proof Proposition 7.22 for $m = 3$.

Remark 7.23 (The $n = 2$ case) *The previous regularity argument fails when $n = 2$. This is due to two topological obstructions. First, the patching procedure across concentric annuli relies on Lemma 5.5 in [LT13], which requires the singular set to have codimension at least 2. This is impossible when $n = 2$. Second, the final step invokes the maximum principle on a punctured neighborhood; however, a punctured 1-dimensional curve is not connected.*

This failure for $n = 2$ is not merely a technical artifact of the proof, but a limitation of the method itself. While the stable regularity theory developed in previous chapters still holds for $n = 2$ (ensuring regularity of geodesics arising from stable critical points), this regularity does not extend to critical points with Morse index 1. The optimal existence result obtainable from the Allen–Cahn method on surfaces is a geodesic that is smooth everywhere except possibly at a single point with a transverse self-intersection of two components; as proved by Mantoulidis [Man21].

Nevertheless, the existence of a non-trivial closed (everywhere smooth) geodesic on any closed Riemannian 2-manifold is a classical result, proved by Birkhoff in 1917 [Bir17]. This existence result also holds for closed manifolds of arbitrary dimension, as shown by Lyusternik and Fet in 1951 [Fet52].

7.5 Closing Remarks and Further Developments

Guaraco's proof establishes the Allen–Cahn phase transition framework as an alternative to the classical Almgren–Pitts theory. However, the construction of a single minimal hypersurface leaves open questions regarding the multiplicity of the limit, the exact correspondence of the Morse index, and the existence of infinitely many such surfaces.

While Guaraco's theorem guarantees the existence of a minimal hypersurface, we have seen that the limit varifold V is an integer multiple of a smooth surface, $V = N\Sigma$. Ruling out higher multiplicities ($N > 1$) is a standard difficulty in both geometric and PDE-based min-max theories. Furthermore, while Guaraco bounded the geometric Morse index of the limit by the PDE Morse index of the sequence ($m(\Sigma) \leq \liminf m(u_\varepsilon)$), exact equality was not established. Chodosh and Mantoulidis [CM20] resolved these issues for generic metrics on 3-manifolds. By establishing curvature and strong sheet separation estimates for stable solutions, they proved that Allen–Cahn min-max limits occur as two-sided embedded minimal surfaces with multiplicity $N = 1$. They also established that the Morse index of the resulting minimal surface exactly equals the Morse index of the Allen–Cahn critical points.

The existence of a single minimal surface naturally leads to the question of whether closed manifolds admit an infinite spectrum of such surfaces. This was formulated by Yau in 1982.

Conjecture 7.24 (Yau, 1982) *Every closed Riemannian 3-manifold contains infinitely many geometrically distinct, closed minimal surfaces.*

Using the classical Almgren–Pitts machinery, this conjecture was proven by the combined works of Irie, Marques, and Neves [IMN18] for generic metrics, Marques and Neves [MN17] for manifolds with positive Ricci curvature, and Song [Son23] for arbitrary metrics.

By extending Guaraco's 1-parameter min-max argument to k-parameter sweepouts, Gaspar and Guaraco [GG19] developed an Allen–Cahn version of the volume spectrum used in [IMN18]. Combined with the multiplicity-one and index bounds of Chodosh and Mantoulidis, this phase-transition method provides an independent PDE proof of Yau's conjecture for generic 3-manifolds. Furthermore, this resolution confirms the Marques–Neves index lower bound and multiplicity one conjectures, yielding the conclusion that for every integer $m \geq 1$, a generic 3-manifold contains a two-sided embedded minimal surface with Morse index m and area scaling as $m^{1/3}$.

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Appendix A

The Coarea Formula

This appendix states the Coarea formula, which generalizes Fubini's theorem to Lipschitz maps between spaces of different dimensions.

Theorem A.1 *Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be a Lipschitz map with $n \geq m$. For any Lebesgue measurable set $A \subset \mathbb{R}^n$ and any $g \in L^1(A)$, we have:*

$$\int_A g(x) J_f(x) \, \mathrm{d}x = \int_{\mathbb{R}^m} \left(\int_{A \cap f^{-1}(y)} g(x) \, \mathrm{d}\mathcal{H}^{n-m}(x) \right) \, \mathrm{d}y,$$

where $J_f(x) = \sqrt{\det\left(df_x \circ (df_x)^T\right)}$ is the m-dimensional Jacobian of f.

The same holds for Riemannian manifolds.

Theorem A.2 *Let (M, g_M) and (N, g_N) be smooth Riemannian manifolds of dimensions n and m respectively, with $n \geq m$. Let $f : M \to N$ be a Lipschitz map. For any measurable set $A \subset M$ and $g \in L^1(A)$,*

$$\int_A g(x) J_f(x) \, \mathrm{d}\mu_M(x) = \int_N \left(\int_{A \cap f^{-1}(y)} g(x) \, \mathrm{d}\mathcal{H}^{n-m}(x) \right) \, \mathrm{d}\mu_N(y),$$

where $J_f(x) = \sqrt{\det\left(df_x \circ (df_x)^*\right)}$ and $(df_x)^*$ denotes the adjoint of the differential with respect to the metrics g_M and g_N.

Particularly relevant is the case when the target is $N = \mathbb{R}$ and $f : M \to \mathbb{R}$ is a smooth function, the Jacobian simplifies to the norm of the gradient, $J_f(x) = |\nabla f(x)|$. The formula then reduces to integrating over the level sets $\Sigma_y = \{x \in M \mid f(x) = y\}$:

$$\int_M g(x) |\nabla f(x)| \, \mathrm{d}\mu_M(x) = \int_{-\infty}^{\infty} \left(\int_{\Sigma_y} g(x) \, \mathrm{d}\mathcal{H}^{n-1}(x) \right) \, \mathrm{d}y.$$

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